WHO INFLUENZA CENTRE

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Southern Hemisphere Vaccine Recommendation Meeting for 2015 influenza season.

WHO CC, London: Final Report for WHO meeting: 22-25 September 2014.

Influenza Activity in the WHO European Region, October 2013 - July 2014

Weekly reporting on influenza activity started in week 40/2013 and terminated in week 20/2014. The 2013-2014 season began late compared to the 2012-2013 season with levels of transmission increasing slowly following week 50/2013 (**Figure 1**). The numbers of weekly influenza detections were considerably lower than the numbers of detections reported in the previous season and remained so throughout **(Figure 1B)**. The proportion of respiratory specimens from ILI and ARI patients testing positive peaked at ~40% over weeks 03-08/2014 (**Figure 1C**), over which period many countries reported increasing trends in ILI and ARI and peaks were reached in most countries of the region.

More than 48,000 influenza detections were reported to EuroFlu from across the whole WHO European Region with type A detections (93%) predominating over type B (7%) **(Table 1A)**. Within type A, the A(H1NI)pdm09 subtype has been slightly more prevalent (56%) compared to A(H3N2) viruses (44%) **(Figure 1 & Table 1A)**. However, in EU/EEA countries over the period weeks 21-35/2014, of the small number of type A viruses detected and subtyped, H3N2 viruses have predominated by a ratio of ~2.8:1 (**Table 1B**). Within type B, the great majorioty of viruses for which lineage-determination was performed were of the Yamagata lineage.

Even during week 08/2014, when the greatest number of influenza detections was reported, no country was reporting very high influenza activity and 'dominant' virus reporting varied between countries (**Figure 2**). At the same time, in terms of geographic spread, widespread/regional activity was largely restricted to western and southern Europe.

Within individual countries, while influenza A virus detections greatly outnumbered those of influenza B, levels of influenza A subtype detections (H1/H3) have shown considerable variation **(Figures 2 & 3)**. Overall, influenza activity across the Region was substantially lower than that seen in the 2012-2013 season, particularly for influenza B. Indeed, in a number of countries with established ILI/ARI epidemic thresholds these were either not (e.g. United Kingdom [England]) or barely reached (e.g. Sweden and the Russian Federation) while it was greatly exceeded only in Spain (**Figure 3**).

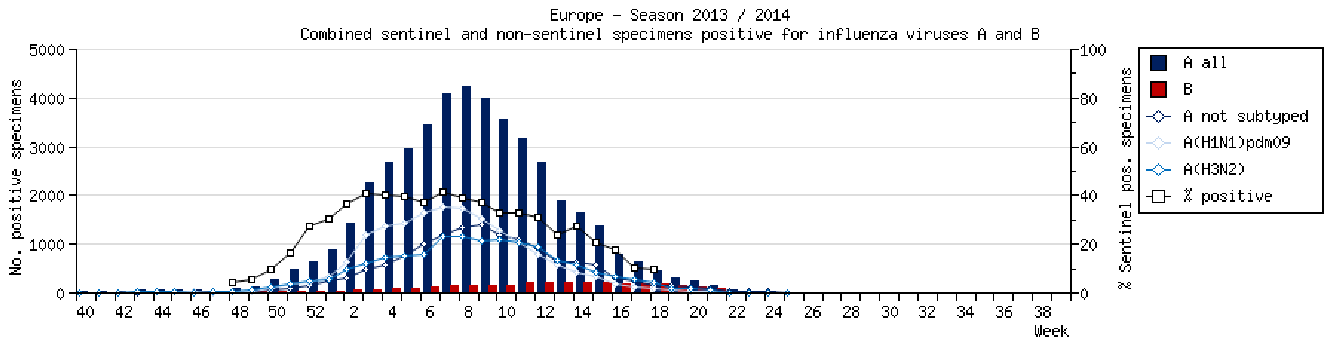
There are two systems for reporting severe disease associated with influenza infection in the WHO European Region. Severe Acute Respiratory Infections (SARI) from surveillance at sentinel hospitals have been reported to EuroFlu by 11 countries located in south-eastern European and west-central Asian subregions: since week 40/2014, 1011 cases yielding influenza virus detections have been reported of which 965 (95%) were influenza A and 46 (5%) influenza B. Subtyping was performed for 931 influenza A viruses: 330 (35%) were

A(H1N1)pdm09 and 601 (65%) A(H3N2). EU/EEA countries have reported on hospitalization of severe laboratory-confirmed influenza cases to the European Centre for Disease Prevention and Control (ECDC). Up to week 35/2014, 4779 cases had been reported, 4714 (99%) were associated with influenza A and 65 (1%) with influenza B infection: 3231 influenza A viruses were subtyped, 2392 (74%) as A(H1N1)pdm09 and 839 (26%) as A(H3N2). Of the 4779 infections, 422 resulted in deaths with 5 being related to type B and 417 to type A influenza infection: 303 type A viruses were subtyped, 247 (82%) as A(H1N1)pdm09 and 56 (18%) as A(H3N2). The differences in proportions of type A and B viruses, and A(H1N1)pdm09 and A(H3N2) viruses, between the two reporting systems may be due, in part, to differences in circulating virus populations across the Region in the 2013-2014 season and differences in health care seeking in individual countries.

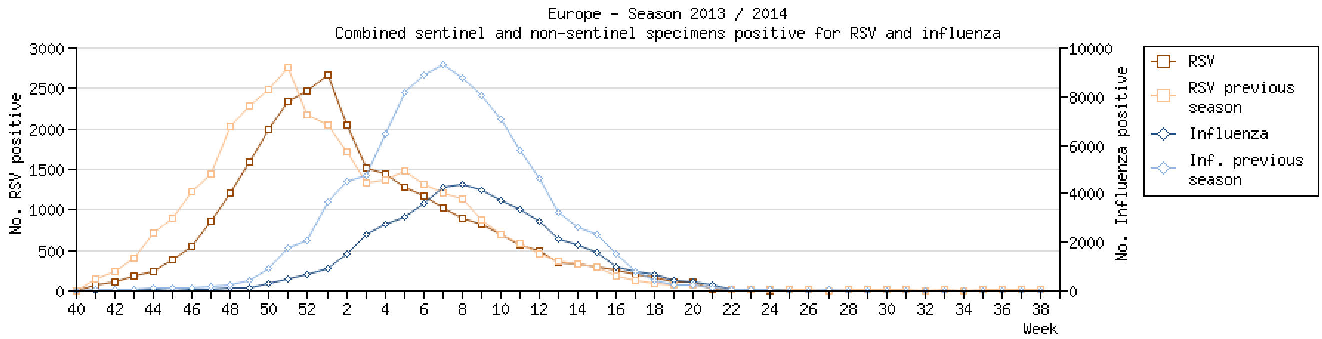
Up to week 20/2014, 13 countries had reported results to EuroFlu on susceptibility to antiviral drugs, oseltamivir and zanamivir - covering 1212 A(H1N1)pdm09, 410 A(H3N2) and 72 type B influenza viruses. Sixteen A(H1N1)pdm09 viruses carrying the neuraminidase H275Y amino acid substitution, resulting in highly reduced inhibition by oseltamivir, were detected with 15 being in hospitalized patients in the UK. A single influenza A(H3N2) virus showing reduced susceptibility to NA inhibitors was reported; it carried the neuraminidase E119V amino acid substitution, was detected in a hospitalized patient treated with oseltamivir in the UK, and showed reduced inhibition by oseltamivir but normal inhibition by zanamivir. All influenza B viruses tested showed normal susceptibility to oseltamivir and zanamivir. Overall, there has been no indication of increased resistance to the neuraminidase inhibitors oseltamivir and zanamivir during the 2013-2014 season in the WHO European Region. All A(H1N1)pdm09 and A(H3N2) influenza viruses screened for susceptibility to adamantanes were found to to carry the S31N resistance marker.

Figure 1. Influenza Activity in WHO European Region: 2013-2014 Season (weeks 40-25) – *data from Euroflu.org*

A. Total Influenza Detections



B. Past Season RSV and Influenza



C. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season

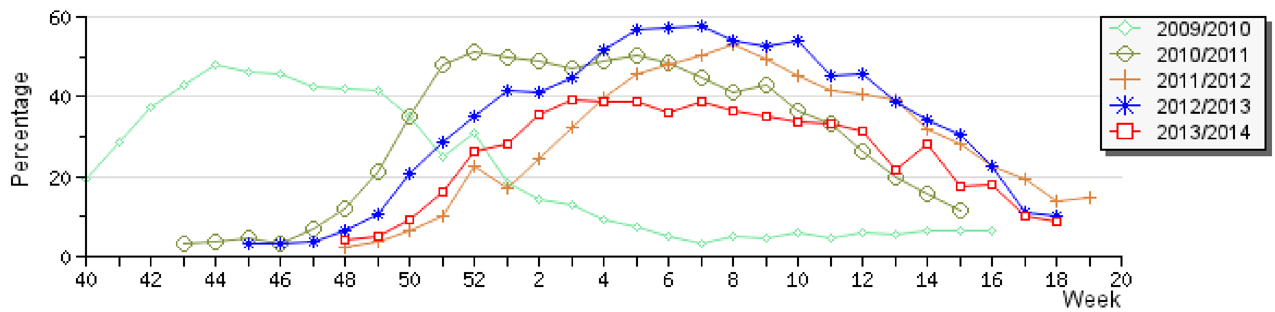
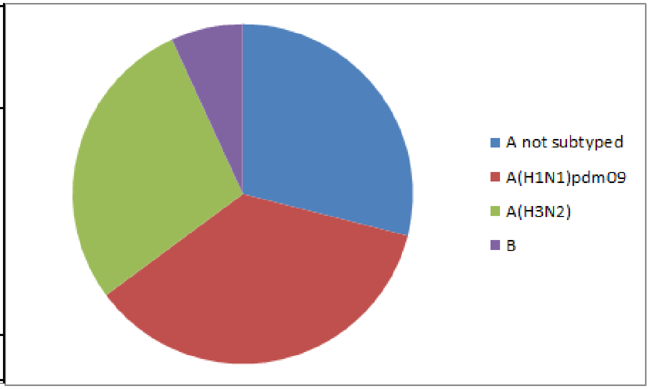


Table 1. Influenza Typing and Subtyping in the WHO European Region: 2013-2014 Season (weeks 40-25) – *data compiled in Euroflu.org (2014-06-25) and ECDC (week 30/2014)*

A. Influenza Detections in the WHO European Region Weeks 40/2013 – 20/2014

Influenza	Number Detected	%	A Subtype Ratio
A Total	44982	93.2	
A not subtyped	13980	29.0	
A(H1N1)pdm09	17334	35.9	1.3
A(H3N2)	13668	28.3	1
B	3301	6.8	
Total	48283		



B. Influenza Detections in EU/EEA Countries Weeks 21-35/2014

Virus type / subtype	Sentinel wks 21-35/2014	Non Sentinel wks 21-35/2014
Influenza A	8	374
A not subtyped	1	216
A(H1N1)pdm09	0	44
A(H3N2)	7	114
Influenza B	2	131
B lineage unknown	0	115
B/Vic lineage	0	2
B/Yam lineage	2	14
Total influenza	10	505

Figure 2. The Situation in the WHO European Region in the Week of Peak Number of Influenza Detections (week 08/2014) - ***data compiled in Euroflu.org (2014-06-25)***

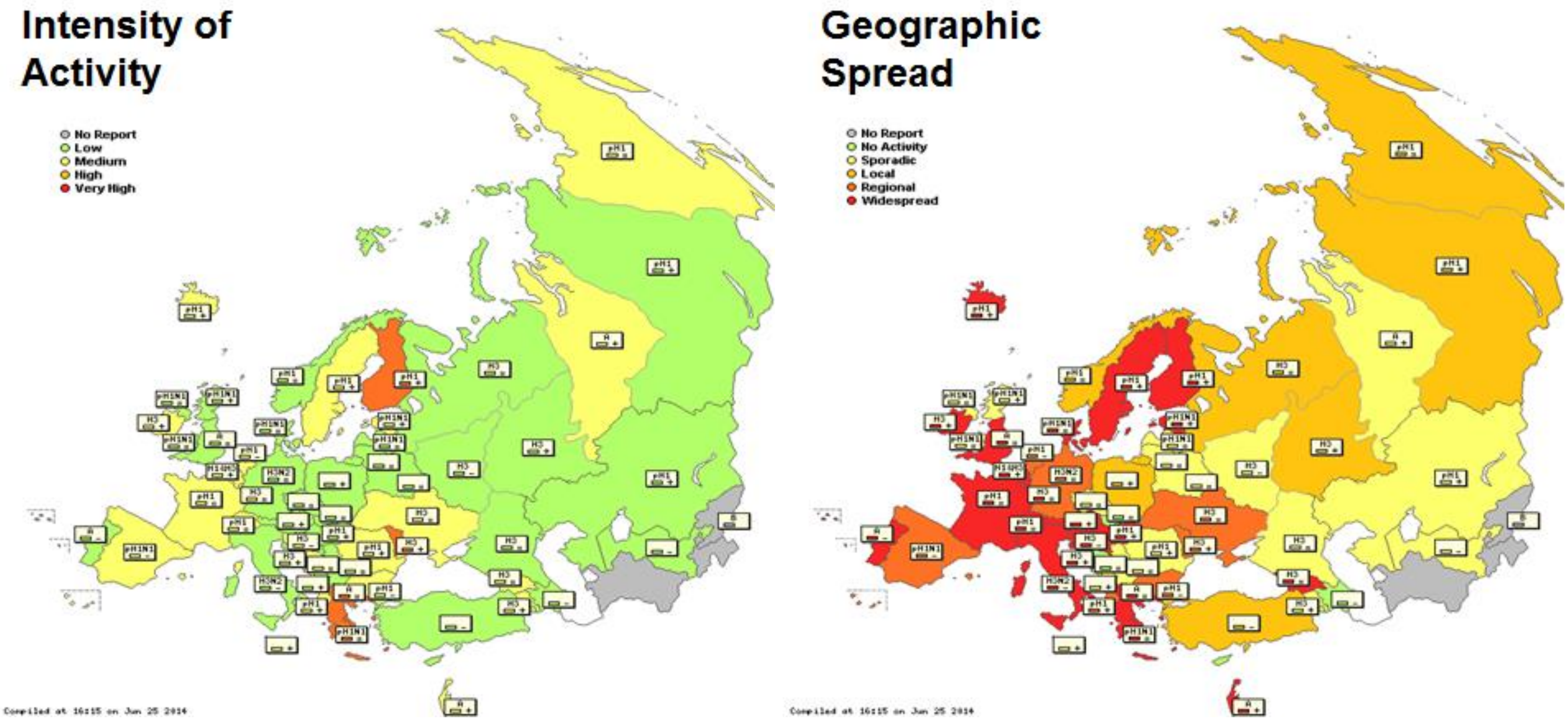
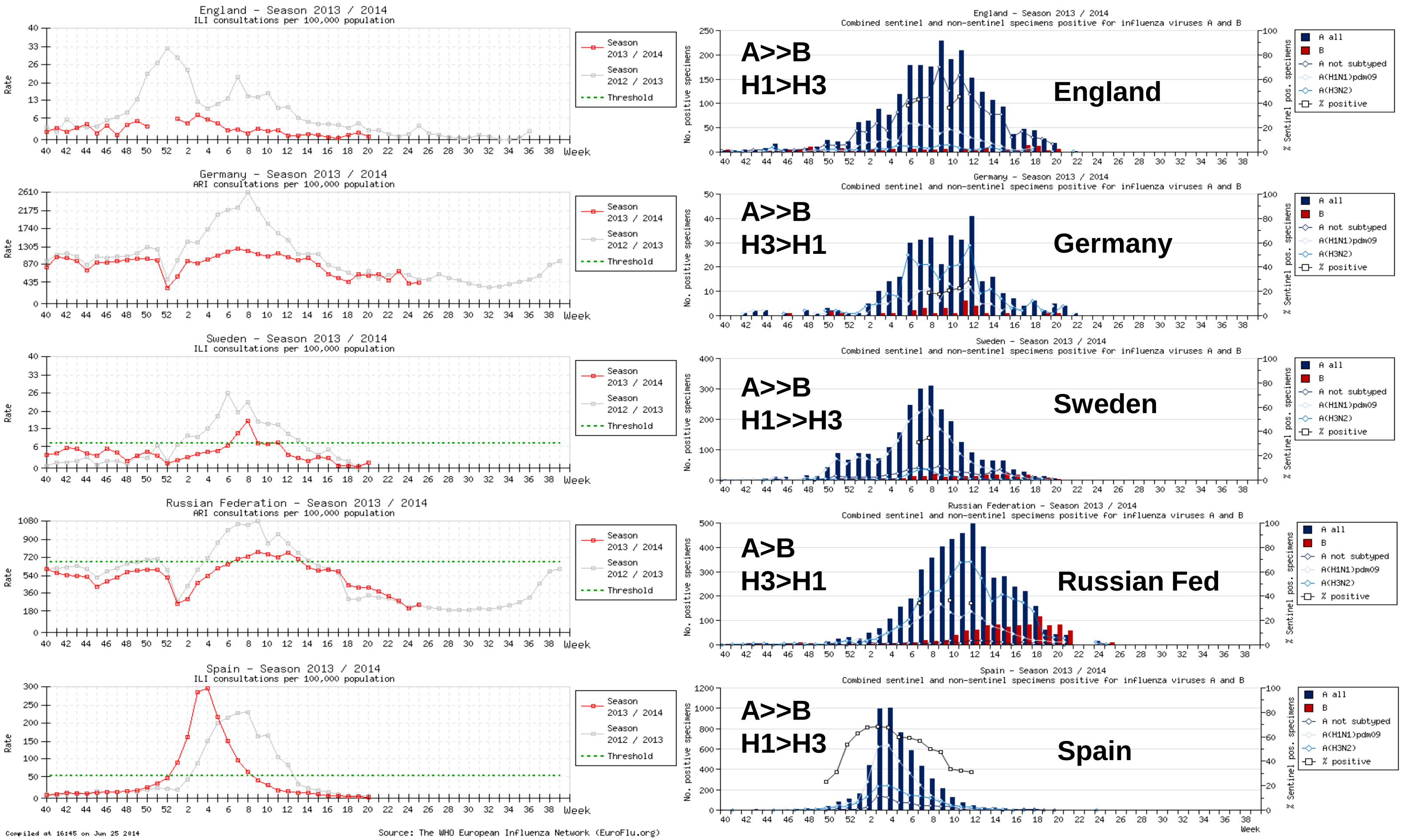


Figure 3. Influenza ILI/ARI Consulations and Type/Subtype Circulation Patterns in the WHO European Region - *data compiled in Euroflu.org (2014-06-25)*

ILI/ARI consultations **Influenza detections (total)**



= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥5

Influenza specimens received at NIMR

Samples, viruses or clinical specimens, with collection dates after 2014-01-31 have been received from 39 countries in Europe, Africa, the Indian Ocean (La Réunion), the Middle East, Central Asia, the Far East and South America (Argentina). The large majority (~85%) were type A viruses, with A(H3N2) viruses predominating over A(H1N1)pdm09 viruses by a ratio of ~2:1 (**Table 2**). Of the type B viruses (just over 15% of all specimens received) B/Yamagata lineage viruses predominated over those of the B/Victoria lineage by a ratio of ~2:1. These proportions of influenza types/subtypes/lineages are generally similar to those reported within the WHO European Region (**Figure 1 & Table 1A**). **Table 3** shows the isolation rates from specimens received with collection dates after 2014-01-31. Isolates from each type/subtype/lineage were recovered efficiently. **Table 4** summarises the number of viruses of the A(H1N1)pdm09 and A(H3N2) subtypes, and the two B/lineages, that showed ≤2-, 4- and ≥8-fold reductions in HI titres with post-infection ferret antisera compared with the titres against their respective homologous vaccine/reference viruses. For A(H3N2) viruses a summary of the results of plaque-reduction neutralisation assays is also shown.

Table 2. Summary of clinical samples and virus isolates received, with collection dates after 2014-01-31, by country

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]
2014											
FEBRUARY											
Albania	10		7	7	3	2					
Austria	3				3	3					
Belgium	6		3	1	3	2					
Bulgaria	26		20	20	6	5					
Cameroon	2						2				
Cyprus	12	1	11	11							
Finland	8		3	3	5	4					
Georgia	9		1	1	5	5				3	3
Germany	11		4	4	3	3		2	2	2	2
Hong Kong	2				2	2					
Hungary	7		5	5	2	1					
Iceland	6		3	3	3	3					
Ireland	4		4	4							
Israel	22				13	11	7			2	2
Italy	28		12	11	14	14				2	2
Kazakhstan	23		8	3	15	7					
Latvia	1		1	1							
Lithuania	4	1	1	1	2	2					
Macedonia	25		13	0	6	1	6				
Norway	13		5	4	8	7					
Poland	9		2	1	7	5					
Portugal	6		4	4	1	1				1	1
Russia	5		4	4				1	1		
Senegal	5		5	5							
Serbia	23	1	4	3	18	15					
Slovakia	5		3	3	2	2					
Slovenia	20		6	6	14	11					
South Africa	3		2	2				1	1		
Spain	1				1	0					
Sweden	7		1	1	4	4				2	2
Switzerland	2		1	1	1	1					
Ukraine	52	3	2	0	44	30	1			2	2
United Kingdom	3				2	1		1	1		
MARCH											
Austria	10		2	2	5	5				3	3
Belarus	2		1	1	1	1					
Belgium	7		4	2	3	1					
Bulgaria	1		1	1							
Cameroon	6							4	4	2	2
Finland	3		1	1	2	2					
France	15		10	10	4	4				1	1
Georgia	7		1	1	5	4				1	1
Hong Kong	1				1	1					
Iceland	7		4	4	1	1		2	2		
Ireland	2		2	2							
Israel	9						5			4	4
Italy	3		2	2	1	1					
Kazakhstan	12		7	7	5	5					
Latvia	11		7	5	3	3				1	1
Lithuania	10				10	10					
Macedonia	10		7	0	2	0	1				
Norway	7				6	4		1	1		
Poland	26	1	2	0	23	9					
Portugal	2		1	1	1	1					
Romania	13		3	3	9	8				1	1
Russia	14		4	4	7	7		1	1	2	2
Senegal	1		1	1							
Serbia	17		5	3	12	7					
Slovakia	3		1	1	1	1				1	1
Slovenia	3		2	2	1	1					
Spain	2		1	1						1	1
Switzerland	7				5	4				2	2
Ukraine	39	2	1	1	30	25	4			2	2
United Kingdom	8		4	4	2	2		1	1	1	1

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]
APRIL											
Austria	3		1	1	2	2					
Belarus	3		2	2	1	1					
Belgium	10		2	1	7	4		1	1		
Cameroon	3		1	0			1	1	1		
Finland	3				3	3					
France	8		3	3	5	4					
Ghana	4		1	1	2	2		1	1		
Hong Kong	5				5	5					
Iceland	3		2	2				1	1		
Ireland	7		1	1	6	6					
Latvia	2				2	2					
Lithuania	8				8	8					
Norway	1				1	1					
Poland	3		2	0	1	1					
Portugal	1				1	1					
Reunion	1		1	1							
Russia	16		9	9	2	1		2	2	3	in process
Senegal	1				1	1					
Serbia	1				1	1					
Slovakia	3		1	1	2	2					
Slovenia	2		1	1	1	1					
South Africa	1				1	1					
Spain	7				7	7					
MAY											
Belarus	2									2	2
Cameroon	3				1	1	2				
France	1									1	1
Ghana	13				11	11		2	2		
Hong Kong	3							3	3		
Iceland	3				2	2				1	1
Lithuania	1				1	1					
Norway	5		2	1	1	1				2	2
Reunion	3		1	1						2	2
Portugal	1									1	1
South Africa	6		1	1	5	5					
United Kingdom	1				1	1					
JUNE											
Argentina	16				11	in process	1	2	in process	2	in process
Cameroon	12				10	9				2	2
France	1		1	1							
Ghana	8		3	2	1	1	2	2	2		
Hong Kong	6		3	3						3	3
Iceland	1				1	1					
Norway	4				3	2				1	1
Reunion	4		2	2						2	2
Senegal	2				2	2					
South Africa	7		1	1	5	5				1	1
Spain	1				1	1					
United Kingdom	1				1	1					
JULY											
Argentina	34				24	in process	1	3	in process	6	in process
Cameroon	5				2	1	3				
Ghana	5		2	2	3	3					
Senegal	6				6	6					
South Africa	2				2	2					
Spain	4				1	1	1			2	2
AUGUST											
France	1				1	in process					
	879	9	252	205	482	360	37	32	27	67	56
			28.7%		54.8%			3.6%		7.6%	
39 Countries			84.5%				15.5%				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Table 3. Isolation rates for specimens collected after 2014-01-31

	Clinical sample			Virus isolate		
	number processed	number propagated[1]	percent recovered	number processed	number propagated[1]	percent recovered
H1	75	33	44%	180	174	97%
H3	166	108	65%	296	258	87%
B Victoria	3	3	100%	30	30	100%
B Yamagata	16	16	100%	41	41	100%

1. Propagated to sufficient titre to perform HI assay

Table 4. Summary of isolates with collection dates after 2014-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays

	Number tested	≤2-fold[a]	4-fold[a]	≥8-fold[a]
H1[1]	205	198 (96.6%)	6 (2.9%)	1 (0.5%)
H3[2]	354	2 (0.6%)	45 (12.7%)	307 (86.7%)
H3[c]	11	1 (9.1%)	2 (18.2%)	8 (72.7%)
B/Victoria[3]	30	13 (43.4%)	10 (33.3%)	7 (23.3%)
B/Yamagata[4]	57	12 (21.1%)	26 (45.6%)	19 (33.3%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:
1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009
2. Compared to homologous titres for ferret antisera raised against vaccine virus A/Texas/50/2012
c. Compared to homologous titres for ferret antisera raised against vaccine virus A/Texas/50/2012 in plaque-reduction neutralisation assays
3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)
4. Compared to homologous titres for ferret antisera raised against vaccine virus B/Massachusetts/02/2012

Influenza A(H1N1) virus analysis.

H1N1pdm09 viruses were received from NICs in Europe, Africa/Indian Ocean (Cameroon, Ghana, Senegal, South Africa and La Réunion), Central Asia (Kazakhstan) and Hong Kong SAR. **Table 5** summarises the antigenic (HI) results for viruses that were propagated successfully at NIMR.

The vast majority (>96%) of the H1N1 viruses collected after 2014-01-31 were antigenically similar to the vaccine virus (**Table 5**); only six test viruses (2.9% of the total tested) showed 4-fold reductions and one virus showed an 8-fold reduction in HI titre with post-infection ferret antiserum raised against the vaccine virus (A/California/7/2009) compared to the homologous titres. **Tables 6-1 to 6-16** show the results of HI assays of H1N1 viruses carried out since the February 2014 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2014-01-31 are shown below red lines in **Tables 6-2**, and **6-4 to 6-16**. Of these 205 test viruses seven showed reduced reactivity with the panel of post-infection ferret antisera; these were from Bulgaria, Hungary, Iceland (genetic subgroup 6B), Ghana and Senegal (genetic subgroup 6C). No marked sequence differences were observed in the HAs of A/Ghana/DILI-0618/2014 and A/Ghana/DILI-0620/2014 (**Table 6-15**) while the HA of A/Dakar/03/2014 showed amino acid polymorphism at position 155 of HA1 (**G155E>G**; **Table 6-14**) and that of A/Dakar/06/2014 (**Table 6-15**) showed HA1 K119N substitution resulting in the gain of a potential N-linked glycosylation site. The HA sequences of two viruses from Iceland, A/Iceland/61/2014 and A/Iceland/48/2014 (**Table 6-11**), carried the substitutions **E224A** in **HA1** and **V193A** in **HA2**, both showed low reactivity to the panel of antisera but only A/Iceland/61/2014 showed reduced recognition by the antiserum raised against A/California/7/2009. The remaining two viruses, A/Bulgaria/775/2014 (**Table 6-5**) and A/Hungary/400/2014 (**Table 6-15**), each carried single rare amino acid substitutions in **HA1**, **H273N** and **S183P** respectively.

Figure 4 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last five years the HA genes have evolved and eight genetic groups have been designated, with A/California/7/2009 representing group 1. Viruses collected after 2014-01-31 fell into genetic group 6. The viruses clustered into two of the three subgroups of group 6, subgroups 6B and 6C (**Figure 4**). These two subgroups carry amino acid substitutions **D97N**, **S185T, S203T** and **K283E** in **HA1** and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009. The two subgroups, 6B and 6C, are defined by the following amino acid substitutions in **HA1** with none in **HA2**:

- Viruses in **subgroup 6B** carry substitutions **K163Q** and **A256T** in **HA1**, e.g. A/South Africa/3626/2013
- Viruses in **subgroup 6C** carry the substitution **V234I** in **HA1;** some, notably a group of viruses from Ghana, also carry the substitutions **V30A, I116M,**

A186T and **M257V** in **HA1**, e.g. A/Ghana/DARI-0095/2014. A group of viruses from Senegal carry the substitution **A141T** in **HA1**, e.g. A/Dakar/02/2014.

An amino acid sequence alignment of the HAs for a selected group of viruses is shown in **Figure 5**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 6 illustrates the location on an H1-HA structure of amino acid residues defining genetic subgroups 6B and 6C.

Figure 7 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 8** shows an amino acid alignment for a selected group of these viruses. The NA gene of viruses from West Africa with HA genes in genetic group 6C also cluster into two subgroups. Overall, the HA and NA phylogenetic trees are generally congruent.

Antiviral resistance.

Figure 24 shows a phylogenetic tree of representative H1N1 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the **S31N** substitution in the **M2** ion-channel protein, known to confer resistance to adamantanes.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on close to 200 viruses recovered from clinical samples with collection dates after 2014-01-31. All but two viruses showed normal inhibition (NI) to both neuraminidase inhibitors (**Table 14**). Neuraminidase (NA) amino acid substitutions associated with the reductions in inhibition are shown on the table. A/Cyprus/F79/2014 carries **S247N** and is associated with reduced susceptibility to zanamivir and oseltamivir but the fold-reductions against both drugs are below the cut-off (10-fold) to place the virus in the reduced inhibition (RI) category. A/Paris/1823/2014 carries **H275Y** and shows highly reduced inhibition (HRI) by oseltamivir. One virus, A/Hungary/366/2014 collected in January, also carries the **H275Y** substitution but has yet to be analysed in the sialidase inhibition assay.

Conclusion.

The vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus, A/California/7/2009, and retain susceptibility to oseltamivir and zanamivir.

Table 5. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2014-01-31

MONTH / Country	H1N1pdm09 Number received	Number propagated[1]	Fold reduction in HI titre ≤2 fold[2]	4 fold[2]	≥8 fold[2]
FEBRUARY					
Albania	7	7	7		
Belgium	3	1	1		
Bulgaria	20	20	19	1	
Cyprus	11	11	11		
Finland	3	3	3		
Georgia	1	1	1		
Germany	4	4	4		
Hungary	5	5	4	1	
Iceland	3	3	3		
Ireland	4	4	4		
Italy	12	11	11		
Kazakhstan	8	3	3		
Latvia	1	1	1		
Lithuania	1	1	1		
Norway	5	4	4		
Poland	2	1	1		
Portugal	4	4	4		
Russia	4	4	4		
Senegal	5	5	3	2	
Serbia	4	3	3		
Slovakia	3	3	3		
Slovenia	6	6	6		
South Africa	2	2	2		
Sweden	1	1	1		
Switzerland	1	1	1		
MARCH					
Austria	2	2	2		
Belarus	1	1	1		
Belgium	4	2	2		
Bulgaria	1	1	1		
Finland	1	1	1		
France	10	10	10		
Georgia	1	1	1		
Iceland	4	4	3	1	
Ireland	2	2	2		
Italy	2	2	2		
Kazakhstan	7	7	7		
Latvia	7	5	5		
Portugal	1	1	1		
Romania	3	3	3		
Russia	4	4	4		
Senegal	1	1	1		
Serbia	5	3	3		
Slovakia	1	1	1		
Slovenia	2	2	2		
Spain	1	1	1		
Ukraine	1	1	1		
United Kingdom	4	4	4		
APRIL					
Austria	1	1	1		
Belarus	2	2	2		
Belgium	2	1	1		
France	3	3	3		
Ghana	1	1	1		
Iceland	2	2	2		
Ireland	1	1	1		
Reunion	1	1	1		
Russia	9	9	9		
Slovakia	1	1	1		
Slovenia	1	1	1		
MAY					
Norway	2	in process	1		
Reunion	1	1	1		
South Africa	1	1	1		
JUNE					
France	1	1	1		
Ghana	3	2	2		
Hong Kong	3	3	3		
Reunion	2	2	2		
South Africa	1	1	1		
JULY					
Ghana	2	2		1	1
35 Countries	225	204	198 96.6%	6 2.9%	1 0.5%

1. Propagated to sufficient titre to perform HI assays
2. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 6-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-02-19 + 2014-02-26)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	40	40	40	80	80	80	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	1280	2560	2560	2560	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	640	1280	2560	1280	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	640	2560	1280	2560	5120	5120	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	1280	1280	2560	2560	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	320	640	320	640	1280	1280	2560	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	160	640	640	640	1280	1280	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	640	1280	1280	1280	1280	1280	
TEST VIRUSES														
A/Bucuresti/157141/2014	6B	2013-01-06	MDCK1/MDCK2	1280	640	1280	1280	2560	2560	2560	2560	2560	2560	
A/Sichuan-Wuhou/SWL2259/2013		2013-09-30	E3/E1	1280	640	1280	1280	2560	1280	2560	5120	2560	1280	
A/Chongqing-Yuzhong/SWL11434/2013		2013-10-12	E3/E1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Wakayama/153/2013	6B	2013-11-05	MDCK1/MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/1920/2013		2013-11-11	MDCK3	1280	640	1280	1280	2560	2560	2560	2560	2560	2560	
A/Cote d'Ivoire/1926/2013	6C	2013-11-12	MDCK2	320	640	640	160	160	160	320	320	320	160	
A/Sichuan-Yucheng/SWL1693/2013		2013-11-14	E3/E1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/1961/2013		2013-11-18	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	5120	1280	
A/Cote d'Ivoire/1993/2013		2013-11-18	SIAT2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Cote d'Ivoire/2033/2013		2013-11-26	MDCK2	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/2093/2013	6B	2013-12-03	SIAT2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Cote d'Ivoire/2089/2013		2013-12-03	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/2149/2013		2013-12-04	SIAT3	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/2115/2013		2013-12-05	MDCK2	1280	640	2560	1280	2560	2560	2560	5120	2560	2560	
A/Cote d'Ivoire/2198/2013		2013-12-13	MDCK3	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Switzerland/9772556/2013	6B	2013-12-23	SIAT2	1280	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Malta/MV745469/2013	6B	2013-12-30	MDCK2	2560	2560	2560	2560	5120	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/2279/2013		2013-12-30	MDCK2	2560	640	2560	1280	2560	2560	5120	5120	5120	2560	
A/Cote d'Ivoire/2273/2013	6C	2013-12-31	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Madagascar/00022/2014		2014-01-02	MDCK2	1280	640	1280	1280	2560	5120	1280	2560	2560	1280	
A/Switzerland/9836540/2014	6B	2014-01-06	MDCK2	1280	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Cote d'Ivoire/0012/2014	6C	2014-01-06	MDCK2	1280	640	1280	640	2560	2560	2560	5120	2560	1280	
A/Madagascar/00023/2014		2014-01-06	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Malta/MV035034/2014	6B	2014-01-07	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Switzerland/9836685/2014	6B	2014-01-08	MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	2560	2180	
A/Switzerland/9851425/2014		2014-01-08	MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Madagascar/00141/2014		2014-01-13	SIAT2	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Malta/MV35493/2014	6B	2014-01-14	MDCK2	1280	320	640	640	1280	2560	1280	2560	2560	1280	
A/Malta/MV35488/2014		2014-01-14	MDCK2	1280	640	1280	1280	2560	1280	1280	2560	2560	1280	
A/Switzerland/9884275/2014	6B	2014-01-14	MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Switzerland/9884048/2014	6B	2014-01-15	MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	5120	1280	
A/Switzerland/9884265/2014		2014-01-15	MDCK2	1280	1280	2560	1280	2560	2560	2560	5120	5120	1280	
A/Malta/MV746793/2014	6B	2014-01-19	MDCK2	2560	1280	2560	2560	5120	5120	5120	5120	5120	2560	

Sequences in phylogenetic trees   Vaccine

Table 6-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-03-05)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	640	640	640	640	640	640	640	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	1280	1280	1280	320	160	160	320	160	640	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	5120	2560	5120	5120	5120	5120	5120	5120	5120	5120	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	1280	320	1280	640	2560	2560	1280	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	2560	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	1280	1280	2560	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	1280	1280	1280	1280	2560	2560	2560	2560	1280	
TEST VIRUSES														
A/Catalonia/6590S/2013	6B	2013-11-29	C0/MDCK1	640	160	320	640	1280	640	640	1280	1280	640	
A/Catalonia/6634S/2013	6B	2013-12-12	C0/MDCK1	1280	1280	2560	2560	5120	2560	5120	5120	5120	2560	
A/Catalonia/2142245NS/2013	6C	2013-12-20	C0/MDCK2	640	640	320	160	80	160	320	160	320	160	G155G=E
A/Valladolid/145/2013		2013-12-23	MDCK1/SIAT1/EGG2/MDCK1	2560	2560	5120	2560	5120	5120	5120	5120	5120	5120	
A/Catalonia/2145903NS/2013		2013-12-30	C0/MDCK1	1280	640	1280	1280	5120	2560	2560	5120	2560	2560	
A/Valencia/621679S/2014		2014-01-02	C0/MDCK1	1280	1280	2560	2560	5120	2560	2560	5120	2560	2560	
A/Burgos/11/2014	6B	2014-01-08	MDCK1/SIAT1/MDCK1	5120	2560	5120	2560	5120	5120	5120	5120	5120	5120	
A/Athens.GR/14/2014		2014-01-10	MDCK2	2560	2560	2560	5120	5120	5120	5120	5120	5120	2560	
A/Athens.GR/18/2014		2014-01-10	MDCK2	1280	640	1280	2560	2560	2560	2560	5120	2560	1280	
A/Catalonia/2148757NS/2014		2014-01-11	C0/MDCK1	1280	640	1280	1280	5120	1280	2560	5120	1280	1280	
A/Astros.GR/39/2014		2014-01-12	MDCK3	2560	2560	2560	5120	5120	5120	5120	5120	5120	2560	
A/Astros.GR/42/2014	6B	2014-01-12	MDCK3	1280	640	1280	2560	2560	2560	2560	5120	5120	2560	
A/Arkadia.GR/55/2014	6B	2014-01-14	MDCK3	1280	640	1280	1280	2560	1280	2560	2560	2560	1280	
A/Athens.GR/71/2014	6B	2014-01-14	MDCK3	640	320	640	1280	1280	1280	1280	2560	2560	640	
A/Lamia.GR/77/2014	6B	2014-01-14	MDCK3	320	640	320	320	40	80	320	160	320	160	G155E
A/Athens.GR/68/2014		2014-01-15	MDCK3	1280	2560	2560	2560	2560	2560	5120	5120	5120	2560	
A/Athens.GR/70/2014		2014-01-15	MDCK3	2560	2560	2560	2560	2560	5120	5120	5120	5120	2560	
A/Lamia.GR/80/2014		2014-01-15	MDCK2	1280	1280	2560	2560	2560	2560	5120	5120	5120	2560	
A/Valladolid/46/2014		2014-01-15	MDCK1/SIAT1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	2560	2560	
A/Palencia/55/2014		2014-01-15	MDCK1/SIAT1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	2560	2560	
A/Athens.GR/89/2014		2014-01-16	MDCK2	2560	2560	2560	2560	2560	5120	5120	5120	5120	2560	
A/Catalonia/2150477NS/2014		2014-01-17	C0/MDCK1	2560	1280	2560	2560	5120	2560	2560	5120	2560	2560	
A/Valencia/624382S/2014	6B	2014-01-17	C0/MDCK1	1280	640	1280	1280	5120	1280	2560	5120	2560	2560	
A/Athens.GR/109/2014		2014-01-19	MDCK2	1280	640	640	1280	1280	2560	1280	5120	2560	1280	
A/Athens.GR/116/2014	6B	2014-01-20	MDCK2	5120	1280	2560	2560	2560	5120	2560	5120	5120	2560	
A/Burgos/80/2014		2014-01-21	MDCK1/SIAT1/MDCK1	5120	2560	5120	2560	5120	5120	5120	5120	5120	5120	
A/Leon/10020/2014	6B	2014-01-22	MDCK1/SIAT1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	2560	2560	
A/Berlin/7/2014	6B	2014-01-22	C2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Berlin/11/2014		2014-01-22	C2	2560	1280	2560	2560	5120	5120	5120	5120	5120	2560	
A/Valladolid/116/2014		2014-01-27	MDCK1/SIAT1/MDCK1	2560	1280	2560	5120	5120	5120	5120	5120	5120	2560	
A/Hessen/5/2014	6B	2014-02-03	C2/MDCK1	1280	1280	1280	1280	5120	2560	2560	5120	5120	2560	
A/Hamburg/2/2014	6B	2014-02-04	C2/MDCK1	2560	1280	2560	2560	5120	2560	2560	5120	5120	2560	
A/Brandenburg/2/2014		2014-02-04	C2/MDCK1	1280	1280	1280	1280	5120	2560	2560	5120	5120	2560	
A/Sachsen/1/2014	6B	2014-02-05	C1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	5120	2560	

Vaccine

Table 6-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-03-19 + 2014-03-25)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera A/Cal 7/09 F30/11	*A/Cal 7/09 F05/14	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	*A/DR 7293/13 F06/14 6C	
REFERENCE VIRUSES																
A/California/7/2009		2009-04-09	E1/E2	640	640	640	640	640	640	640	640	640	640	640	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	80	160	80	40	40	40	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	320	1280	640	160	80	160	320	160	320	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	1280	5120	1280	2560	1280	2560	2560	1280	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	640	320	640	640	1280	1280	1280	2560	2560	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	320	640	640	1280	1280	2560	2560	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	640	1280	640	1280	2560	2560	2560	1280	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	640	1280	1280	1280	2560	2560	5120	1280	1280	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	320	320	320	320	640	640	640	1280	1280	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	640	1280	1280	1280	1280	1280	1280	1280	
A/Dominican Republic/7293/2013	6C	2013-05-01	E3/E1	640	640	640	1280	640	1280	2560	1280	2560	2560	1280	2560	
TEST VIRUSES																
A/Dakar/03/2013	6C	2013-09-26	MDCK2	640	640	640	640	640	2560	2560	1280	2560	2560	1280	1280	
A/Dakar/04/2013		2013-09-26	MDCK2	2560	1280	1280	1280	1280	5120	5120	2560	5120	5120	2560	2560	
A/Dakar/01/2013		2013-09-30	MDCK2	1280	1280	640	1280	1280	2560	5120	2560	5120	5120	2560	2560	
A/Dakar/05/2013	6C	2013-10-30	MDCK4	1280	ND	640	640	640	2560	2560	1280	5120	2560	1280	ND	
A/Dakar/17/2013	6C	2013-11-01	MDCK3	320	ND	320	320	320	640	640	640	1280	2560	640	ND	
A/Dakar/16/2013		2013-11-03	MDCK2	1280	1280	640	1280	640	2560	2560	2560	2560	2560	1280	2560	
A/Dakar/08/2013		2013-11-12	MDCK2	1280	2560	1280	2560	2560	5120	5120	2560	5120	5120	2560	5120	
A/Cameroon/6450/2013	6C	2013-11-16	SIAT2	640	320	1280	1280	1280	1280	1280	1280	5120	2560	640	1280	
A/Dakar/09/2013		2013-11-17	MDCK2	1280	640	640	1280	640	2560	2560	2560	2560	2560	1280	1280	
A/Jordan/30653/2013	6B	2013-11-25	MDCK2	1280	ND	640	1280	1280	2560	2560	2560	5120	5120	2560	ND	
A/Dakar/10/2013		2013-11-27	MDCK3	640	640	640	1280	640	2560	2560	1280	2560	2560	640	1280	
A/Jordan/30663/2013	6B	2013-11-28	SIAT1/MDCK1	160	ND	160	160	160	320	320	160	160	320	160	ND	
A/Dakar/11/2013	6C	2013-12-02	MDCK3	1280	1280	640	1280	1280	2560	5120	2560	5120	5120	1280	2560	
A/Dakar/15/2013	6C	2013-12-17	MDCK2	1280	1280	640	1280	640	2560	2560	2560	5120	2560	1280	1280	
A/Athens.GR/23/2014	6B	2014-01-07	MDCK2	1280	1280	1280	1280	1280	5120	2560	2560	5120	2560	2560	2560	
A/Athens.GR/19/2014		2014-01-08	MDCK3	1280	1280	1280	1280	1280	5120	5120	2560	5120	2560	2560	2560	

* New antisera; ND = Not Done

Vaccine

Table 6-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-04-01 + 2014-04-08)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre / Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	320	320	320	320	640	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	320	160	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	2560	5120	5120	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	320	1280	1280	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	1280	1280	1280	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	5120	2560	5120	5120	5120	5120	5120	5120	5120	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	1280	1280	1280	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	320	640	640	1280	1280	1280	5120	5120	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	640	1280	640	2560	1280	1280	
TEST VIRUSES														
A/Cameroon/5524/2013		2013-10-07	MDCK3	320	160	320	320	640	1280	640	1280	1280	640	
A/Cameroon/6297/2013		2013-11-12	MDCK3	2560	1280	2560	1280	5120	5120	2560	5120	5120	2560	
A/Cameroon/6457/2013	6C	2013-11-15	MDCK3	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Togo/LNG448/2013	6C	2013-11-26	MDCK3	320	160	320	320	640	640	640	1280	1280	640	
A/Cameroon/6804/2013		2013-11-29	MDCK3	1280	640	1280	640	2560	2560	2560	5120	5120	1280	
A/Cameroon/6814/2013		2013-12-03	MDCK3	1280	640	2560	1280	2560	2560	2560	5120	5120	2560	
A/Cyprus/F5/2014		2014-01-21	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Cyprus/F10/2014	6B	2014-01-23	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F19/2014		2014-01-27	MDCK2	1280	320	1280	1280	2560	2560	5120	5120	2560	1280	
A/Cyprus/F20/2014		2014-01-27	MDCK2	640	320	640	640	1280	1280	640	2560	2560	1280	
A/Cyprus/F22/2014	6B	2014-01-27	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Cyprus/F24/2014	6B	2014-01-27	MDCK3	640	320	640	320	640	640	640	2560	1280	640	
A/Cyprus/F18/2014	6B	2014-01-27	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Cyprus/F37/2014		2014-02-03	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F38/2014		2014-02-03	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560	1280	
A/Cyprus/F39/2014		2014-02-03	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560	1280	
A/Cyprus/F45/2014		2014-02-04	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F51/2014		2014-02-05	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F60/2014	6B	2014-02-10	MDCK2	1280	640	1280	1280	2560	2560	1280	5120	5120	1280	
A/Cyprus/F76/2014		2014-02-12	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Cyprus/F79/2014	6B	2014-02-17	MDCK2	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F81/2014		2014-02-17	MDCK2	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Cyprus/F88/2014	6B	2014-02-18	MDCK2	640	640	1280	1280	2560	2560	1280	5120	2560	1280	
A/Cyprus/F94/2014	6B	2014-02-20	MDCK3	640	320	640	640	1280	1280	1280	2560	2560	1280	

Vaccine

Table 6-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-05-07 + 2014-05-14)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	2560	5120	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	1280	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK6	1280	640	1280	1280	2560	2560	2560	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	320	1280	1280	640	1280	1280	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	320	640	640	1280	1280	640	1280	1280	2560	
TEST VIRUSES														
A/Tehran/53422/2013		2013-07-29	MDCK2/MDCK1	640	320	640	640	640	1280	1280	2560	1280	1280	
A/Poland/1480/2014	6B	unknown	MDCK3	1280	640	2560	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/047/2014	6B	2014-01-09	SIAT2/MDCK1	640	320	1280	1280	1280	1280	2560	2560	2560	1280	
A/Bulgaria/475/2014	6B	2014-01-30	SIAT1/MDCK1	1280	1280	2560	2560	2560	5120	2560	5120	5120	2560	
A/Bulgaria/478/2014		2014-01-30	SIAT1/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	1280	
A/Switzerland/10015059/2014	6B	2014-02-03	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Bulgaria/665/2014		2014-02-05	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Bulgaria/659/2014		2014-02-06	SIAT1/MDCK1	640	640	1280	1280	2560	1280	2560	2560	2560	1280	
A/Bulgaria/706/2014		2014-02-07	SIAT2/MDCK1	640	640	1280	1280	2560	2560	2560	2560	2560	1280	
A/Bulgaria/819/2014		2014-02-08	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	1280	1280	1280	
A/Bulgaria/821/2014		2014-02-08	SIAT2/MDCK1	640	320	640	640	1280	2560	2560	2560	2560	1280	
A/Bulgaria/823/2014	6B	2014-02-08	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	2560	2560	1280	
A/Bulgaria/812/2014	6B	2014-02-09	SIAT2/MDCK1	640	640	1280	1280	1280	2560	1280	2560	2560	1280	
A/Bulgaria/700/2014		2014-02-10	SIAT2/MDCK1	1280	320	1280	1280	2560	1280	2560	2560	2560	2560	
A/Bulgaria/723/2014		2014-02-10	SIAT2/MDCK1	1280	640	2560	2560	2560	2560	2560	5120	2560	2560	
A/Bulgaria/750/2014	6B	2014-02-10	SIAT2/MDCK1	640	320	640	1280	1280	1280	1280	2560	2560	1280	
A/Bulgaria/775/2014	6B	2014-02-11	SIAT2/SIAT2	320	160	640	320	640	640	640	1280	1280	640	
A/Bulgaria/726/2014	6B	2014-02-11	SIAT2/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/774/2014		2014-02-11	SIAT1/MDCK1	1280	640	2560	2560	2560	2560	2560	5120	5120	2560	
A/Bulgaria/747/2014		2014-02-12	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Bulgaria/824/2014	6B	2014-02-12	SIAT1/MDCK1	1280	640	2560	2560	2560	5120	2560	5120	5120	2560	
A/Perugia/28/2014	6B	2014-02-14	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	1280	
A/Bulgaria/793/2014	6B	2014-02-14	SIAT2/MDCK1	640	320	640	640	1280	2560	1280	2560	2560	2560	
A/Bulgaria/805/2014		2014-02-14	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Georgia/275/2014	6B	2014-02-14	MDCK2	640	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/795/2014		2014-02-16	SIAT2/MDCK1	1280	320	1280	1280	2560	2560	2560	5120	5120	2560	
A/Milano/117/2014		2014-02-17	SIAT1/MDCK1	1280	640	2560	2560	2560	2560	2560	5120	5120	2560	
A/Bulgaria/797/2014	6B	2014-02-17	SIAT2/MDCK1	640	160	320	320	1280	1280	640	1280	1280	1280	
A/Milano/112/2014	6B	2014-02-18	SIAT1/MDCK1	640	640	1280	640	1280	1280	2560	5120	2560	1280	
A/Bulgaria/837/2014	6B	2014-02-19	SIAT1/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	2560	2560	
A/Poland/16K/2014	6B	2014-02-21	MDCK3	1280	640	1280	1280	1280	1280	1280	5120	2560	1280	
A/Perugia/38/2014	6B	2014-02-26	MDCK1/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	2560	2560	
A/Bulgaria/942/2014	6B	2014-03-03	SIAT2/MDCK1	1280	640	2560	1280	5120	2560	2560	5120	2560	2560	
A/Perugia/44/2014		2014-03-10	MDCK1/MDCK1	640	640	1280	1280	1280	2560	2560	2560	5120	1280	
A/Perugia/45/2014	6B	2014-03-14	MDCK2/MDCK1	640	320	640	640	1280	1280	640	1280	2560	1280	
A/Georgia/622/2014	6B	2014-03-18	MDCK2	1280	640	1280	2560	2560	2560	2560	5120	2560	2560	

Vaccine

Table 6-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-05-21)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	320	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	640	2560	1280	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	640	320	640	1280	640	1280	2560	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	1280	1280	2560	2560	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	1280	2560	2560	1280	5120	5120	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	320	640	1280	640	2560	2560	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	1280	1280	1280	
TEST VIRUSES														
A/Bulgaria/1042/2013	6B	2013-12-20	SIAT1/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	1280	
A/Bulgaria/009/2014		2014-01-05	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Bulgaria/015/2014		2014-01-07	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/017/2014		2014-01-07	SIAT2/MDCK1	640	320	1280	1280	1280	1280	1280	5120	2560	1280	
A/Bulgaria/039/2014		2014-01-08	SIAT2/MDCK1	640	320	1280	640	1280	1280	1280	5120	2560	1280	
A/Bulgaria/042/2014	6B	2014-01-08	SIAT2/MDCK1	1280	640	1280	1280	1280	1280	1280	2560	2560	1280	
A/Bulgaria/093/2014		2014-01-13	SIAT2/MDCK1	640	320	1280	640	1280	1280	1280	5120	2560	1280	
A/Bulgaria/121/2014	6B	2014-01-13	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	2560	
A/Parma/9/2014		2014-01-15	MDCK1/MDCK1	1280	640	1280	640	1280	2560	2560	5120	5120	1280	
A/Bulgaria/171/2014		2014-01-16	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/177/2014		2014-01-16	SIAT2/MDCK1	640	640	640	640	1280	2560	1280	2560	2560	1280	
A/Bulgaria/211/2014		2014-01-16	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	5120	5120	5120	2560	
A/Bulgaria/217/2014	6B	2014-01-16	SIAT2/MDCK1	640	640	1280	640	1280	2560	1280	5120	2560	1280	
A/Bulgaria/218/2014		2014-01-16	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	5120	2560	1280	
A/Bulgaria/219/2014		2014-01-16	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Parma/7/2014		2014-01-20	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Bulgaria/513/2014	6B	2014-01-22	SIAT2/MDCK1	640	320	1280	640	1280	2560	2560	5120	2560	1280	
A/Milano/24/2014		2014-01-22	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/348/2014		2014-01-23	SIAT2/MDCK1	640	320	640	640	640	1280	1280	2560	1280	1280	
A/Bulgaria/386/2014		2014-01-23	SIAT2/MDCK1	1280	640	1280	640	1280	2560	1280	5120	2560	1280	
A/Ukraine/28/2014	6B	2014-01-24	SIAT2/MDCK1	640	320	1280	640	2560	1280	1280	5120	2560	1280	
A/Bulgaria/408/2014		2014-01-27	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Bulgaria/435/2014	6B	2014-01-28	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Parma/15/2014		2014-01-28	MDCK1/MDCK1	1280	640	1280	640	5120	2560	2560	5120	5120	1280	
A/Bulgaria/447/2014		2014-01-29	SIAT1/MDCK1	1280	640	1280	2560	2560	2560	2560	5120	5120	2560	
A/Bulgaria/449/2014		2014-01-29	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	640	
A/Bulgaria/451/2014		2014-01-29	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	5120	2560	640	
A/Bulgaria/483/2014		2014-01-29	SIAT2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Firenze/5/2014	6B	2014-01-30	SIAT1/MDCK1	640	320	640	320	1280	1280	1280	2560	2560	640	
A/Parma/23/2014		2014-01-31	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Milano/69/2014	6B	2014-02-03	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	1280	
A/Roma/9/2014	6B	2014-02-03	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	640	
A/Genova/04/2014		2014-02-07	SIAT1/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280	
A/Genova/6/2014	6B	2014-02-07	SIAT1/MDCK1	640	320	320	320	640	1280	640	1280	1280	640	
A/Roma/10/2014	6B	2014-02-09	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Genova/01/2014	6B	2014-02-10	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Parma/31/2014	6B	2014-02-11	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	2560	

Vaccine

Table 6-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-06-04)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	640	320	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	320	160	160	320	160	640	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	1280	5120	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	1280	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	1280	640	640	640	1280	2560	1280	1280	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	1280	2560	1280	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	1280	2560	2560	1280	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	640	640	1280	2560	1280	2560	5120	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	1280	640	1280	1280	1280	1280	
TEST VIRUSES														
A/Finland/402/2014		2014-01-20	MDCK1/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280	
A/Poprad/104/2014		2014-01-30	MDCK1/MDCK1	1280	640	2560	1280	2560	5120	2560	5120	5120	2560	
A/Prievidza/109/2014	6B	2014-02-06	MDCK1/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	5120	2560	
A/Finland/416/2014		2014-02-11	MDCK1/MDCK1	640	640	640	640	2560	1280	1280	2560	2560	1280	
A/Finland/417/2014		2014-02-12	MDCK1/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Lubica/131/2014		2014-02-20	MDCK1/MDCK1	640	640	1280	640	2560	2560	1280	2560	2560	1280	
A/Kosice/132/2014	6B	2014-02-20	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Latvia/02-053060/2014	6B	2014-02-24	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Finland/420/2014	6B	2014-02-26	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	2560	5120	5120	2560	
A/Latvia/03-020560/2014	6B	2014-03-10	MDCK1/MDCK1	1280	640	2560	1280	2560	1280	1280	2560	2560	1280	
A/Finland/422/2014	6B	2014-03-10	MDCK1/MDCK1	1280	640	2560	1280	2560	2560	5120	5120	5120	2560	
A/England/348/2014		2014-03-11	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Latvia/03-023857/2014		2014-03-11	MDCK1/MDCK1	640	640	1280	1280	5120	2560	2560	5120	5120	5120	
A/Latvia/03-028999/2014		2014-03-12	MDCK1/MDCK1	640	640	640	640	2560	1280	1280	2560	2560	2560	
A/England/378/2014	6B	2014-03-17	SIAT1/MDCK1	2560	2560	2560	2560	5120	5120	2560	5120	5120	2560	
A/Latvia/03-043266/2014	6B	2014-03-19	MDCK2/MDCK1	1280	320	640	320	1280	1280	640	1280	1280	1280	
A/Latvia/03-050084/2014	6B	2014-03-20	MDCK1/MDCK1	1280	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/England/395/2014		2014-03-25	SIAT1/MDCK1	640	640	1280	1280	2560	1280	2560	2560	2560	1280	
A/England/403/2014	6B	2014-03-26	SIAT1/MDCK1	640	640	1280	1280	2560	2560	2560	2560	2560	1280	
A/Nove Zamky/202/2014	6B	2014-03-26	MDCK1/MDCK1	1280	1280	2560	2560	2560	5120	2560	5120	5120	2560	
A/Trencin/207/2014	6B	2014-04-02	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-06-11)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	40	40	40	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	80	160	320	640	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	640	1280	2560	2560	5120	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	640	320	640	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	320	320	640	640	640	1280	2560	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	640	640	640	1280	1280	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	640	1280	1280	1280	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	640	640	1280	1280	2560	5120	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	320	640	640	640	640	640	1280	1280	1280	
TEST VIRUSES														
A/Stockholm/37/2013	6B	2013-12-23	MDCK0/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Stockholm/3/2014	6B	2014-01-06	MDCK2/MDCK1	640	640	1280	320	640	1280	640	1280	1280	1280	
A/Serbia/NS-601/2014	6B	2014-01-08	MDCK1	1280	640	1280	1280	1280	2560	2560	5120	5120	2560	
A/Uppsala/3/2014	6B	2014-01-20	MDCK2/MDCK1	1280	640	1280	640	1280	2560	2560	5120	2560	1280	
A/Slovenia/215/2014		2014-01-22	MDCKx/MDCK1	1280	640	1280	640	2560	2560	2560	5120	2560	2560	
A/Slovenia/220/2014	6B	2014-01-22	MDCK1/MDCK1	640	320	640	320	1280	1280	1280	2560	2560	1280	
A/Slovenia/401/2014		2014-02-04	MDCK1/MDCK1	1280	640	1280	640	2560	1280	1280	5120	2560	2560	
A/Slovenia/423/2014		2014-02-05	MDCKx/MDCK1	1280	640	1280	640	2560	2560	2560	5120	5120	2560	
A/Slovenia/461/2014		2014-02-07	MDCK1/MDCK1	640	320	640	320	1280	1280	640	2560	1280	1280	
A/Ireland/9559/2014	6B	2014-02-09	MDCK1/MDCK1	640	640	1280	640	1280	2560	2560	5120	2560	2560	
A/Slovenia/608/2014		2014-02-13	MDCKx/MDCK1	1280	640	1280	640	2560	2560	1280	5120	2560	1280	
A/Stockholm/10/2014	6B	2014-02-18	MDCK0/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Ireland/11861/2014	6B	2014-02-19	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Slovenia/748/2014		2014-02-21	MDCKx/MDCK1	1280	640	1280	640	2560	2560	1280	5120	2560	1280	
A/Ireland/13388/2014		2014-02-21	MDCK1/MDCK1	1280	640	1280	640	1280	2560	1280	2560	2560	2560	
A/Ireland/13753/2014	6B	2014-02-24	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Slovenia/808/2014	6B	2014-02-25	MDCKx/MDCK1	1280	640	1280	640	2560	2560	2560	5120	2560	2560	
A/Serbia/NS-703/2014	6B	2014-02-28	MDCK2	1280	640	1280	640	2560	2560	2560	5120	2560	2560	
A/Slovenia/896/2014		2014-03-04	MDCKx/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Belgium/14S0252/2014	6B	2014-03-04	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Serbia/NS-735/2014	6B	2014-03-06	MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Ireland/17883/2014	6B	2014-03-14	MDCK3/MDCK1	1280	640	1280	640	1280	2560	2560	2560	2560	2560	
A/Serbia/NS-772/2014	6B	2014-03-18	MDCK1	1280	640	1280	640	1280	2560	2560	5120	2560	2560	
A/Ireland/18909/2014	6B	2014-03-22	MDCK1/MDCK1	1280	640	1280	1280	1280	2560	2560	5120	2560	2560	
A/Slovenia/1137/2014	6B	2014-03-24	MDCK1/MDCK1	1280	640	1280	640	1280	2560	1280	2560	2560	1280	
A/Ireland/21869/2014	6B	2014-04-06	MDCK1/MDCK1	640	640	1280	640	1280	2560	1280	5120	2560	1280	
A/Slovenia/1263/2014	6B	2014-04-10	MDCK1/MDCK1	1280	640	1280	1280	1280	2560	2560	5120	2560	2560	
A/Belgium/14G0500/2014	6B	2014-04-10	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-06-18)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	160	160	160	160	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	40	40	80	40	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	1280	1280	2560	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	320	320	1280	1280	640	1280	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	640	320	640	640	640	1280	1280	640	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	1280	1280	1280	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	320	640	640	640	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	320	1280	320	320	640	320	640	640	640	
TEST VIRUSES														
A/Gambia/G0122238/2013		2013-08-22	MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280	
A/Gambia/G0123236/2013		2013-08-22	MDCK1	1280	640	5120	1280	2560	2560	2560	5120	2560	2560	
A/Gambia/G0123336/2013	6C	2013-08-22	MDCK1	640	320	1280	1280	1280	1280	1280	2560	1280	1280	
A/Gambia/G0123436/2013		2013-08-23	MDCK1	640	160	640	320	1280	1280	1280	2560	1280	1280	
A/Gambia/G0128336/2013		2013-09-07	MDCK1	640	320	1280	640	2560	1280	1280	5120	2560	1280	
A/Gambia/G0131136/2013		2013-09-17	MDCK1	640	320	1280	320	1280	1280	1280	2560	1280	1280	
A/Gambia/G0136336/2013	6C	2013-09-28	MDCK1	640	160	640	320	640	1280	640	2560	1280	640	
A/Gambia/G0137036/2013	6C	2013-10-01	MDCK1	1280	320	2560	640	2560	2560	2560	5120	2560	2560	
A/Czech Republic/3/2014	6B	unknown	MDCK4/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280	
A/Czech Republic/4/2014	6B	unknown	MDCK5/MDCK1	640	320	640	320	1280	1280	1280	2560	1280	1280	
A/Czech Republic/5/2014	6B	unknown	MDCK5/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	2560	
A/Arad/161825/2014	6B	2014-03-07	MDCK1/MDCK1	320	160	320	320	640	640	640	1280	1280	1280	
A/Belgium/14S0350/2014	6B	2014-03-13	MDCK2	1280	640	1280	1280	5120	2560	1280	2560	2560	1280	
A/Arges/161972/2014		2014-03-13	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	2560	
A/Sibiu/162492/2014	6B	2014-03-19	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	2560	
A/Irkutsk/2/2014	6B	2014-03-25	C1/MDCK1	640	320	640	320	1280	1280	640	1280	640	1280	
A/Astrakhan/2/2014	6B	2014-03-25	C1/MDCK1	640	320	640	320	640	640	640	1280	1280	640	
A/Kaliningrad/3/2014	6B	2014-03-25	C1/MDCK1	640	640	1280	640	1280	1280	1280	5120	2560	1280	
A/Khabarovsk/6/2014	6B	2014-04-14	C1/MDCK2	1280	640	1280	640	1280	1280	1280	2560	1280	1280	

Sequences in phylogenetic trees

Vaccine

Table 6-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-06-25)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	160	160	320	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	40	40	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	320	640	320	1280	1280	1280	2560	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	1280	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	1280	1280	1280	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	320	1280	1280	1280	2560	2560	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	1280	1280	1280	
TEST VIRUSES														
A/Sarajevo/1/2013		2013-01-03	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Sarajevo/5/2013		2013-01-09	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Sarajevo/26/2013	6C	2013-01-14	MDCK2	320	640	320	160	160	320	320	320	320	160	K154E>K
A/Sarajevo/59/2013		2013-01-18	MDCK2	640	640	1280	320	1280	1280	1280	2560	1280	1280	
A/Sarajevo/94/2013		2013-01-24	MDCK1	1280	1280	2560	2560	2560	2560	2560	5120	5120	2560	
A/Sarajevo/111/2013		2013-01-30	MDCK1	1280	640	1280	640	2560	2560	2560	5120	5120	2560	
A/Sarajevo/151/2013		2013-02-18	MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/Gambia/G0122240/2013		2013-08-18	MDCK2	1280	640	1280	640	2560	2560	2560	5120	2560	1280	
A/Gambia/G0123240/2013		2013-08-22	SIAT2/SIAT1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Gambia/G0123340/2013	6C	2013-08-22	SIAT1	1280	640	640	160	320	320	320	640	640	320	N156D>N
A/Gambia/G0123440/2013	6C	2013-08-23	SIAT1	640	320	640	320	1280	2560	1280	2560	2560	1280	
A/Gambia/G0128340/2013	6C	2013-09-07	SIAT1	2560	640	2560	1280	5120	5120	2560	5120	5120	2560	
A/Gambia/G0136840/2013	6C	2013-10-01	SIAT1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Gambia/G0141436/2013		2013-10-18	SIAT2/SIAT1	1280	640	1280	640	2560	2560	1280	5120	2560	2560	
A/Gambia/G0141440/2013	6C	2013-10-18	MDCK3	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Czech Republic/2/2014	6B	unknown	MDCK7/MDCK2	320	160	80	40	<	40	80	40	40	80	N156D
A/Czech Republic/4/2014	6B	unknown	C2/E3/MDCK1	1280	640	1280	640	2560	1280	1280	5120	2560	1280	
A/Czech Republic/5/2014	6B	unknown	C2/E2/MDCK1	640	320	320	320	640	640	1280	2560	1280	640	
A/Belgium/14G0006/2014	6B	2014-01-04	MDCK1	1280	640	1280	2560	2560	2560	2560	5120	2560	2560	
A/Belgium/14H0004/2014		2014-02-04	MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	2560	
A/Odessa/260/2014		2014-03-06	SIAT2/SIAT1	1280	640	1280	640	2560	2560	1280	5120	2560	2560	

1. < = <40

Vaccine

Table 6-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-07-02 + 2014-07-09)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
				Haemagglutination inhibition titre – Post infection ferret antisera										
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	160	160	160	160	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	40	40	80	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	640	640	160	80	80	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	640	1280	1280	2560	1280	1280	1280	2560	1280	640	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	160	1280	640	640	1280	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	640	320	640	1280	640	2560	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	1280	640	1280	1280	1280	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	1280	1280	1280	2560	1280	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	80	320	160	320	640	640	1280	1280	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	320	320	160	320	320	320	640	640	640	
TEST VIRUSES														
A/Sarajevo/86/2013		2013-01-23	MDCK3	640	320	640	320	1280	1280	1280	2560	1280	1280	
A/Armenia/406/2013		2013-03-16	MDCK1	1280	1280	2560	1280	5120	2560	5120	5120	5120	1280	
A/Armenia/461/2013	6C	2013-03-21	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Austria/774096/2014	6C	unknown	SIAT0/MDCK1	1280	1280	1280	1280	1280	1280	2560	5120	2560	1280	
A/Iceland/6/2014	6B	2014-01-22	MDCK1/MDCK1	320	160	320	320	640	640	640	1280	1280	640	T72K, S162I
A/Iceland/10/2014		2014-01-27	MDCK1/MDCK1	640	640	1280	640	1280	1280	1280	5120	2560	1280	
A/Iceland/22/2014		2014-02-04	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Iceland/27/2014		2014-02-11	MDCK0/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Iceland/32/2014		2014-02-16	MDCK0/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Iceland/48/2014	6B	2014-03-01	MDCK1/MDCK1	320	160	320	160	320	320	640	640	640	640	E224A, V193A
A/Austria/781667/2014	6B	2014-03-03	SIAT1/MDCK1	640	640	1280	640	1280	1280	1280	5120	2560	1280	
A/Iceland/61/2014	6B	2014-03-13	MDCK1/MDCK1	160	160	160	160	640	320	320	640	640	320	E224A, V193A
A/Iceland/64/2014		2014-03-16	MDCK0/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Iceland/72/2014		2014-03-23	MDCK0/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Austria/787591/2014	6B	2014-03-31	SIAT2/MDCK1	1280	1280	2560	1280	2560	2560	2560	5120	2560	2560	
A/Iceland/75/2014	6B	2014-04-03	MDCK0/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Austria/789192/2014	6B	2014-04-05	SIAT1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Iceland/76/2014	6B	2014-04-08	MDCK0/MDCK1	640	320	640	640	640	1280	1280	2560	1280	1280	
A/St Petersburg/48/2014	6B	2014-04-08	E3/E1	640	320	640	320	640	640	640	1280	1280	2560	
A/Krasnoyarsk/2/2014	6B	2014-04-14	E2/E1	640	640	1280	640	1280	1280	1280	2560	2560	1280	
A/St Petersburg/60/2014		2014-04-14	E2/E1	640	640	640	640	1280	1280	1280	1280	1280	1280	
A/Chita/25/2014	6B	2014-04-28	E1/E1	320	320	640	320	640	640	1280	2560	1280	640	
A/Chita/5/2014		2014-04-28	E1/E1	640	320	640	320	640	640	1280	1280	1280	1280	
A/Hong Kong/6000/2014	6B	2014-06-08	MDCK1/MDCK2	1280	640	1280	1280	2560	1280	2560	2560	2560	2560	
A/Hong Kong/6110/2014	6B	2014-06-08	MDCK1/MDCK1	1280	1280	2560	1280	2560	2560	2560	5120	2560	2560	
A/Hong Kong/5999/2014	6B	2014-06-09	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-07-23 + 2014-07-29)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre / Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	*A/Chch 16/10 F15/14 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES															
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	320	160	320	320	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	320	160	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	5120	1280	2560	2560	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	320	320	320	640	640	640	1280	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	320	320	640	640	1280	640	1280	640	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	640	2560	1280	2560	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	1280	1280	1280	2560	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	320	320	640	640	640	2560	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	640	640	1280	1280	2560	1280	1280	
TEST VIRUSES															
A/Gambia/G0070536/2012	WAO[1]	2012-11-12	MDCK1	1280	640	1280	1280	ND	2560	2560	2560	5120	2560	2560	
A/Lisboa/132/2014	6B	2014-02-05	SIAT1/MDCK1	1280	640	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Tomsk/18/2014	6B	2014-02-10	MDCK1/MDCK1	640	640	1280	640	ND	1280	1280	1280	2560	2560	1280	
A/Tomsk/17/2014		2014-02-10	MDCK1/MDCK1	1280	640	1280	1280	ND	2560	2560	2560	2560	2560	1280	
A/Lisboa/120/2014	6B	2014-02-12	SIAT1/MDCK1	1280	640	1280	640	1280	1280	2560	2560	5120	2560	1280	
A/IIV-Moscow/207/2014	6B	2014-02-17	MDCK1/MDCK1	640	320	640	640	ND	1280	640	1280	2560	1280	1280	
A/Lisboa/127/2014	6B	2014-02-24	SIAT1/MDCK1	1280	640	1280	1280	1280	1280	2560	2560	5120	2560	1280	
A/Tomsk/3/2014	6B	2014-02-26	MDCK1/MDCK1	640	640	1280	1280	ND	2560	1280	2560	5120	2560	2560	
A/Lisboa/131/2014	6B	2014-02-26	SIAT1/MDCK1	1280	1280	2560	1280	1280	2560	2560	2560	5120	5120	2560	
A/Vladivostok/164/2014	6B	2014-03-03	MDCK2/MDCK1	640	320	640	80	ND	80	80	160	160	80	160	
A/Lisboa/130/2014	6B	2014-03-05	SIAT1/MDCK1	1280	1280	2560	1280	1280	2560	2560	2560	5120	5120	2560	
A/IIV-Moscow/78/2014	6B	2014-04-01	MDCK1/MDCK1	1280	1280	2560	2560	ND	5120	5120	2560	5120	5120	5120	
A/Vladivostok/121/2014	6B	2014-04-04	MDCK1/MDCK1	1280	640	1280	1280	ND	2560	2560	1280	5120	2560	2560	
A/Mogilev/1570/2014	6B	2014-04-05	MDCK1/MDCK1	640	320	640	640	ND	1280	1280	1280	2560	1280	1280	
A/Mogilev/1428/2014	6B	2014-04-07	MDCK2/MDCK1	640	320	640	640	ND	1280	1280	1280	2560	2560	1280	
A/Brest/1161/2014	1	2014-04-25	MDCK2/MDCK1	320	160	320	320	ND	640	640	640	1280	640	640	

Vaccine

* New antiserum; ND = Not Done

[1] WAO = West African Outlier group

Sequences in phylogenetic trees

Table 6-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-08-06)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/13 5	A/St. P 27/11 F23/11 6	*A/St. P 27/11 F26/14 6	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
REFERENCE VIRUSES															
A/California/7/2009		2009-04-09	E1/E2	640	640	640	160	160	160	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	640	160	40	40	80	80	40	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	2560	2560	1280	5120	5120	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	320	640	320	1280	1280	1280	640	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	1280	640	1280	640	1280	1280	1280	640	5120	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	1280	640	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	1280	2560	2560	1280	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	320	1280	1280	1280	640	2560	2560	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	1280	1280	640	1280	1280	1280	640	2560	1280	1280	
TEST VIRUSES															
A/South Africa/292/2014		2014-02-03	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	2560	5120	5120	5120	
A/Belgrade/1072/2014	6B	2014-02-12	C2/MDCK1	1280	640	1280	640	2560	2560	2560	1280	5120	2560	2560	
A/South Africa/606/2014	6B	2014-02-17	MDCK2/MDCK1	2560	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/Belgrade/1329/2014	6B	2014-02-19	MDCK1	2560	1280	2560	2560	5120	5120	5120	2560	5120	5120	5120	
A/Kazakhstan/3586/2014	6B	2014-02-23	SIAT1	2560	2560	5120	5120	5120	5120	5120	2560	5120	5120	5120	
A/Kazakhstan/3669/2014		2014-02-25	MDCK1	2560	1280	2560	2560	2560	5120	5120	2560	5120	5120	2560	
A/Kazakhstan/3670/2014	6B	2014-02-25	MDCK1	1280	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/Kazakhstan/3659/2014	6B	2014-03-04	MDCK1	2560	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/Kazakhstan/3660/2014		2014-03-04	SIAT1	1280	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/Kazakhstan/3662/2014		2014-03-04	SIAT1	1280	640	1280	1280	2560	2560	2560	1280	5120	2560	1280	
A/Kazakhstan/3664/2014		2014-03-04	MDCK1	1280	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/Kazakhstan/3665/2014	6B	2014-03-04	MDCK1	2560	1280	2560	2560	5120	5120	5120	2560	5120	5120	2560	
A/Kazakhstan/3777/2014	6B	2014-03-04	SIAT1	2560	1280	2560	1280	2560	2560	2560	2560	5120	5120	2560	
A/IIV-Moscow/138/2014	6B	2014-03-05	E1/E1	1280	640	1280	640	2560	1280	2560	1280	2560	2560	2560	
A/Kazakhstan/3668/2014	6B	2014-03-11	MDCK1	1280	1280	1280	1280	2560	2560	2560	1280	5120	5120	1280	
A/Extremadura/1756/2014	6B	2014-03-17	SIAT1/MDCK1	1280	1280	1280	1280	2560	2560	2560	1280	5120	5120	2560	
A/Kragujevac/2377/2014	6B	2014-03-26	C2/MDCK1	1280	640	1280	640	2560	2560	2560	1280	5120	2560	2560	
A/South Africa/4042/2014	6B	2014-05-17	MDCK1/MDCK1	1280	1280	2560	1280	2560	2560	2560	1280	5120	5120	2560	
A/South Africa/4764/2014	6B	2014-06-11	MDCK2/MDCK1	1280	640	1280	640	2560	2560	2560	1280	5120	2560	2560	

* New antiserum

Vaccine

Sequences in phylogenetic trees

Table 6-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-08-14)

				Haemagglutination inhibition titre										
				Post infection ferret antisera										
Viruses		Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	EP1/E2	1280	1280	1280	320	320	320	320	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	640	320	40	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E1	1280	2560	2560	5120	2560	2560	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK2	320	160	640	320	1280	1280	640	2560	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	1280	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	640	1280	2560	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	640	320	1280	1280	1280	2560	2560	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	1280	1280	1280	
TEST VIRUSES														
A/Dakar/02/2014	6C	2014-01-23	C3/MDCK1	1280	320	320	80	80	80	160	160	160	160	
A/Dakar/03/2014	6C	2014-02-03	C3/MDCK1	320	640	320	80	80	80	320	160	320	160	G155E>G
A/Dakar/07/2014	6C	2014-02-16	C1/MDCK1	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Lithuania/5569/2014	6B	2014-02-24	MDCK1/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	5120	2560	
A/Dakar/08/2014	6C	2014-03-27	C1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Norway/1916/2014	6B	2014-05-15	MDCK1/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-08-20)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6B	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	1280	640	1280	160	160	160	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	160	40	40	40	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E1	1280	1280	160	5120	640	2560	2560	5120	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	320	320	640	640	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	1280	640	1280	1280	640	2560	2560	5120	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	1280	320	320	1280	1280	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	640	1280	1280	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	640	640	1280	1280	2560	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	320	640	320	320	640	640	1280	1280	
TEST VIRUSES													
A/Hungary/366/2014	6B	2014-01-02	MDCK2/E2/MDCK1	320	160	320	320	640	640	640	1280	1280	
A/Dakar/01/2014		2014-01-22	C1/MDCK3	1280	640	1280	1280	2560	2560	2560	5120	1280	
A/Hungary/386/2014		2014-01-27	MDCK2/E2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Hungary/399/2014	6B	2014-01-27	MDCK2/E2/MDCK1	640	640	640	640	1280	1280	1280	2560	1280	
A/Albania/4851/2014		2014-02-02	MDCK2/MDCK1	640	320	640	640	1280	2560	2560	2560	2560	
A/Albania/4853/2014	6B	2014-02-02	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Hungary/400/2014	6B	2014-02-04	MDCK2/E2/MDCK1	320	160	640	320	640	640	640	1280	640	
A/Dakar/04/2014	6C	2014-02-05	C2/MDCK3	1280	640	320	80	640	160	160	320	160	
A/Hungary/403/2014		2014-02-06	MDCK2/E2/MDCK1	1280	320	640	640	1280	1280	1280	2560	1280	
A/Hungary/410/2014	6B	2014-02-06	MDCK2/E2/MDCK1	640	320	640	320	1280	640	1280	2560	1280	
A/Dakar/06/2014	6C	2014-02-07	C1/MDCK3	320	160	320	160	640	640	640	640	640	
A/Dakar/05/2014	6C	2014-02-10	C1/MDCK3	1280	320	1280	640	1280	1280	1280	5120	1280	
A/Hungary/418/2014	6B	2014-02-10	MDCK2/E2/MDCK1	1280	640	640	640	1280	1280	1280	2560	1280	
A/Norway/681/2014		2014-02-11	MDCK3/MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Albania/4937/2014	6B	2014-02-13	MDCK3/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Norway/667/2014	6B	2014-02-17	MDCK2/MDCK2	2560	1280	2560	1280	1280	2560	5120	5120	2560	
A/Albania/4978/2014	6B	2014-02-17	MDCK3/MDCK1	2560	1280	2560	1280	2560	2560	2560	5120	2560	
A/Norway/778/2014	6B	2014-02-19	MDCK2/MDCK2	2560	1280	2560	2560	1280	2560	2560	5120	2560	
A/Albania/5022/2014	6B	2014-02-21	C2/MDCK1	640	640	1280	640	1280	1280	1280	2560	1280	
A/Albania/5023/2014	6B	2014-02-22	C2/MDCK1	640	640	1280	640	1280	1280	1280	2560	1280	
A/Norway/686/2014	6B	2014-02-24	MDCK3/MDCK2	1280	320	1280	640	640	1280	2560	2560	1280	
A/Albania/5073/2014	6B	2014-02-25	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/La Reunion/2058/2014	6B	2014-04-15	MDCK2/MDCK1	640	320	640	320	640	1280	1280	2560	1280	
A/Ghana/DARI-0095/2014	6C	2014-04-30	C1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/La Reunion/2052/2014		2014-05-30	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/La Reunion/2049/2014	6B	2014-06-02	MDCK2/MDCK1	1280	1280	2560	1280	5120	2560	2560	5120	5120	
A/La Reunion/2040/2014	6B	2014-06-14	MDCK2/MDCK1	640	320	640	320	640	1280	1280	2560	1280	
A/Ghana/DILI-0567/2014	6C	2014-06-16	C1/MDCK1	1280	640	1280	1280	1280	2560	2560	5120	2560	
A/Ghana/DILI-0582/2014	6C	2014-06-23	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Ghana/DILI-0618/2014	6C	2014-07-04	C2/MDCK1	160	160	320	160	160	640	160	1280	320	
A/Ghana/DILI-0620/2014	6C	2014-07-07	C2/MDCK1	320	80	80	80	320	320	320	640	320	

Sequences in phylogenetic trees

Vaccine

Table 6-16. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-08-26)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	EP1/E2	1280	1280	2560	320	320	640	640	640	640	640	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	640	320	80	40	80	160	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	5120	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	320	1280	640	1280	1280	1280	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	1280	640	2560	640	2560	2560	2560	5120	5120	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	640	2560	2560	2560	5120	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	2560	640	2560	2560	2560	5120	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	1280	320	1280	1280	1280	5120	1280	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	1280	1280	640	1280	1280	1280	2560	1280	2560	
TEST VIRUSES														
A/Hungary/367/2014		2014-01-08	MDCK1/E2/MDCK1	640	640	1280	640	1280	1280	2560	5120	1280	2560	
A/Hungary/406/2014	6B	2014-02-06	MDCK2/E2/MDCK1	640	640	1280	640	1280	1280	1280	2560	1280	1280	
A/Paris/976/2014	6B	2014-03-06	MDCK1/MDCK1	1280	1280	2560	1280	5120	5120	5120	5120	5120	5120	
A/Paris/1050/2014	6B	2014-03-10	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	
A/Haute Normandie/1058/2014	6B	2014-03-11	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Paris/1045/2014	6B	2014-03-13	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	
A/Pays de Loire/1086/2014	6B	2014-03-14	MDCK1/MDCK1	2560	640	2560	2560	2560	2560	2560	5120	5120	5120	
A/Lorraine/1078/2014	6B	2014-03-17	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Haute Normandie/1098/2014	6B	2014-03-17	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	
A/Picardie/1221/2014	6B	2014-03-17	MDCK1/MDCK1	1280	1280	2560	1280	2560	2560	2560	5120	5120	2560	
A/Paris/1126/2014	6B	2014-03-21	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	
A/Hyogo/3030/2014	6B	2014-03-24	MDCK2/MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Lorraine/1248/2014	6B	2014-03-31	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Centre/1238/2014	6B	2014-04-01	MDCK1/MDCK1	1280	1280	2560	2560	2560	5120	5120	5120	5120	5120	
A/Bretagne/1498/2014	6B	2014-04-14	MDCK1/MDCK1	1280	1280	2560	2560	5120	5120	2560	5120	5120	5120	
A/Centre/1550/2014	6B	2014-04-22	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	2560	5120	5120	5120	
A/Paris/1823/2014	6B	2014-06-11	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	

Sequences in phylogenetic trees

Vaccine

Figure 4. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

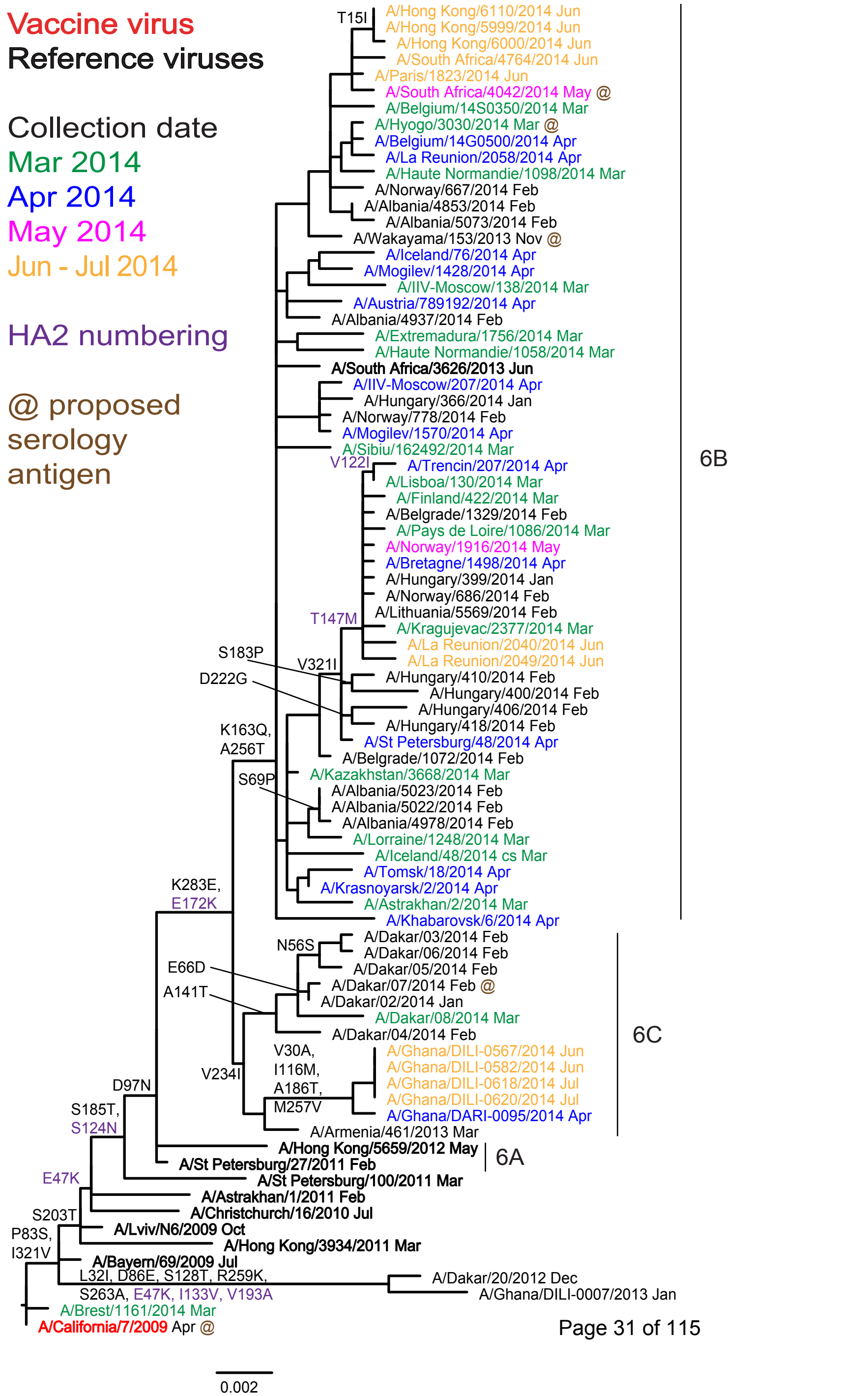


Figure 5. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTASSWSYIVETPSSDNGTCYPGDFIDYEE
A/Bayern/69/2009            ..................................................................................S.................
A/Lviv/N6/2009              ..................................................................................S.................
A/Christchurch/16/2010      ..................................................................................S..........N...... 4
A/Hong Kong/3934/2011       ..................................................................................S................. 3
A/Astrakhan/1/2011          ..................................................................................S.............N... 5
A/St Petersburg/100/2011    ..................................................................................S................. 7
A/St Petersburg/27/2011     ..................................................................................S.............N... 6
A/Hong Kong/5659/2012       ..................................................................................S.............N... 6A
A/South Africa/3626/2013    ..................................................................................S.............N... 6B
A/Ghana/DILI-0618/2014      .............................A....................................................S.............N... 6C
A/Dakar/07/2014             .................................................................D................S.............N... 6C
A/Dakar/06/2014             .......................................................S..........................S.............N... 6C
A/Wakayama/153/2013         ..................................................................................S.............N... 6B
A/Astrakhan/2/2014          ..............................................I...................................S.............N... 6B
A/Tomsk/18/2014             ..................................................................................S.............N... 6B
A/Lorraine/1248/2014        ..................................................................................S.............N... 6B
A/St Petersburg/48/2014     ..................................................................................S.............N... 6B
A/La Reunion/2049/2014      ..................................................................................S.............N... 6B
A/Bretagne/1498/2014        ..................................................................................S.............N... 6B
A/Norway/1916/2014          ..................................................................................S.............N... 6B
A/Trencin/207/2014          ..................................................................................S.............N... 6B
A/Austria/789192/2014       ..................................................................................S.............N... 6B
A/Iceland/76/2014           ..................................................................................S.............N... 6B
A/Belgium/14G0500/2014      ..................................................................................S.............N... 6B
A/Hyogo/3030/2014           ..................................................................................S.............N... 6B
A/South Africa/4042/2014    ..................................................................................S.............N... 6B
A/Paris/1823/2014           ..................................................................................S.............N... 6B
A/South Africa/4764/2014    ..................................................................................S.............N... 6B
A/Hong Kong/6110/2014       ..............I...................................................................S.............N... 6B

                                   110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLSKSYINDKGKEVLVLWGIHHPSTSADQQSLYQNADAYVF
A/Bayern/69/2009            ......................................................E.............................................
A/Lviv/N6/2009              ....................................................................................................
A/Christchurch/16/2010      ........................D..................................T........................................
A/Hong Kong/3934/2011       ..................X..............T......S.........................................P.................
A/Astrakhan/1/2011          ....................................................................................................
A/St Petersburg/100/2011    ..........................................G.........................................T...........T...
A/St Petersburg/27/2011     ....................................................................................T...............
A/Hong Kong/5659/2012       ..................N..................R..............................................T...............
A/South Africa/3626/2013    ..............................................................Q.....................T...............
A/Ghana/DILI-0618/2014      ...............M....................................................................TT..............
A/Dakar/07/2014             ........................................T...........................................T...............
A/Dakar/06/2014             ..................N.....................T...........................................T...............
A/Wakayama/153/2013         ..........................X...................................Q.....................T...............
A/Astrakhan/2/2014          ..............................................................Q.....................T...............
A/Tomsk/18/2014             ..............................................................Q.....................T...............
A/Lorraine/1248/2014        ..............................................................Q.....................T...............
A/St Petersburg/48/2014     ..........................X...................................Q.....................T...............
A/La Reunion/2049/2014      ..............................................................Q.....................T...............
A/Bretagne/1498/2014        ..............................................................Q.....................T...............
A/Norway/1916/2014          ...................................................I..........Q.....................T...............
A/Trencin/207/2014          ..............................................................Q.....................T...............
A/Austria/789192/2014       ..............................................................Q.....................TT..............
A/Iceland/76/2014           ..............................................................Q.....................T...............
A/Belgium/14G0500/2014      ..............................................................Q.....................T...............
A/Hyogo/3030/2014           ..............................................................Q.....................T...............
A/South Africa/4042/2014    ..............................................................Q.....................T...............
A/Paris/1823/2014           ..............................................................Q.....................T...............
A/South Africa/4764/2014    ..............................................................Q.....................T...............
A/Hong Kong/6110/2014       ..............................................................Q.....................T...............

                                   210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         VGSSRYSKKFKPEIAIRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFAMERNAGSGIIISDTPVHDCNTTCQTPKGAINTSLPFQNIHPITI
A/Bayern/69/2009            ....................................................................................................
A/Lviv/N6/2009              ..T..................G..............................................................................
A/Christchurch/16/2010      ..T..................N...........................A..................................................
A/Hong Kong/3934/2011       ..T..................N.........................................................................V....
A/Astrakhan/1/2011          ..T.K.........XV................................L...................................................
A/St Petersburg/100/2011    ..T...................R.............................................................................
A/St Petersburg/27/2011     ..T...................R.............................................................................
A/Hong Kong/5659/2012       ..T.............................................L...................................................
A/South Africa/3626/2013    ..T..................................................T..........................E.................
A/Ghana/DILI-0618/2014      ..T..............................I......................V.........................E.................
A/Dakar/07/2014             ..T..............................I................................................E.................
A/Dakar/06/2014             ..T..............................I................................................E.................
A/Wakayama/153/2013         ..T..................................................T..........................E.................
A/Astrakhan/2/2014          ..T..................N...............................T..........................E.................
A/Tomsk/18/2014             ..T..................................................T..........................E.................
A/Lorraine/1248/2014        ..T..................................................T..........................E.................
A/St Petersburg/48/2014     ..T..................................................T..........................E.................
A/La Reunion/2049/2014      ..T..................................................T..........................E...........V.....
A/Bretagne/1498/2014        ..T..................................................T..........................E.................
A/Norway/1916/2014          ..T..................................................T..........................E.................
A/Trencin/207/2014          ..T..................................................T..........................E.................
A/Austria/789192/2014       ..T..................................................T..........................E.................
A/Iceland/76/2014           ..T..................................................T..........................E.................
A/Belgium/14G0500/2014      ..T..................................................T..........................E.................
A/Hyogo/3030/2014           ..T.................K................................T..........................E.................
A/South Africa/4042/2014    ..T..................................................T..........................E.................
A/Paris/1823/2014           ..T..................................................T..........................E.................
A/South Africa/4764/2014    ..T..................................................T..........................E.................
A/Hong Kong/6110/2014       ..T..................................................T..........................E.................
```

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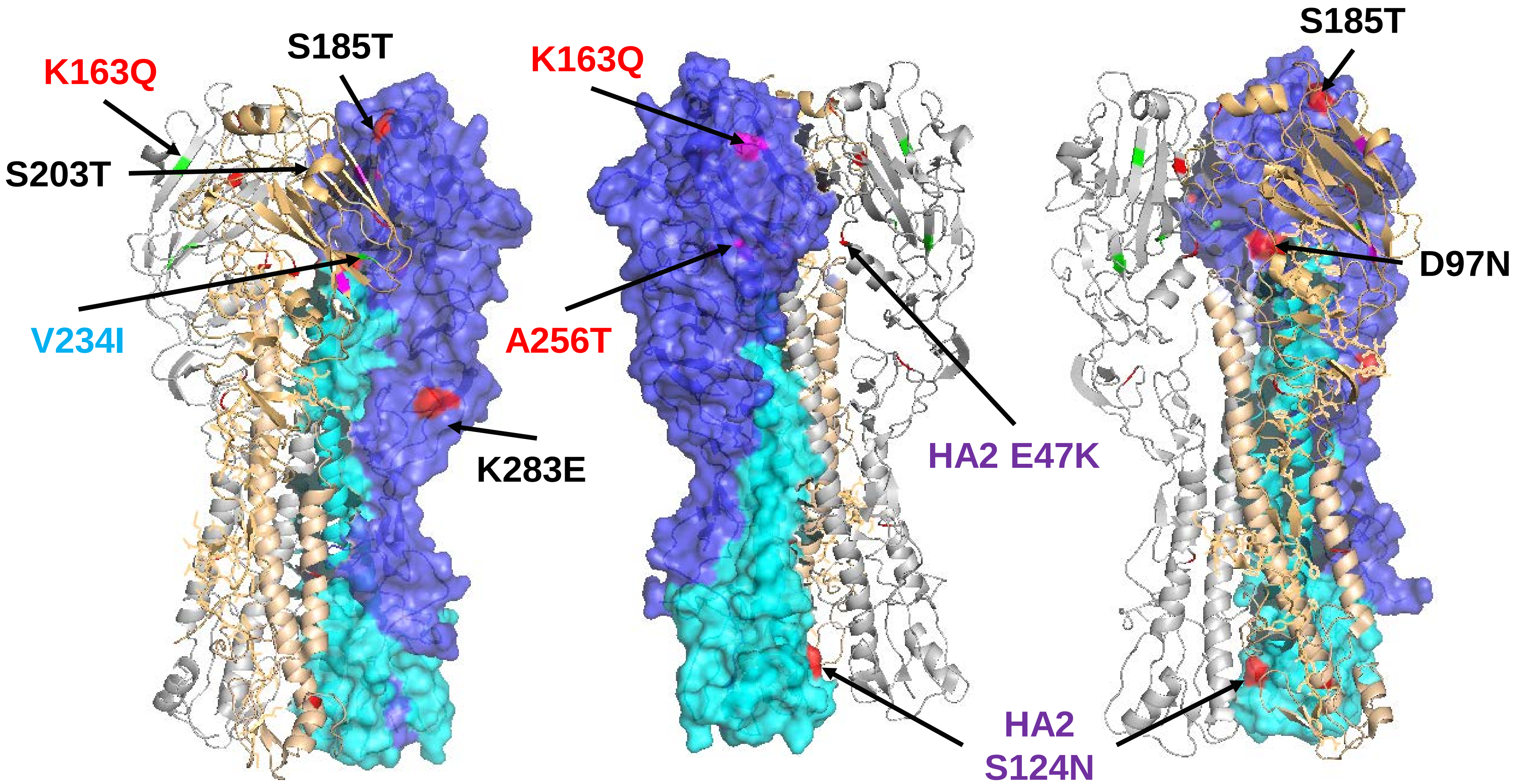
                                310       320       330       340       350       360       370       380       390       400
                        ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009     GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL
A/Bayern/69/2009        ....................V...............................................................................
A/Lviv/N6/2009          ....................V...............................................................................
A/Christchurch/16/2010  ....................V....................................................K..........................
A/Hong Kong/3934/2011   ....................V...............................................................................
A/Astrakhan/1/2011      ....................V....................................................K..........................
A/St Petersburg/100/2011 ....................V....................................................K..........................
A/St Petersburg/27/2011 ....................V....................................................K..........................
A/Hong Kong/5659/2012   ....................V....................................................K..........................
A/South Africa/3626/2013 ....................V....................................................K..........................
A/Ghana/DILI-0618/2014  ....................V....................................................K..........................
A/Dakar/07/2014         ....................V....................................................K..........................
A/Dakar/06/2014         ....................V....................................................K..........................
A/Wakayama/153/2013     ....................V....................................................K..........................
A/Astrakhan/2/2014      ....................V....................................................K..........................
A/Tomsk/18/2014         ....................V....................................................K..........................
A/Lorraine/1248/2014    ....................V....................................................K..........................
A/St Petersburg/48/2014 .........................................................................K..........................
A/La Reunion/2049/2014  .........................................................................K..........................
A/Bretagne/1498/2014    .........................................................................K..........................
A/Norway/1916/2014      .........................................................................K..........................
A/Trencin/207/2014      .........................................................................K..........................
A/Austria/789192/2014   ....................V....................................................K..........................
A/Iceland/76/2014       ....................V....................................................K..........................
A/Belgium/14G0500/2014  ....................V....................................................K..........................
A/Hyogo/3030/2014       ....................V....................................................K..........................
A/South Africa/4042/2014 ....................V....................................................K..........................
A/Paris/1823/2014       ....................V....................................................K..........................
A/South Africa/4764/2014 .........................................................................K..........................
A/Hong Kong/6110/2014   ....................V....................................................K..........................

                                410       420       430       440       450       460       470       480       490       500
                        ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009     EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRSQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI
A/Bayern/69/2009        ....................................................................................................
A/Lviv/N6/2009          ....................................................................................................
A/Christchurch/16/2010  ....................................................................................................
A/Hong Kong/3934/2011   ..............................................................................I.....................
A/Astrakhan/1/2011      ....................................................................................................
A/St Petersburg/100/2011 ..................................................N.................................................
A/St Petersburg/27/2011 ..................................................N.................................................
A/Hong Kong/5659/2012   ..................................................N.................................................
A/South Africa/3626/2013 ..................................................N...............................................K.
A/Ghana/DILI-0618/2014  ..................................................N...............................................K.
A/Dakar/07/2014         ..................................................N...............................................K.
A/Dakar/06/2014         ..................................................N...............................................K.
A/Wakayama/153/2013     ..................................................N...............................................K.
A/Astrakhan/2/2014      ..................................................N...............................................K.
A/Tomsk/18/2014         ..................................................N...............................................K.
A/Lorraine/1248/2014    ..................................................N...............................................K.
A/St Petersburg/48/2014 ..................................................N......Q........................................K.
A/La Reunion/2049/2014  .R................................................N......................M........................K.
A/Bretagne/1498/2014    .........................X........................N......................M.....................X..K.
A/Norway/1916/2014      ..................................................N......................M........................K.
A/Trencin/207/2014      ................................................I.N......................M........................K.
A/Austria/789192/2014   ..................................................N.........................D.....................K.
A/Iceland/76/2014       ..................................................N...............................................K.
A/Belgium/14G0500/2014  ..................................................N...............................................K.
A/Hyogo/3030/2014       ..................................................N...............................................K.
A/South Africa/4042/2014 ..................................................N......E........................................K.
A/Paris/1823/2014       ..................................................N.......................................G.......K.
A/South Africa/4764/2014 ..................................................N...............................................K.
A/Hong Kong/6110/2014   ..................................................N...............................................K.

                                510       520       530       540
                        ....|....|....|....|....|....|....|....|....|....
A/California/7/2009     DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Bayern/69/2009        .................................................
A/Lviv/N6/2009          .................................................
A/Christchurch/16/2010  .................................................
A/Hong Kong/3934/2011   .................................................
A/Astrakhan/1/2011      .................................................
A/St Petersburg/100/2011 .................................................
A/St Petersburg/27/2011 .................................................
A/Hong Kong/5659/2012   .................................................
A/South Africa/3626/2013 .......................V.........................
A/Ghana/DILI-0618/2014  .................................................
A/Dakar/07/2014         .................................................
A/Dakar/06/2014         .................................................
A/Wakayama/153/2013     .................................................
A/Astrakhan/2/2014      .................................................
A/Tomsk/18/2014         .................................................
A/Lorraine/1248/2014    .................................................
A/St Petersburg/48/2014 .................................................
A/La Reunion/2049/2014  .................................................
A/Bretagne/1498/2014    .................................................
A/Norway/1916/2014      .................................................
A/Trencin/207/2014      .................................................
A/Austria/789192/2014   .................................................
A/Iceland/76/2014       .........V.......................................
A/Belgium/14G0500/2014  .................................................
A/Hyogo/3030/2014       .................................................
A/South Africa/4042/2014 .................................................
A/Paris/1823/2014       .................................................
A/South Africa/4764/2014 .................................................
A/Hong Kong/6110/2014   .................................................
```

Vaccine virus: **Reference virus**: **Genetic group**: **HA1/HA2 boundary**: **Potential N-linked glycosylation site**

Figure 6. H1-HA locations of genetic group defining amino acid substitutions



HA1 substitutions D97N, S185T, S203T & K283E and HA2 substitutions E47K, S124N & E172K (not in structure) differentiate subgroup 6B and 6C viruses from A/California/7/2009. HA1 substitutions K163Q and A256T define subgroup 6B and V234I defines subgroup 6C.

Figure 7. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

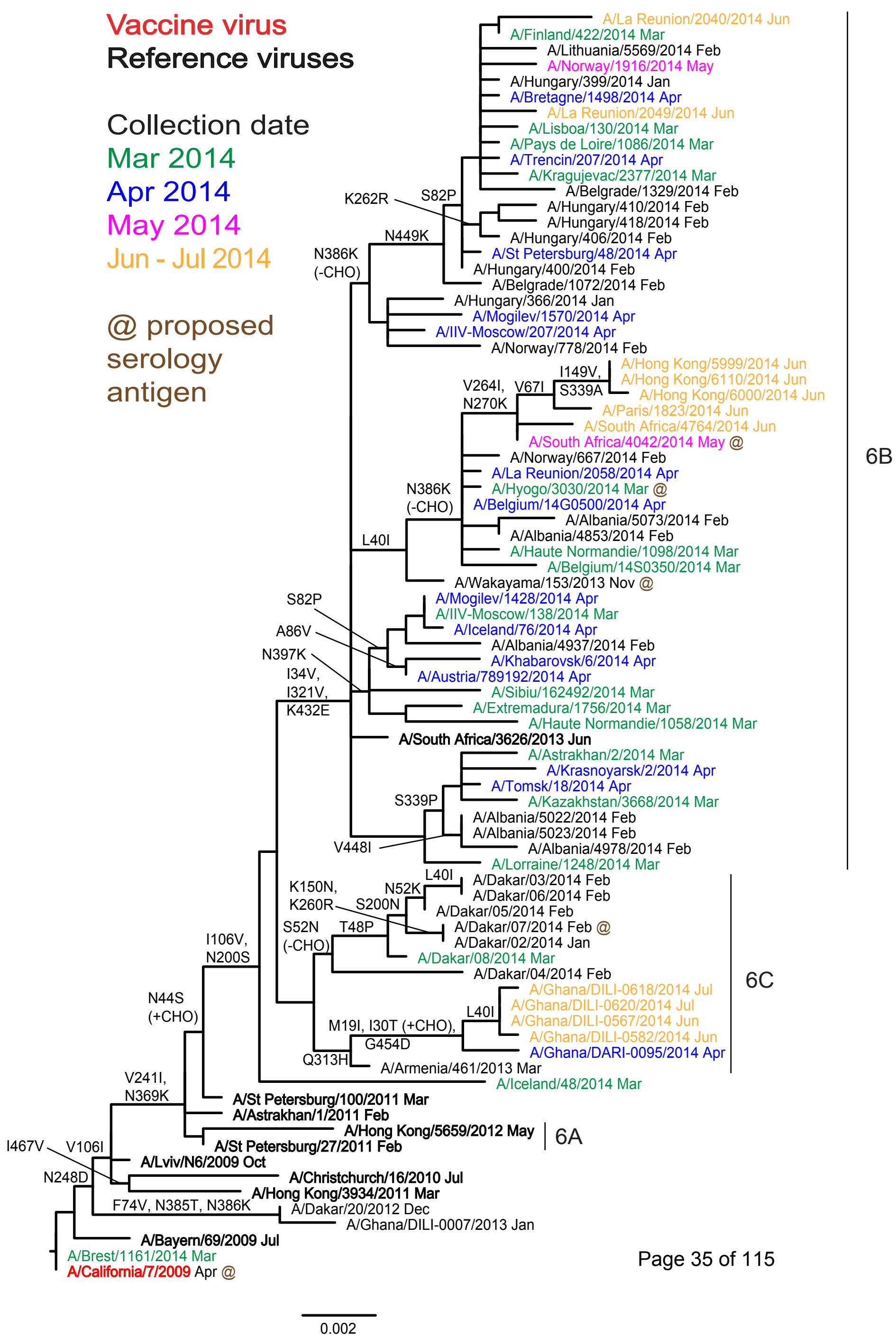
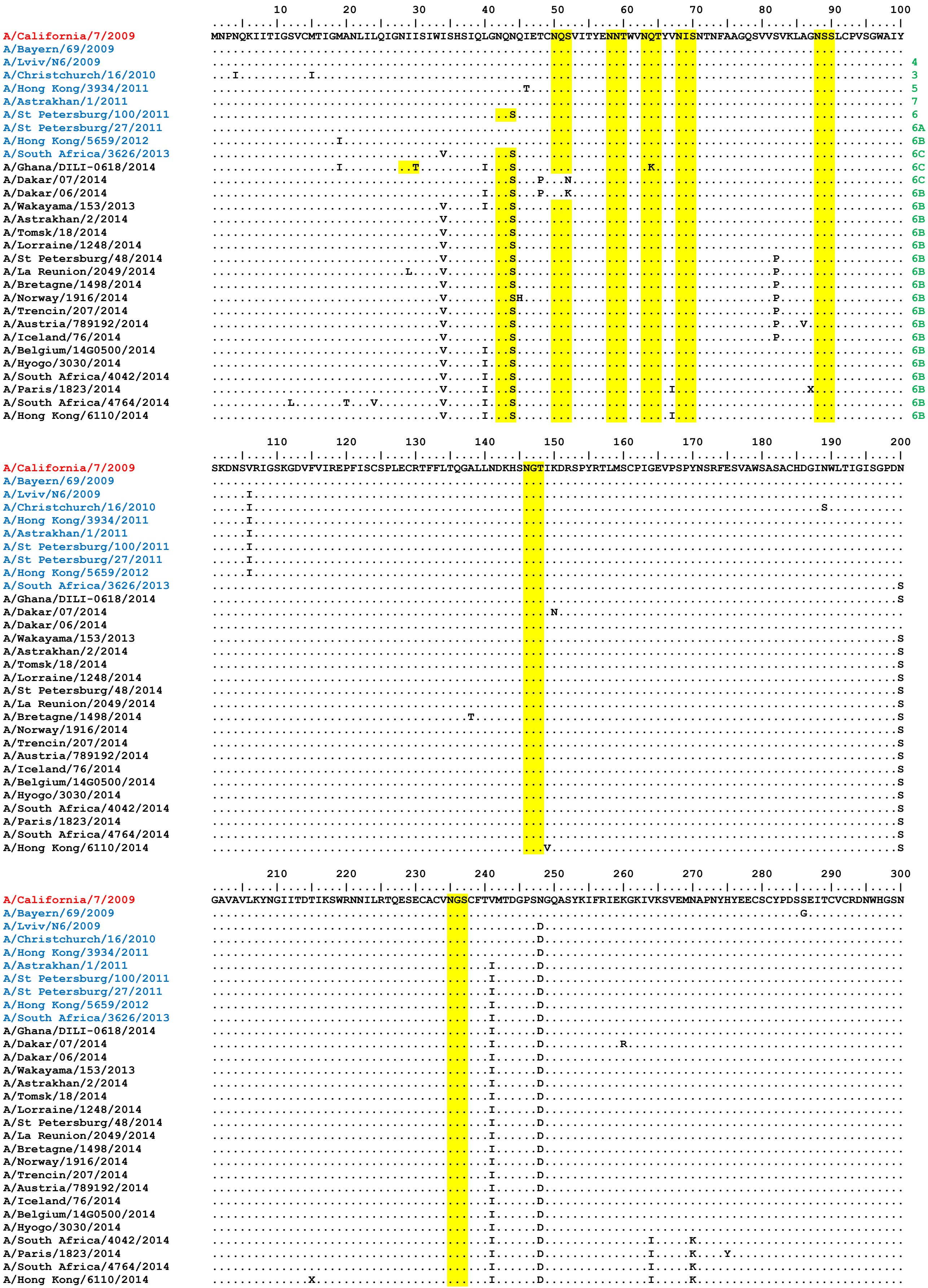


Figure 8. NA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



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                                   310       320       330       340       350       360       370       380       390       400
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009       RPWVSFNQNLEYQIGYICSGIFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRNGFEMIWDPNGWTGTDNNFSIKQDIVGINEWS
A/Bayern/69/2009          ....................................................................................................
A/Lviv/N6/2009            ....................................................................................................
A/Christchurch/16/2010    ........................................................................I...........................
A/Hong Kong/3934/2011     ............R................................................................................I......
A/Astrakhan/1/2011        ....................................................................K...............................
A/St Petersburg/100/2011  ........................................................I...........K...............................
A/St Petersburg/27/2011   ....................................................................K...............................
A/Hong Kong/5659/2012     ....................................................................K................S..............
A/South Africa/3626/2013  ....................V...............................................K...............................
A/Ghana/DILI-0618/2014    ............H.......................................................K...............................
A/Dakar/07/2014           ....................................................................K...............................
A/Dakar/06/2014           ....................................................................K...............................
A/Wakayama/153/2013       ....................VL..............................................K...............................
A/Astrakhan/2/2014        ....................V.................P.............................K...............................
A/Tomsk/18/2014           ....................V.................P.............................K...............................
A/Lorraine/1248/2014      ....................V...................K...........................K...............................
A/St Petersburg/48/2014   ....................V...............................................K.......L........K..............
A/La Reunion/2049/2014    ....................V..................F.......................G....K................K..............
A/Bretagne/1498/2014      ....................V...............................................K................K..............
A/Norway/1916/2014        ....................V...............................................K................K..............
A/Trencin/207/2014        ....................V...............................................K................K..............
A/Austria/789192/2014     ....................V...............................................K...........................K...
A/Iceland/76/2014         ....................V...............................................K...........................K...
A/Belgium/14G0500/2014    ....................V...............................................K................K..............
A/Hyogo/3030/2014         ....................V...............................................K...............YK..............
A/South Africa/4042/2014  ....................V...............................................K................K..............
A/Paris/1823/2014         ....................V...............................................K................K..............
A/South Africa/4764/2014  ....................V...............................................K................K..............
A/Hong Kong/6110/2014     ....................V.................A.............................K................K..............

                                   410       420       430       440       450       460
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/California/7/2009       GYSGSFVQHPELTGLDCIRPCFWVELIRGRPKENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK
A/Bayern/69/2009          .....................................................................
A/Lviv/N6/2009            .....................................................................
A/Christchurch/16/2010    ..................................................................V..
A/Hong Kong/3934/2011     ..................................................................V..
A/Astrakhan/1/2011        .....................................................................
A/St Petersburg/100/2011  .....................................................................
A/St Petersburg/27/2011   .....................................................................
A/Hong Kong/5659/2012     ..................................................G..................
A/South Africa/3626/2013  ...............................E.....................................
A/Ghana/DILI-0618/2014    .....................................................D...............
A/Dakar/07/2014           .....................................................................
A/Dakar/06/2014           .....................................................................
A/Wakayama/153/2013       ...............................E.....................................
A/Astrakhan/2/2014        ...............................E.....................................
A/Tomsk/18/2014           ...............................E.....................................
A/Lorraine/1248/2014      ...............................E.....................................
A/St Petersburg/48/2014   ...............................E................K....................
A/La Reunion/2049/2014    ...............................E................K....................
A/Bretagne/1498/2014      ...............................E................K....................
A/Norway/1916/2014        ...............................E................K....................
A/Trencin/207/2014        ...............................E................K....................
A/Austria/789192/2014     ...............................E.....................................
A/Iceland/76/2014         ...............................E.....................................
A/Belgium/14G0500/2014    ...............................E.....................................
A/Hyogo/3030/2014         ...............................E.....................................
A/South Africa/4042/2014  ...............................E.....................................
A/Paris/1823/2014         ...............................E.....................................
A/South Africa/4764/2014  ...............................E.....................................
A/Hong Kong/6110/2014     ...............................E.....................................
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses collected since 2014-01-31 have been received from NICs in Europe, Africa (Cameroon, Ghana, Senegal and South Africa), the Middle East (Israel), Central Asia (Kazakhstan), South America (Argentina) and from Hong Kong SAR. As described in previous reports, these viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. All but 7 viruses (817 of 824) had sufficient HA titre in assays conducted using guinea pig red blood cells in the presence of 20nM oseltamivir, added to circumvent any NA-mediated binding of H3N2 viruses to the red blood cells, to be analysed by HI assay (**Figure 9**). Virus neutralisation assays were used to complement HI tests using the plaque reduction neutralisation assay (PRNA).

A summary of the HI results is shown in **Table 7** and the HI results are shown in **Tables 8-1 to 8-29**. Viruses with collection dates after 2014-01-31 are shown in **Tables 8-4** and **8-6 to 8-29.** Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2014-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees presented here are highlighted.

Over 85% of test viruses with collection dates after 2014-01-31 that were propagated in MDCK-SIAT1 cells reacted poorly in HI assays (≥8-fold decrease) with post-infection ferret antiserum raised against the egg-propagated vaccine virus, A/Texas/50/2012, compared with the titre of the antiserum with the homologous virus (**Table 7**). Similar low levels of reactivity were seen with antisera raised against three other egg-propagated reference viruses A/Serbia/NS-210/2013 (**Tables 8-1 to 8-20**), A/Hong Kong/146/2013 (**Tables 8-1 to 8-29**) and A/Almaty/2958/2013 (**Tables 8-1 to 8-14**), approximately 35%, 22% and 33% of test viruses, respectively, reacted within 4-fold of the titre with the homologous egg-propagated viruses. Better reactivity was seen with test viruses when analysed with an antiserum raised against the exclusively egg propagated A/Stockholm/1/2014 clone 36-18 (**Tables 8-10 to 8-26**), a virus belonging to genetic subgroup 3C.2. This antiserum showed a low titre for the homologous virus but recognised over 80% of test viruses at titres within 4-fold of this homologous titre.

Two antisera raised against egg-propagated viruses that belonged to the new genetic group 3C.3a also displayed low homologous titres. Antisera raised against A/Norway/466/2014 clone 58 (**Tables 8-21 to 8-26**), and A/Tasmania/11/2014 (**Table 8-28**) recognised 85% and 100%, respectively, of test viruses at titres within 4-fold of the titre for the homologous virus. An antiserum raised against the egg-propagated virus A/Switzerland/9715293/2013 clone 123 (**Tables 8-26 to 8-29**), also belonging to genetic subgroup 3C.3a, had a higher homologous titre but recognised only a minority (~25%) of test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus.

Ferret antisera raised against reference viruses propagated in tissue culture cells A/Stockholm/18/2011 (**Tables 8-1 to 8-9**), A/Victoria/361/2011 (**Tables 8-1 to 8-29**), A/Athens/112/2012 (**Tables 8-1 to 8-11**) and A/Samara/73/2013 (**Tables 8-1 to 8-29**) generally recognised the test viruses effectively. Between ~80% and ~95% of test viruses propagated in MDCK-SIAT1 cells were recognised at titres within 4-fold of the titres for these antisera with their corresponding homologous viruses. These reference viruses had HA genes from genetic subgroups 3A, 3B, 3C.1 and 3C.3. Antisera were raised against reference viruses belonging to genetic subgroup 3C.3a that had been exclusively propagated in cell culture: antisera raised against A/Stockholm/6/2014 (**Tables 8-12 to 8-29**), A/Norway/466/2014 (**Tables 8-14 to 8-29**) and A/Switzerland/9715293/2013 (**Tables 8-19 to 8-21** and **8-26 to 8-29**) recognised 97%, 93% and 82%, respectively, of test viruses at titres within 4-fold of those with the corresponding homologous viruses.

Thirty seven test viruses belonging to genetic subgroup 3C.3a were received. Of these cell-propagated 3C.3a viruses only 20-25% were recognised by antisera raised against the cell-propagated reference viruses A/Victoria/361/2011 and A/Samara/73/2013 at titres within 4-fold of the titre with the respective homologous viruses. All test viruses belonging to genetic subgroup 3C.3a were recognised at titres within 4-fold of the titre for the homologous virus by antisera raised against the 3C.3a viruses A/Stockholm/6/2014 and A/Norway/466/2014, as were the smaller number of test viruses analysed with an antiserum raised against the cell-propagated cultivar of A/Switzerland/9715293/2013, another 3C.3a virus. Only eight test viruses belonging to the emerging genetic subgroup 3C.2a were received. These viruses were not recognised well by antisera raised against the cell-propagated reference viruses A/Victoria/361/2011 and A/Samara/73/2013. They were better recognised by antisera raised against reference cell-propagated viruses in genetic subgroup 3C.3a - A/Stockholm/6/2014 and A/Norway/466/2014.

The results of PRNA performed with MDCK-SIAT1 cells are shown in **Table 9**. A total of 11 test viruses with collection dates after 2014-01-31 were analysed with a panel of antisera raised against both cell culture-propagated and egg-propagated viruses from genetic subgroups 3C.1 and 3C.3a. The genetic groups to which the HA genes of the reference viruses and the test viruses belong are shown. The titres given are to the nearest 2-fold dilution factor with the numbers in parentheses being interpolated directly from the results using image-processing software. The results of PRNA were consistent with the results from HI assays performed in the presence of oseltamivir. By PRNA the test viruses in genetic subgroup 3C.3a were recognised poorly by the antisera raised against the cell-propagated reference virus A/Victoria/361/2011 and the egg-propagated vaccine virus A/Texas/50/2012. Antisera raised against cell-propagated 3C.3a reference viruses (A/Stockholm/6/2014, A/Norway/466/2014, A/Switzerland/9715293/2013 and

A/Palau/6759/2014) recognised test viruses in genetic subgroups 3C.3 or 3C.3a equally well at titres within 4-fold of the titres with the homologous viruses.

Table 9 also shows results of PRNA using antisera raised against egg-propagated reference viruses. The amino acid substitutions acquired by the reference viruses on adaptation to propagation in eggs are shown. Test viruses were recognised at a titre within 4-fold of the titre for the homologous virus with the antisera raised against A/Switzerland/9715293/2013 clone 123 and A/Norway/466/2014 clone 58 but were recognised less well by antisera raised against A/Palau/6759/2014 or A/Norway/466/2014 clone 32. The latter antiserum had a higher homologous titre than those of the antisera raised against the other viruses. This high titre is possibly related to the loss of a glycosylation motif at residue 246 located at the top of the HA glycoprotein trimer.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 10**. The HA genes fell within the A/Victoria/208 genetic clade, predominantly within genetic subgroup 3C of group 3. This subgroup has three subdivisions: 3C.1, 3C.2 and 3C.3. Two new genetic groups have emerged in recent months – one in subdivision 3C.2, 3C.2a, and one in 3C.3, 3C.3a.

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subgroup 3C.1. Most of the viruses received since 2014-01-31 fall into subgroups 3C.2, 3C.2a, 3C.3 and 3C.3a. Amino acid substitutions that define these subgroups are:

(**3C.2**) **N145S** and **V186G**@ in **HA1**, and **D160N** in **HA2**, e.g. A/Hong Kong/146/2013;

(**3C.2a**) as 3C.2 plus **L3I**, **N144S** (resulting in the loss of a potential glycosylation site), **F159Y**, **K160T** (resulting in the gain of a potential glycosylation site), **N225D** and **Q311H** in **HA1** e.g. A/Hong Kong/3579/2014

(**3C.3**) **T128A** (resulting in the loss of a potential glycosylation site), **R142G, N145S** and **V186G**@ in **HA1,** e.g. A/Samara/73/2013.

(**3C.3a**) as 3C.3 plus **A138S, F159S** and **N225D**, e.g. A/Switzerland/9715293/2013.

@Note: **G186V** substitutions in HA1 occur during adaptation of H3N2 viruses to propagation in hens' eggs.

Another genetic subgroup has been designated 3C.3b (**Figure 10**) which carries the 3C.3 amino acid substitutions plus **E62K, K83R, N122D** (resulting in the loss of a potential glycosylation site), **L157S** and **R261Q** in **HA1** with **M18K** in **HA2**. However, viruses in this subgroup do not appear to be antigenic drift variants.

Figure 11 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 10**). The vaccine virus A/Texas/50/2012 is shown in red, glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group to which each of the sequences belongs is also shown.

Figure 12 shows the location on a 2005 H3 HA structure of amino acid substitutions that define genetic subgroups 3C.2a (**Figure 12A**), 3C.3a (**Figure 12B**) and 3C.3b (**Figure 12C**) each compared with the vaccine virus A/Texas/50/2012.

Figure 13 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 14** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of viruses with HA genes in genetic subgroup 3C do not cluster in the same manner as the HA sequences. The NA substitutions **Y155F, D251V and S315G** define two major divisions among recently collected viruses and viruses carrying these NA substitutions have HA genes in genetic subgroups 3C.3 and 3C.3b. NA genes of viruses with HA genes in the 3C.3a subgroup fall into two distinct clusters. One cluster is defined by the NA substitutions **E221D** and **I392T**, the other has viruses from West Africa which carry **I77V, A110D, V313A** and **V412I**. **Figure 15** indicates the positions of amino acid substitutions, on the structure of an N2 NA, that define viruses with HA genes falling into 3C.2a and 3C.3a genetic subgroups and differentiate the 3C.3a viruses from West Africa.

Antiviral resistance

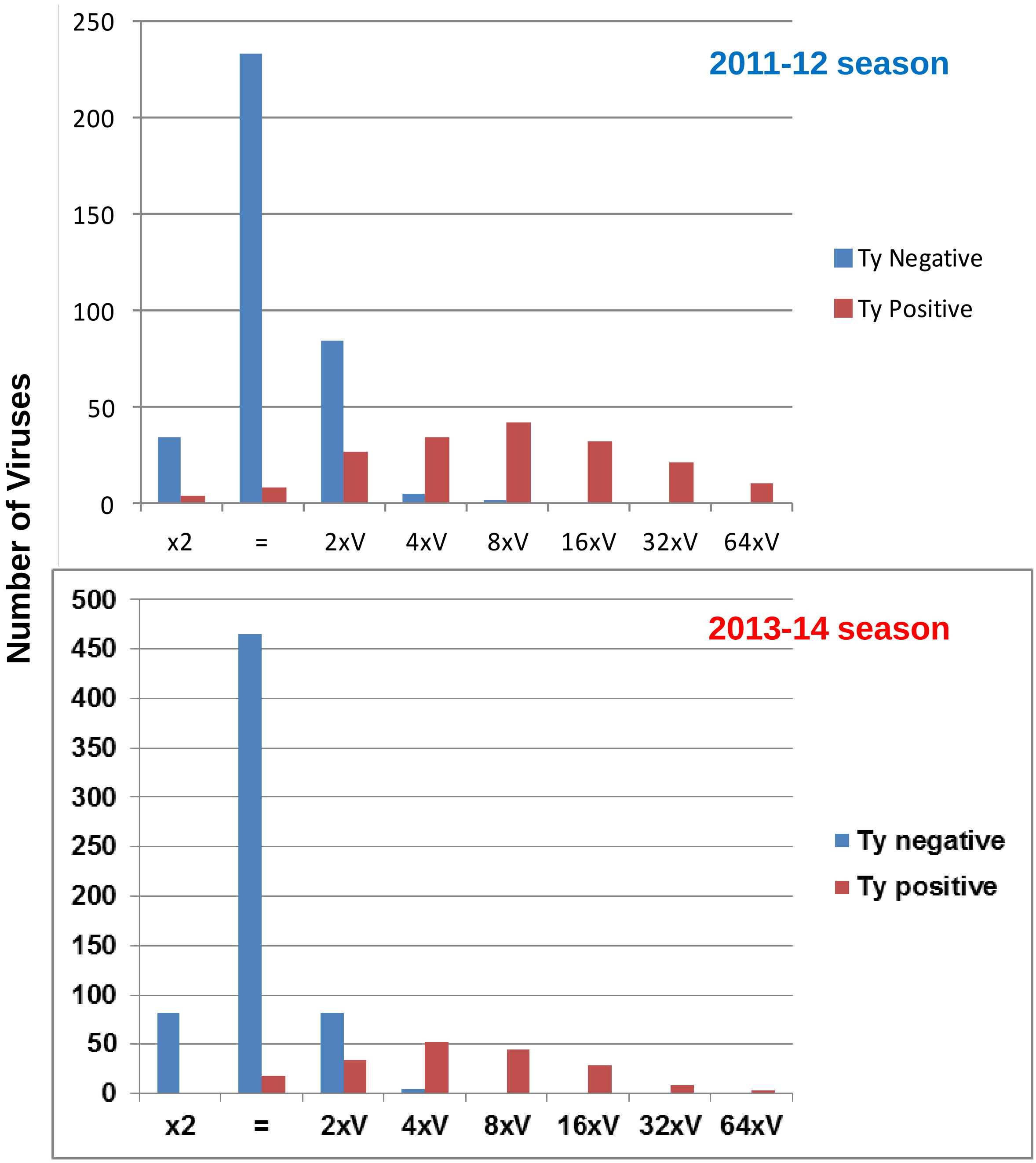
Figure 25 shows a phylogenetic tree of representative H3N2 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the **S31N** substitution in the **M2** ion-channel protein, known to confer resistance to adamantanes.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on over 280 viruses recovered from samples with collection dates after 2014-01-31. All but four showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 14**). The four viruses showing reduced inhibition (RI) by oseltamivir had 10- to 26-fold increases in IC_{50} values compared to the mean IC_{50} of viruses showing normal inhibition (NI), but all showed NI by zanamivir with IC_{50} fold increases of <10. NA amino acid substitutions associated with the reductions in inhibition are shown on the table. The **S331R** substitution, alone or in combination with others, was observed in all four viruses showing RI by oseltamivir collected since 2014-01-31.

Conclusion.

The vast majority of recent test viruses analysed showed poor reactivity with antisera raised against the egg-propagated vaccine virus A/Texas/50/2012 in HI assays. Two new genetic subgroups of viruses have emerged and are increasing in prevalence; they are poorly recognised by these antisera. Viruses in these emergent genetic groups were also poorly recognised by antisera raised against viruses in genetic subgroup 3C.3 that had been solely propagated in cell culture. Viruses in the new genetic subgroups, 3C.2a and 3C.3a, appear to be antigenically district from other recently circulating viruses.

Figure 9. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Oseltamivir



Fold shift in HA activity in the presence of 20nM Oseltamivir:
[Enhancement (x2) Equivalence (=) Reduction (2x-64x)]

For A(H3N2) viruses recovered in SIAT1 cells we were able to perform HI assays on:
~75% in 2011-12
>99% in 2013-14 (817 of 824)

Table 7. Summary of influenza A(H3N2) viruses analysed, collected after 2014-01-31

MONTH	H3N2				
Country	Number received	Number propagated[1]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
FEBRUARY					
Albania	3	2			2
Austria	3	3			3
Belgium	3	2			2
Bulgaria	6	5			5
Finland	5	4			4
Georgia	5	5		2	3
Germany	3	3			3
Hong Kong	2	2			2
Hungary	2	1			1
Iceland	3	3			3
Israel	13	11		4	7
Italy	14	14		1	13
Kazakhstan	15	7			7
Lithuania	2	2			2
Macedonia	6	1			1
Norway	8	7			7
Poland	7	5	1		4
Portugal	1	1			1
Serbia	18	15		9	6
Slovakia	2	2			2
Slovenia	14	11		1	10
Sweden	4	4			4
Switzerland	1	1		1	
Ukraine	44	30		4	26
United Kingdom	2	1			1
MARCH					
Austria	5	5			5
Belarus	1	1			1
Belgium	3	1			1
Finland	2	2			2
France	4	4			4
Georgia	5	4			4
Hong Kong	1	1			1
Iceland	1	1			1
Italy	1	1			1
Kazakhstan	5	5			5
Latvia	3	3		2	1
Lithuania	10	10		1	9
Norway	6	4			4
Poland	23	9			9
Portugal	1	1			1
Romania	9	8		1	7
Russia	7	7			7
Serbia	12	7			7
Slovakia	1	1			1
Slovenia	1	1			1
Switzerland	5	4		2	2
Ukraine	30	25		8	17
United Kingdom	2	2			2
APRIL					
Austria	2	2			2
Belarus	1	1			1
Belgium	7	4		1	3
Finland	3	3			3
France	5	4			4
Ghana	2	2			2
Hong Kong	5	5		3	2
Ireland	6	6			6
Latvia	2	2		2	
Lithuania	8	8		1	7
Norway	1	1			1
Poland	1	1			1
Portugal	1	1			1
Russia	2	2			2
Senegal	1	1			1
Serbia	1	1			1
Slovakia	2	2			2
Slovenia	1	1			1
South Africa	1	1			1
Spain	7	7	1	2	4
MAY					
Cameroon	1	in process			
Ghana	11	11			11
Iceland	2	2			2
Lithuania	1	1			1
Norway	1	1			1
South Africa	5	5			5
United Kingdom	1	1			1
JUNE					
Argentina	11	in process			
Cameroon	10	in process			3
Ghana	1	1			1
Iceland	1	1			1
Norway	3	2			2
Senegal	2	2			2
South Africa	5	5			5
Spain	1	1			1
United Kingdom	1	1			1
JULY					
Argentina	24	in process			
Cameroon	2	1			1
Ghana	3	3			3
Senegal	6	6			6
South Africa	2	2			2
Spain	1	1			1
AUGUST					
France	1	in process			
	479	351	2	45	307
37 Countries			0.6%	12.7%	86.7%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
2. Compared to homologous titres for ferret antisera raised against egg grown A/Texas/50/2012 vaccine virus

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-02-12

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F28/11 3A	A/Iowa 19/10 F15/11 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	320	640	160	160	160	320	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	320	320	640	640	160	640	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	2560	2560	2560	1280	1280	1280	1280	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	160	160	320	640	160	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	320	640	1280	320	640	320	320	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	2560	1280	1280	1280	1280	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	320	640	1280	320	1280	640	1280	1280
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	1280	1280	1280	1280	1280	2560
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	2560	2560	1280	2560	640	1280	1280	2560	2560
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	2560	1280	2560	1280	1280	2560	2560	2560
TEST VIRUSES													
A/Tokyo/31512/2013		2013-11-01	MDCK2/MDCK2/SIAT1	<	160	80	320	640	160	320	320	160	320
A/Fujian-Yanping/2707/2013		2013-11-04	C4/SIAT1	40	160	160	320	320	160	640	320	320	320
A/Jiangxi-Xunyang/1790/2013		2013-11-08	C3/SIAT1	<	160	80	160	320	160	320	160	160	160
A/Chongqing-Yuzhong/11653/2013		2013-11-10	C3/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Ankara/3804/2013		2013-11-26	SIAT1/SIAT1	80	320	320	640	1280	320	1280	320	640	640
A/Ankara/3851/2013		2013-11-29	SIAT1/SIAT1	40	320	320	640	640	320	640	320	640	640
A/Malatya/3854/2013		2013-11-29	SIAT1/SIAT1	80	320	320	640	1280	320	1280	320	640	640
A/Samsun/3860/2013		2013-11-29	SIAT1/SIAT1	40	320	320	640	1280	320	640	320	640	640
A/Malatya/3872/2013	3C.2	2013-12-03	SIAT1/SIAT1	<	160	40	160	320	80	320	80	160	320
A/Ankara/3879/2013		2013-12-03	SIAT1/SIAT1	<	160	160	320	640	160	320	160	160	320
A/Trabzon/3892/2013		2013-12-05	SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Ankara/1324/2013		2013-12-06	SIAT1/SIAT1	40	160	160	320	1280	320	640	160	320	320
A/Pais Vasco/131532/2013		2013-12-07	SIAT1/MDCK2	40	320	320	640	1280	160	640	320	640	320
A/Malatya/3946/2013		2013-12-10	SIAT1/MDCK2	80	320	320	640	1280	320	1280	320	1280	640
A/Erzurum/4024/2014		2013-12-17	SIAT1/SIAT1	<	80	80	160	640	80	640	160	320	320
A/Ankara/4084/2013	3C.3	2013-12-20	SIAT1/SIAT1	<	80	80	160	320	80	320	160	160	160
A/Diyarbakır/4107/2013	3C.2	2013-12-21	SIAT1/SIAT1	80	320	160	640	640	160	640	320	640	640
A/Diyarbakır/4109/2013		2013-12-21	SIAT1/SIAT1	<	160	160	320	640	160	640	160	320	320
A/Malatya/4118/2013		2013-12-23	SIAT1/SIAT1	<	80	40	160	320	80	320	80	160	160
A/Ankara/4142/2013		2013-12-25	SIAT1/SIAT2	40	320	160	640	640	160	640	320	640	640
A/Yalova/4145/2013	3C.2	2013-12-25	SIAT1/SIAT2	<	80	80	160	320	80	320	80	160	160
A/Samsun/4151/2013		2013-12-25	SIAT1/SIAT2	<	160	80	160	320	80	320	80	160	160
A/Malatya/4163/2013		2013-12-26	SIAT1/SIAT2	40	160	160	320	320	160	640	160	80	320
A/Adana/4168/2013		2013-12-26	SIAT1/SIAT2	40	320	160	320	640	160	320	160	320	320
A/Samsun/4191/2013	3C.2	2013-12-27	SIAT1/SIAT2	<	160	80	160	320	40	160	160	80	80
A/Adana/4211/2013		2013-12-27	SIAT1/SIAT2	40	160	160	320	640	160	320	320	320	320
A/Diyarbakır/4212/2013	3C.2	2013-12-27	SIAT1/SIAT2	<	80	80	160	320	40	160	80	160	160
A/Malatya/24/2014		2014-01-02	SIAT1/SIAT3	80	160	80	320	640	160	640	160	320	320
A/Samsun/17/2014		2014-01-02	SIAT1/SIAT3	<	80	80	160	320	80	320	80	160	320
A/Samsun/35/2014		2014-01-03	SIAT1/SIAT3	80	320	160	320	320	160	640	320	320	320
A/Trabzon/33/2014		2014-01-03	SIAT1/SIAT3	80	160	160	320	640	160	640	160	320	320
									Vaccine				

1. < = <40

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-02-18 + 2014-02-21

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
	Genetic group				3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	160	320	160	80	80	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	320	160	640	640	160	640	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	2560	2560	1280	640	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	160	160	320	640	160	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	160	160	320	640	160	320	160	160	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	1280	640	1280	1280	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	1280	320	1280	640	1280	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	640	1280	1280	1280	640	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	1280	1280	1280	1280	640	640	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	320	1280	1280	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Cote d'Ivoire/1667/2013		2013-10-08	SIAT2	40	320	320	320	640	320	640	320	320	320
A/Cote d'Ivoire/1666/2013		2013-10-08	SIAT2	<	160	160	320	640	160	640	320	320	320
A/Cote d'Ivoire/1846/2013		2013-10-30	SIAT2	40	320	320	320	640	160	640	320	320	320
A/Izmir/1170/2013		2013-11-02	SIAT1/SIAT1	40	320	320	320	640	160	1280	320	320	320
A/Malatya/3608/2013		2013-11-07	SIAT1/SIAT1	40	320	320	320	640	160	640	320	320	320
A/Cote d'Ivoire/1922/2013		2013-11-11	SIAT2	<	160	160	160	320	80	320	160	160	160
A/Ankara/3653/2013		2013-11-13	SIAT1/SIAT1	80	320	320	320	1280	160	640	320	320	320
A/Cote d'Ivoire/1954/2013		2013-11-14	SIAT2	40	320	320	320	640	320	640	320	320	320
A/Cote d'Ivoire/1955/2013		2013-11-14	SIAT2	<	160	80	160	320	80	160	80	160	160
A/Cote d'Ivoire/1973/2013		2013-11-18	SIAT2	40	320	320	320	640	160	640	320	320	320
A/Cote d'Ivoire/1984/2013	3C.3	2013-11-20	SIAT2	<	80	40	160	160	80	160	40	160	80
A/Ankara/3727/2013		2013-11-20	SIAT1/SIAT1	80	320	320	640	1280	320	1280	320	640	640
A/Cote d'Ivoire/1988/2013		2013-11-21	SIAT2	<	160	160	320	640	160	320	160	320	160
A/Cote d'Ivoire/2020/2013		2013-11-25	SIAT2	<	160	160	320	640	160	320	160	320	160
A/Ankara/3805/2013		2013-11-26	SIAT1/SIAT1	<	160	160	160	640	80	320	160	160	160
A/Cote d'Ivoire/2038/2013	3C.3	2013-11-27	SIAT2	<	80	80	160	320	80	320	160	160	80
A/Cote d'Ivoire/2043/2013		2013-11-27	SIAT2	<	160	80	320	320	80	320	160	160	160
A/Cote d'Ivoire/2086/2013		2013-12-02	SIAT2	<	80	80	160	320	160	320	160	160	160
A/Cote d'Ivoire/2138/2013		2013-12-06	SIAT2	<	80	80	160	320	80	320	80	160	80
A/Cote d'Ivoire/2150/2013		2013-12-09	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Cote d'Ivoire/2190/2013		2013-12-09	SIAT2	<	160	80	320	640	160	320	320	320	320
A/Cote d'Ivoire/2163/2013		2013-12-11	SIAT2	<	80	80	160	320	80	320	80	160	80
A/Cote d'Ivoire/2165/2013	3C.3	2013-12-11	SIAT2	40	160	160	320	640	160	320	160	320	160
A/Cote d'Ivoire/2183/2013		2013-12-12	SIAT2	<	160	40	160	320	80	160	80	80	80
A/Cote d'Ivoire/2202/2013	3C.3	2013-12-12	SIAT2	<	80	40	160	320	40	80	40	80	40
A/Cote d'Ivoire/2207/2013	3C.3	2013-12-13	SIAT2	<	80	160	320	640	160	320	160	320	320
A/Cote d'Ivoire/2208/2013		2013-12-13	SIAT2	<	160	160	320	640	160	320	160	320	320
A/Adiyaman/230/2014		2014-01-11	SIAT2	<	160	320	320	1280	160	640	320	320	320
A/Giresun/231/2014		2014-01-11	SIAT2	<	160	160	160	640	80	320	160	160	160
A/Van/251/2014		2014-01-13	SIAT2	<	160	160	320	1280	80	320	160	160	80
A/Ankara/266/2014	3C.3	2014-01-14	SIAT2	<	80	80	160	320	80	160	80	160	80
A/Samsun/310/2014		2014-01-16	SIAT2	<	80	80	160	320	80	160	160	160	160
A/Trabzon/338/2014		2014-01-16	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Hatay/367/2014	3C.3	2014-01-17	SIAT2	<	160	160	160	640	160	320	160	160	160
A/Diyarbakir/371/2014		2014-01-17	SIAT2	40	160	160	320	640	160	640	320	320	320
									Vaccine				

1. < = <40

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-02-25 + 2014-02-28

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
	Genetic group				3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	160	320	160	80	80	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	160	160	320	640	160	640	160	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	2560	2560	2560	1280	1280	1280	1280	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	160	320	320	640	320	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	160	160	320	640	160	320	160	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	1280	640	1280	640	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	640	320	640	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	640	1280	640	640	1280	640	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	1280	1280	1280	1280	640	640	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	2560	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Cote d'Ivoire/1847/2013		2013-10-31	SIAT3	<	80	80	160	160	80	320	160	80	160
A/Ankara/34/2014		2013-10-31	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Ankara/1373/2013		2013-10-31	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Ankara/1110/2013		2013-10-31	SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Cote d'Ivoire/1900/2013		2013-11-04	SIAT3	<	160	160	160	320	80	320	160	160	160
A/Tokat/239/2014		2013-11-04	SIAT2	<	80	160	160	320	40	320	80	80	80
A/Ankara/1396/2013		2013-11-04	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Ankara/1356/2013		2013-11-04	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Madagascar/00147/2014	3C.3	2013-11-04	SIAT2	<	160	160	160	320	80	320	160	320	160
A/Catalonia/2141012NS/2013	3C.3	2013-12-06	C0/SIAT1	<	80	160	160	320	80	320	160	160	80
A/Catalonia/6623S/2013	3C.3	2013-12-10	C0/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Catalonia/2141628NS/2013		2013-12-10	C0/SIAT1	<	160	160	160	320	160	320	160	160	160
A/Catalonia/2143786NS/2013		2013-12-19	C0/SIAT1	<	160	160	160	320	160	320	160	160	160
A/Catalonia/2144572NS/2014		2013-12-22	C0/SIAT1	<	80	40	160	320	80	320	160	160	160
A/Valencia/624026S/2014	3C.3	2014-01-06	C0/SIAT1	<	160	160	160	320	160	320	160	160	160
A/Switzerland/9851671/2014	3C.2	2014-01-06	SIAT4	<	160	160	160	320	80	320	160	160	160
A/Adana/234/2014		2014-01-06	SIAT2	<	160	160	160	320	160	320	320	160	160
A/Ankara/1410/2013		2014-01-06	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Ankara/1404/2013		2014-01-06	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Ankara/1366/2013		2014-01-06	SIAT2	<	80	80	160	320	80	320	160	160	160
A/Antalya/1083/2013	3C.3	2014-01-06	SIAT1/SIAT1	<	40	80	80	160	80	160	80	80	40
A/Madagascar/00153/2014	3C.3	2014-01-06	SIAT2	80	320	640	640	1280	320	1280	640	640	640
A/Salamanca/27/2014		2014-01-10	MDCK1/SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	160
A/Catalonia/6095851NS/2014		2014-01-13	C0/SIAT1	<	80	160	160	320	80	320	160	160	160
A/Salamanca/44/2014	3C.3	2014-01-14	MDCK1/SIAT1/SIAT1	40	160	160	160	320	160	320	160	320	160
A/Segovia/81/2014	3C.3	2014-01-16	MDCK1/SIAT1/SIAT1	40	160	160	320	320	160	320	320	320	320
A/Salamanca/92/2014	3C.3	2014-01-22	MDCK1/SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Catalonia/2152420NS/2014	3C.3	2014-01-25	C0/SIAT1	40	80	80	160	320	80	320	160	160	160
									Vaccine				

1. < = <40

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-03-04 + 2014-03-07 + 2014-03-12

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
	Genetic group				3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	320	640	160	160	160	320	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	<	160	160	160	320	80	320	160	160	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	2560	2560	2560	1280	1280	1280	1280	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	160	320	320	640	320	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	320	320	640	160	640	320	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	1280	640	1280	640	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	320	320	640	320	640	320	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	1280	1280	1280	640	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	1280	1280	1280	1280	640	1280	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	2560	1280	1280	1280	1280	1280	2560	1280
TEST VIRUSES													
A/Madagascar/04245/2013	3C.3	2013-12-27	SIAT2	40	160	160	320	640	160	640	320	320	320
A/Jordan/30661/2013		2013-11-28	SIAT1/SIAT1	<	80	80	160	320	80	160	160	160	160
A/Jordan/30677/2013	3C.3	2013-11-29	SIAT1/SIAT1	160	160	320	320	640	160	640	320	320	320
A/Jordan/20539/2013	3C.3	2013-12-06	SIAT1/SIAT1	<	160	160	320	320	160	320	160	160	320
A/Cote d'Ivoire/2205/2013		2013-12-17	SIAT2	40	160	320	320	640	160	640	320	320	160
A/Cote d'Ivoire/2210/2013		2013-12-17	SIAT2	40	160	320	320	640	160	640	320	320	160
A/Madagascar/04228/2013	3C.3	2013-12-26	SIAT2	<	160	160	160	320	160	320	320	320	160
A/Cote d'Ivoire/2259/2013		2013-12-26	SIAT2	40	160	320	320	640	160	640	320	320	320
A/Ankara/23/2014		2014-01-08	SIAT2	<	160	160	320	640	160	640	160	320	320
A/Jordan/20006/2014	3C.3	2014-01-16	SIAT1/SIAT1	40	160	160	320	320	160	640	160	160	160
A/Hamburg/1/2014	3C.3	2014-01-20	C2/SIAT1	<	80	80	80	160	80	160	80	80	80
A/Berlin/8/2014		2014-01-21	C2/SIAT1	<	160	160	320	640	160	640	320	320	320
A/Hessen/4/2014	3C.3	2014-01-27	C2/SIAT1	<	40	80	80	160	80	160	80	80	80
A/Berlin/12/2014		2014-01-27	C2/SIAT1	<	80	80	160	160	80	160	80	80	80
A/Hessen/1/2014		2014-01-27	C2/SIAT1	40	160	160	320	320	160	320	160	160	160
A/Baden-Wurttemberg/7/2014		2014-01-29	C2/SIAT1	40	160	160	160	320	160	320	160	160	160
A/Bayern/3/2014		2014-01-30	C2/SIAT1	<	160	160	320	640	160	320	320	320	320
A/Berlin/20/2014	3C.3	2014-02-04	C2/SIAT1	<	80	80	160	320	80	160	80	80	80
A/Bremen/1/2014	3C.3	2014-02-07	C2/SIAT1	<	80	160	160	320	80	160	160	80	80
A/Baden-Wurttemberg/14/2014	3C.3	2014-02-10	C2/SIAT1	<	160	160	320	640	160	320	160	160	160
									Vaccine				

1. < = <40

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-03-14 + 2014-03-26 + 2014-03-28

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] / Post infection ferret antisera: A/Perth 16/09 F35/11	A/Stock 18/11 F12/13	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	*A/Texas 50/12 Egg F08/14	A/Samara 73/13 F24/13	Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
	Genetic group				3A	6	3C.1	3B	3C.1	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	320	160	640	160	160	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	320	80	160	320	80	160	320	160	160	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	2560	1280	1280	640	640	640	640	640	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	40	160	160	160	320	80	160	160	80	160	160
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	160	160	320	640	160	320	320	160	320	160
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	320	640	1280	1280	1280	1280	1280	640	1280	640	1280
A/Texas/50/2012 (8)	3C.1	2012-04-15	MDCK1/C2/SIAT2	80	320	320	320	640	320	320	1280	320	320	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	320	320	640	320	320	640	320	640	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	640	1280	1280	1280	640	1280	640	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	1280	2560	1280	2560	640	1280	1280	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	320	640	2560	640	640	1280	1280	1280	1280	1280	1280
TEST VIRUSES														
A/Jordan/30541/2013	3C.3	2013-10-29	SIAT2	<	320	160	320	640	160	ND	640	160	320	160
A/Jordan/30517/2013		2013-10-31	SIAT3	<	160	160	160	320	80	ND	320	160	160	160
A/Jordan/30577/2013		2013-11-06	SIAT2	<	160	160	160	320	80	ND	320	160	160	160
A/Jordan/30586/2013	3C.3	2013-11-11	SIAT2	<	160	80	160	320	80	ND	320	160	160	160
A/Jordan/30609/2013		2013-11-16	SIAT2	<	320	160	320	640	160	ND	640	320	320	320
A/Cameroon/6468/2013	3C.3	2013-11-18	SIAT3	<	160	80	160	320	160	ND	320	160	160	160
A/Cameroon/6634/2013		2013-11-19	SIAT3	40	160	160	160	320	160	ND	320	160	160	320
A/Jordan/30654/2013		2013-11-25	SIAT2	<	160	80	160	320	80	ND	320	160	160	160
A/Jordan/30657/2013		2013-11-27	SIAT3	<	160	80	160	320	160	ND	320	160	160	160
A/Jordan/30669/2013		2013-11-30	SIAT2	<	160	80	160	320	80	ND	320	160	160	160
A/Jordan/20524/2013		2013-12-03	SIAT3	<	160	160	160	320	160	ND	320	160	160	160
A/Jordan/30694/2013		2013-12-04	SIAT3	40	320	320	320	640	160	ND	640	160	320	160
A/Jordan/30698/2013		2013-12-05	SIAT3	40	160	160	160	320	160	ND	320	160	160	160
A/Jordan/30707/2013		2013-12-09	SIAT2	<	160	160	160	320	80	ND	320	160	320	160
A/Cameroon/7012/2013	3C.3	2013-12-10	SIAT4	<	160	160	160	320	80	ND	320	160	160	160
A/Cameroon/7186/2013	3C.3	2013-12-17	SIAT4	40	320	160	320	320	160	ND	640	320	320	160
A/Jordan/20551/2013		2013-12-21	SIAT2	<	320	160	160	640	160	ND	640	160	160	160
A/Jordan/30714/2013	3C.3	2013-12-21	SIAT2	<	160	80	160	320	80	ND	320	160	160	160
A/Jordan/20565/2013		2013-12-24	SIAT2	<	320	160	320	640	160	ND	640	160	160	160
A/Jordan/20561/2013	3C.3	2013-12-24	SIAT2	<	160	160	160	320	160	160	320	320	160	160
A/Jordan/20564/2013	3C.3	2013-12-24	SIAT2	<	160	160	160	320	80	160	320	160	160	160
A/Algeria/277/2013	3C.3	2013-12-25	SIAT2	40	160	160	320	640	160	160	320	320	160	320
A/Jordan/30014/2014		2013-12-28	SIAT2	<	80	40	160	320	80	160	320	160	160	160
									Vaccine					

1. < = <40; * new antiserum

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-04-01 + 2014-04-07 +2014-04-14

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F12/13 3A	A/Iowa 19/10 F15/11 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	320	320	320	640	160	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	640	160	320	640	160	640	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	5120	2560	2560	1280	1280	1280	1280	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	160	320	640	160	640	160	320	640
A/Athens/112/2012	3B	2012-02-01	SIAT5	80	320	160	320	640	160	640	160	320	640
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	1280	2560	2560	2560	2560	1280	1280	2560	1280	2560
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	640	320	320	640	320	1280	320	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	2560	2560	1280	1280	1280	2560
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	1280	2560	1280	1280	640	1280	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	2560	1280	2560	2560	1280	2560	2560	2560
TEST VIRUSES													
A/Cameroon/6624/2013	3C.3	2013-11-26	SIAT4	40	320	160	160	640	160	640	320	320	320
A/Cameroon/6793/2013		2013-12-03	SIAT4	40	320	160	160	320	160	640	320	320	320
A/Cameroon/7010/2013	3C.3	2013-12-05	SIAT4	<	160	80	160	320	80	320	80	160	160
A/Cameroon/7002/2013		2013-12-09	SIAT4	40	320	160	320	320	160	640	160	320	320
A/Jordan/30022/2014	3C.3	2014-01-02	SIAT2	<	320	80	160	320	80	320	160	160	160
A/Madagascar/00031/2014	3C.3	2014-01-03	SIAT3	<	320	80	160	320	80	320	80	160	160
A/Greece/12/2014	3C.3	2014-01-05	SIAT4	<	160	40	80	160	40	160	80	80	80
A/Madagascar/00041/2014	3C.3	2014-01-06	SIAT2	<	320	160	80	320	80	320	160	160	160
A/Slovenia/87/2014		2014-01-09	SIAT3	<	320	80	160	320	80	320	160	80	160
A/Israel/Z774/2014	3C.3	2014-01-10	SIAT2	<	320	160	160	320	160	320	320	320	160
A/Switzerland/9851384/2014	3C.3	2014-01-10	SIAT2	<	160	160	80	320	80	320	160	160	80
A/Jordan/30031/2014		2014-01-14	SIAT2	<	320	160	160	320	160	320	160	160	160
A/Lamia.GR/79/2014	3C.3	2014-01-15	SIAT4	<	320	80	160	320	80	640	160	160	160
A/Switzerland/9884248/2014	3C.3	2014-01-15	SIAT2	<	160	160	160	320	160	320	320	320	160
A/Van/354/2014	3C.3	2014-01-17	SIAT2	<	320	160	160	320	160	320	160	160	160
A/Samsun/357/2014		2014-01-17	SIAT2	<	320	80	160	320	160	320	160	160	160
A/Adana/363/2014		2014-01-17	SIAT2	<	320	160	160	640	160	320	320	320	160
A/Jordan/30062/2014	3C.3	2014-01-19	SIAT2	<	320	160	160	320	160	320	320	160	160
A/Jordan/20053/2014	3C.3	2014-01-21	SIAT2	<	320	160	160	320	160	320	160	160	160
A/Cyprus/F4/2014	3C.3	2014-01-21	SIAT3	<	160	160	80	320	80	320	160	160	80
A/Cyprus/F15/2014		2014-01-24	SIAT2	<	160	80	160	320	80	320	160	160	160
A/Cyprus/F30/2014	3C.3	2014-01-28	SIAT2	<	160	80	160	640	80	320	160	160	160
A/Cyprus/F31/2014		2014-01-28	SIAT2	<	320	80	160	320	80	320	160	160	160
A/Macedonia/IK(80)/2014	3C.3	2014-02-11	SIAT4	40	640	320	320	640	160	640	320	320	320
									Vaccine				

1. < = <40

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-05-02

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F12/13 3A	A/Iowa 19/10 F15/11 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	320	320	320	640	160	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	640	160	320	320	160	320	160	160	160
A/Iowa/19/2010	6	2010-12-30	E3/E1	640	2560	5120	2560	2560	2560	1280	1280	1280	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	160	320	320	160	320	160	160	160
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	640	160	320	640	320	320	320	320	320
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	1280	320	1280	1280	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	640	320	640	1280	640	640	320	1280	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	1280	1280	640	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	1280	1280	640	1280	1280	320	320	1280	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	320	1280	1280	1280	1280	1280	640	1280	2560	1280
TEST VIRUSES													
A/Zambia/13/176/2013	3C.3	2013-09-26	SIAT2	<	320	160	320	320	160	640	320	80	320
A/Cameroon/5634/2013		2013-10-04	SIAT2	40	320	80	160	320	80	320	160	160	160
A/Cameroon/5638/2013		2013-10-07	SIAT2	<	320	80	160	320	160	320	160	160	160
A/Cameroon/5793/2013	3C.3	2013-10-11	SIAT2	<	320	80	160	320	80	160	160	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT2	<	40	<	40	80	40	160	80	80	80
A/Bulgaria/283/2014	3C.2	2014-01-21	SIAT2/SIAT1	<	320	80	160	320	160	160	160	160	160
A/Bulgaria/507/2014		2014-01-22	SIAT2/SIAT1	40	320	80	160	640	160	320	160	160	160
A/Bulgaria/477/2014		2014-01-30	SIAT2/SIAT1	40	320	80	160	640	160	320	160	320	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2	<	40	<	40	80	<	80	80	80	80
A/Stockholm/6/2014	3C.3a	2014-02-06	MDCK1/SIAT1	<	80	40	80	160	80	160	80	80	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2	<	40	<	<	80	<	80	80	80	80
A/Bulgaria/652/2014	3C.3	2014-02-06	SIAT1/SIAT1	<	320	80	160	640	80	320	80	160	80
A/Bulgaria/725/2014	3C.3	2014-02-11	SIAT2/SIAT1	<	320	80	160	320	160	320	160	160	160
A/Bulgaria/806/2014		2014-02-13	SIAT2/SIAT1	<	320	80	160	320	160	320	160	160	160
A/Bulgaria/800/2014	3C.3	2014-02-14	SIAT2/SIAT1	<	160	80	160	160	80	320	160	160	160
A/Bulgaria/845/2014	3C.3	2014-02-19	SIAT2/SIAT1	<	320	80	320	320	160	320	160	160	160

1. < = <40

Vaccine

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-05-06

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Stock 18/11 F12/13 3A	A/Iowa 19/10 F18/13 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	320	160	640	640	320	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	80	640	160	640	640	160	640	320	160	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	2560	2560	1280	1280	1280	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	160	320	320	160	320	160	160	160
A/Athens/112/2012	3B	2012-02-01	SIAT5	160	640	320	640	640	160	320	320	320	320
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	320	1280	640	2560	1280	1280	640	1280	640	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	1280	320	1280	640	320	1280	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	2560	640	2560	1280	1280	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	2560	640	2560	1280	640	1280	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	1280	2560	2560	1280	1280	1280	2560	1280
TEST VIRUSES													
A/Cameroon/5516/2013		2013-10-07	SIAT3	40	320	160	320	320	160	320	320	160	320
A/Togo/LNG398/2013	3C.3	2013-10-28	SIAT3	80	320	160	640	320	160	320	320	160	320
A/Togo/LNG333/2013	3C.3	2013-10-01	SIAT3	<	320	80	640	320	160	320	160	160	160
A/Bulgaria/040/2014	3C.3	2014-01-08	SIAT2/SIAT1	40	640	320	1280	640	160	640	320	320	320
A/Bulgaria/127/2014	3C.3	2014-01-14	SIAT2/SIAT1	40	640	320	640	640	320	640	320	320	320
A/Bulgaria/200/2014		2014-01-15	SIAT2/SIAT1	40	640	320	1280	640	160	640	320	320	320
A/Norway/309/2014		2014-01-17	MDCK1/SIAT1	<	320	80	640	640	160	640	320	160	160
A/Norway/211/2014	3C.3	2014-01-18	LLC/MDCK2/MDCK1/SIAT1	40	320	160	640	640	160	640	320	160	320
A/Norway/293/2014		2014-01-20	LLC/MDCK2/MDCK1/SIAT1	40	320	160	640	640	160	640	320	160	320
A/Norway/272/2014	3C.3	2014-01-23	MDCK1/SIAT1	40	640	320	1280	640	320	1280	640	640	640
A/Norway/313/2014	3C.3	2014-01-23	MDCK1/SIAT1	40	640	160	640	640	160	640	320	320	320
A/Norway/371/2014		2014-01-23	MDCK1/SIAT1	160	1280	640	1280	640	320	1280	640	640	640
A/Norway/277/2014	3C.3	2014-01-26	MDCK1/SIAT1	40	640	160	640	640	160	640	320	320	320
A/Norway/369/2014		2014-01-29	MDCK2/SIAT1	40	320	160	640	640	320	640	320	320	320
A/Poland/896/2014	3C.3	2014-01-29	SIAT2	40	320	160	640	320	160	640	320	320	320
A/Norway/409/2014	3C.3	2014-01-30	LLC/MDCK2/MDCK1/SIAT1	40	640	160	640	640	320	640	640	320	320
A/Poland/120/2014	3C.3	unknown	SIAT2	<	320	160	320	320	160	320	160	160	160
A/Poland/154/2014	3C.3	unknown	SIAT2	<	320	160	320	320	160	640	320	320	160
A/Poland/148/2014	3C.3	unknown	SIAT2	40	320	80	320	320	80	320	160	160	80
A/Poland/160/2014	3C.3	unknown	SIAT2	40	320	160	640	640	80	640	320	320	160
A/Poland/179/2014	3C.3	unknown	SIAT2	40	640	160	640	2560	160	640	320	320	320
A/Poland/1861/2014		2014-02-24	SIAT2	<	320	160	640	320	80	320	160	160	160
A/Poland/1955/2014	3C.3	2014-02-26	SIAT2	<	320	160	640	320	160	640	320	320	160
A/Poland/40/2014	3C.3	2014-02-26	SIAT2	<	320	160	640	320	160	320	160	160	160
A/Poland/39/2014	3C.3	2014-02-26	SIAT2	<	320	160	640	320	80	160	160	160	160
A/Poland/758/2014		2014-02-28	SIAT2	<	640	640	1280	640	640	640	640	640	640
A/Poland/2175/2014		2014-03-04	SIAT2	<	320	160	640	640	160	640	320	320	160
A/Poland/1899/2014	3C.3	2014-03-12	SIAT2	<	640	160	640	640	160	640	320	320	160
A/Poland/1905/2014		2014-03-21	SIAT2	80	1280	320	1280	1280	160	1280	320	320	320
A/Poland/3487/2014		2014-03-26	SIAT2	80	640	320	320	640	160	1280	320	640	320
									Vaccine				

1. < = <40

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-05-09

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Stock 18/11 F12/13 3A	A/Iowa 19/10 F18/13 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3
				Haemagglutination inhibition titre[1] – Post infection ferret antisera									
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	320	160	640	640	320	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	80	1280	160	640	640	160	640	320	160	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	2560	2560	1280	1280	1280	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	160	640	320	160	320	160	160	160
A/Athens/112/2012	3B	2012-02-01	SIAT5	160	640	320	640	640	160	320	320	320	320
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	320	1280	640	2560	1280	1280	640	1280	640	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	1280	320	1280	640	320	1280	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	2560	640	2560	1280	1280	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	2560	640	2560	1280	640	1280	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	1280	2560	2560	1280	1280	1280	2560	1280
TEST VIRUSES													
A/Bulgaria/147/2014	3C.3	2014-01-14	SIAT1/SIAT2	<	160	80	320	160	40	160	80	80	160
A/Norway/347/2014	3C.3	2014-01-24	MDCK2/SIAT2	<	320	160	640	320	80	320	160	160	320
A/Norway/328/2014		2014-01-25	MDCK2/SIAT2	<	320	160	320	320	80	320	160	160	160
A/Norway/358/2014		2014-01-27	MDCK1/SIAT2	<	160	80	320	160	40	320	160	80	160
A/Norway/389/2014	3C.3	2014-01-29	MDCK1/SIAT2	<	160	80	320	160	40	320	80	80	160
A/Kharkov/203/2014	3C.3	2014-02-26	SIAT2/SIAT1	<	320	160	640	320	160	320	160	160	160
A/Dnipro/228/2014		2014-02-28	SIAT2/SIAT1	<	320	160	640	320	160	320	160	160	160
A/Dnipro/229/2014	3C.3	2014-02-28	SIAT2/SIAT1	<	320	160	640	160	160	320	160	160	160
A/Poland/1111/2014		2014-03-10	SIAT3	<	640	320	640	640	160	640	320	320	160
A/Poland/1900/2014		2014-03-12	SIAT3	<	160	160	640	320	80	320	160	160	160
A/Dnipro/235/2014	3C.3	2014-03-12	SIAT2/SIAT1	<	320	160	640	320	160	320	160	160	320
A/Dnipro/236/2014		2014-03-14	SIAT2/SIAT1	<	320	160	640	320	80	320	160	160	160
A/Poland/1256/2014		2014-03-18	SIAT3	<	640	160	640	320	160	640	160	160	320
A/Ukraine/218/2014	3C.3	2014-03-21	SIAT2/SIAT1	<	320	160	640	640	80	320	160	320	160
A/Poland/1702/2014	3C.3	2014-04-08	SIAT3	<	320	160	640	640	160	640	160	160	320
									Vaccine				

1. < = <40

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-05-13 + 2014-05-16

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13	*A/Sth Afr 4655/13 F09/14	*A/Sth Afr 4655/13 F10/14	*A/Sth Afr 4655/13 F11/14	*A/Stock 1/13 F12/14
				Haemagglutination inhibition titre1											
				Post infection ferret antisera											
	Genetic group				3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3	3C.3 cl 118-76	3C.3 cl 101-60	3C.3 cl 93-38	3C.2 cl 36-18
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	640	640	160	160	160	160	160	80	40	80	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	640	320	160	320	160	160	160	160	80	40	80
A/Athens/112/2012	3B	2012-02-01	SIAT5	80	640	640	160	320	320	320	320	320	80	160	160
A/Texas/50/2012 (6&7)	3C.1	2012-04-15	E5/E2	640	5120	1280	1280	1280	2560	1280	640	640	160	320	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	1280	1280	320	1280	640	1280	640	640	160	320	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	2560	1280	1280	1280	1280	1280	1280	640	80	320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	2560	1280	640	1280	640	2560	1280	640	160	640	320
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	320	5120	1280	1280	640	1280	2560	1280	640	160	320	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 118-76	80	640	320	160	320	160	320	160	160	320	1280	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	640	320	160	320	160	320	160	80	320	1280	320
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 93-38	80	640	160	160	320	160	320	80	80	320	1280	320
A/Stockholm/1/2013	3C.2	2013-01-13	E6 clone 36-18	80	640	320	160	320	160	320	160	80	320	1280	320
TEST VIRUSES															
A/Tehran/30732/2013		2013-03-22	MDCK3/SIAT1	40	640	320	160	320	320	320	320	320	80	80	160
A/Zambia/13/093/2013	3C.3	2013-08-28	Cx/SIAT2	<	640	320	80	320	<	80	160	160	40	<	40
A/Zambia/13/262/2013		2013-08-07	Cx/SIAT1	40	1280	640	160	640	640	320	320	320	160	160	160
A/Zambia/13/127/2013	3C.3	2013-08-12	SIAT2	80	1280	1280	320	1280	640	1280	640	640	160	160	320
A/Zambia/13/338/2013		2013-09-13	Cx/SIAT1	<	640	320	160	320	320	320	320	320	80	80	160
A/Zambia/13/109/2013	3C.3	2013-09-21	Cx/SIAT1	40	640	640	160	640	320	320	320	320	80	80	160
A/Norway/86/2014	3C.3	2014-01-08	MDCK2/SIAT1	<	320	160	80	320	160	160	160	160	80	80	80
A/Norway/160/2014		2014-01-08	MDCK1/SIAT1	<	640	640	160	320	320	320	320	320	80	80	160
A/Norway/184/2014	3C.3	2014-01-08	MDCK1/SIAT1	<	640	640	320	640	320	640	320	320	80	160	160
A/Norway/163/2014		2014-01-11	MDCK2/SIAT1	40	640	320	160	640	320	320	320	320	80	160	160
A/Norway/161/2014	3C.3	2014-01-12	MDCK2/SIAT1	40	640	320	160	640	320	320	320	320	80	160	160
A/Norway/228/2014		2014-01-12	MDCK2/SIAT1	80	640	320	160	640	320	320	320	320	80	160	160
A/Norway/162/2014		2014-01-13	MDCK1/SIAT1	<	640	320	160	320	320	320	320	320	80	80	160
A/Norway/120/2014	3C.3	2014-01-14	MDCK1/SIAT1	<	640	640	160	320	320	320	320	320	80	80	160
A/Norway/226/2014	3C.3	2014-01-14	MDCK1/SIAT1	<	320	320	80	320	160	160	160	160	80	80	80
A/Norway/208/2014		2014-01-16	MDCK1/SIAT1	<	320	320	160	320	320	320	320	320	80	80	160
A/Switzerland/10030276/2014		2014-02-05	SIAT2	80	1280	1280	320	640	320	320	320	320	160	160	320
A/Dnipro/230/2014		2014-02-28	SIAT2/SIAT3	80	1280	640	320	640	320	640	640	320	160	160	160
A/Dnipro/232/2014	3C.3	2014-03-03	SIAT2/SIAT3	40	1280	1280	320	1280	640	640	640	640	160	160	160
A/Poland/1877/2014		2014-03-03	SIAT2	40	640	640	160	320	320	320	320	320	80	80	160
A/Norway/1003/2014	3C.3	2014-03-04	SIAT2	40	640	640	160	320	160	160	160	320	40	80	80
A/Georgia/495/2014	3C.2	2014-03-04	SIAT2	40	1280	640	160	640	320	640	320	320	80	80	160
A/Dnipro/233/2014		2014-03-04	SIAT2/SIAT1	<	320	160	80	320	160	160	160	160	40	80	80
A/Norway/1020/2014	3C.3	2014-03-07	SIAT2	<	640	640	80	320	80	80	160	160	<	<	80
A/Georgia/542/2014		2014-03-07	SIAT2	40	640	640	160	640	320	320	320	320	80	80	160
A/Georgia/584/2014	3C.2	2014-03-10	SIAT2	<	640	640	160	640	<	320	160	320	80	40	80
A/Norway/1130/2014		2014-03-11	SIAT2	<	640	320	160	640	160	320	160	320	80	80	160
A/Dnipro/234/2014	3C.2	2014-03-11	SIAT2/SIAT3	80	1280	640	160	640	320	640	320	320	80	160	160
A/Norway/1078/2014	3C.3	2014-03-16	SIAT2	<	320	320	80	320	160	160	160	160	40	80	160
A/Poland/3464/2014	3C.2	2014-03-19	SIAT2	<	640	320	160	320	160	320	320	160	80	80	160
A/Switzerland/10261363/2014	3C.3	2014-03-20	SIAT2	<	1280	1280	320	640	640	320	320	320	80	160	160
A/Georgia/699/2014	3C.2	2014-03-26	SIAT2	40	1280	640	160	640	320	320	320	320	80	80	160
A/Switzerland/10295858/2014	3C.3	2014-03-27	SIAT2	80	1280	1280	320	640	640	320	640	640	160	160	320
A/Switzerland/10295823/2014	3C.3	2014-03-28	SIAT2	<	640	640	160	320	320	160	320	320	80	80	160
A/Switzerland/10295759/2014		2014-03-28	SIAT2	40	1280	640	160	320	160	160	160	320	40	40	160
							Vaccine								

1. < = <40; * new antisera

Table 8-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-05-20 + 2014-05-22

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	640	160	160	160	320	160	40	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	320	160	640	160	320	160	160	160
A/Athens/112/2012	3B	2012-02-01	SIAT4	40	160	640	160	640	320	320	320	80	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	2560	1280	1280	1280	2560	1280	1280	80	160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	1280	320	1280	640	1280	640	160	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	640	1280	1280	1280	80	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	160	640	1280	640	1280	640	2560	1280	160	160
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	320	1280	1280	1280	1280	1280	2560	1280	80	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	40	320	160	320	160	320	80	320	160
A/Stockholm/1/2013	3C.2	2013-01-13	E6 clone 36-18	80	40	320	80	160	80	160	80	160	160
TEST VIRUSES													
A/Ukraine/77/2014	3C.3	2014-02-07	SIAT2/SIAT2	<	160	320	80	320	160	160	160	40	80
A/Ukraine/86/2014	3C.3	2014-02-13	SIAT2/SIAT2	<	160	640	160	640	160	320	160	80	160
A/Kharkov/200/2014		2014-02-13	SIAT2/SIAT2	<	160	640	80	320	160	160	160	80	160
A/Ukraine/105/2014		2014-02-15	SIAT2/SIAT2	40	160	320	160	640	320	320	160	80	160
A/Zhitomir/286/2014	3C.3	2014-02-18	SIAT2/SIAT1	<	160	320	80	320	160	160	160	40	80
A/Kharkov/201/2014	3C.3	2014-02-19	SIAT2/SIAT2	<	160	320	80	320	160	160	160	40	160
A/Zhitomir/290/2014	3C.3	2014-02-20	SIAT2/SIAT1	<	160	320	80	320	160	160	160	40	80
A/Dnipro/168/2014		2014-02-21	SIAT2/SIAT2	80	320	640	160	640	320	320	320	80	160

1. < = <40

Vaccine

Table 8-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-06-03

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	*A/Stock 6/14 F14/14 3C.3a
				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	160	160	160	40	80	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	320	640	320	320	320	80	160	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	640	1280	80	160	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	320	640	640	640	640	80	160	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	1280	1280	1280	80	160	320
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	640	640	1280	640	2560	640	80	160	160
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	1280	2560	1280	160	320	320
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	160	320	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	80	160	320	160	320	160	320	320	160
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	40	40	320	160	160	160	40	80	640
TEST VIRUSES													
A/Norway/53/2014	3C.3	2013-12-20	MDCK2/SIAT1	<	160	160	320	160	160	160	80	80	320
A/Norway/3331/2013		2013-12-25	MDCK1/SIAT1	<	160	160	320	160	160	160	80	80	320
A/Norway/104/2014		2013-12-25	MDCK1/SIAT1	<	320	160	640	320	320	320	80	160	640
A/Norway/18/2014	3C.2	2013-12-31	MDCK2/SIAT1	<	80	80	160	40	160	160	<	40	80
A/Norway/58/2014	3C.3	2014-01-02	MDCK2/SIAT1	<	160	160	320	320	160	160	40	80	320
A/Norway/70/2014		2014-01-02	MDCK2/SIAT1	80	320	160	640	320	320	320	160	160	640
A/Norway/115/2014		2014-01-06	MDCK1/SIAT1	<	320	160	640	320	320	320	80	160	320
A/Norway/133/2014		2014-01-06	LLC/MDCK2/ MDCK1/SIAT1	40	320	320	640	320	320	320	80	160	640
A/Norway/67/2014		2014-01-07	MDCK2/SIAT1	80	320	160	640	320	320	640	80	160	640
A/Roma/2/2014		2014-01-13	SIAT1/SIAT1	<	160	160	640	320	320	320	80	160	640
A/Sassari/1/2014		2014-01-14	SIAT2/SIAT1	<	160	160	640	320	160	320	80	80	320
A/Parma/8/2014		2014-01-17	MDCK1/SIAT1	80	320	320	1280	640	640	640	160	160	640
A/Roma/3/2014		2014-01-17	SIAT1/SIAT1	<	160	160	640	320	160	160	80	80	320
A/Trieste/10/2014	3C.3	2014-01-23	SIAT1/SIAT1	<	160	160	320	160	160	160	40	80	320
A/Milano/31/2014		2014-01-23	SIAT1/SIAT1	<	160	160	320	160	320	160	40	80	320
A/Roma/5/2014		2014-01-23	SIAT1/SIAT1	40	320	160	640	320	320	320	80	160	640
A/Roma/7/2014		2014-01-24	SIAT1/SIAT1	<	160	160	640	320	160	320	80	160	320
A/Milano/58/2014	3C.3	2014-01-28	SIAT1/SIAT1	<	160	80	320	160	160	160	40	80	320
A/Firenze/6/2014		2014-02-03	MDCK2/SIAT1	40	160	160	640	320	320	320	80	160	320
A/Sassari/10/2014		2014-02-03	SIAT1/SIAT1	<	160	160	320	160	160	160	80	80	320
A/Parma/28/2014		2014-02-03	MDCK2/SIAT1	40	160	160	320	160	320	320	80	160	320
A/Parma/33/2014		2014-02-03	MDCK2/SIAT1	40	320	320	1280	320	320	320	80	160	640
A/Milano/84/2014	3C.3	2014-02-03	SIAT1/SIAT1	<	320	160	640	320	320	320	80	160	640
A/Roma/11/2014	3C.3	2014-02-04	SIAT1/SIAT1	40	320	160	640	320	320	320	80	160	640
A/Firenze/9/2014	3C.3	2014-02-05	MDCK2/SIAT1	<	80	80	320	160	80	80	40	80	160
A/Sassari/12/2014	3C.2	2014-02-06	SIAT2/SIAT1	40	160	160	320	160	320	160	80	80	320
A/Milano/72/2014	3C.3	2014-02-06	SIAT1/SIAT1	<	80	80	160	160	80	160	40	40	160
A/Sassari/15/2014		2014-02-07	Cx/SIAT1	<	160	160	320	320	320	320	80	80	320
A/Ukraine/67/2014		2014-02-07	SIAT2/SIAT4	<	160	160	640	160	160	160	80	80	320
A/Milano/101/2014		2014-02-10	SIAT1/SIAT1	40	320	320	640	320	320	320	80	160	640
A/Parma/40/2014	3C.3	2014-02-13	SIAT1/SIAT1	<	160	160	320	160	160	160	40	80	320
A/Milano/113/2014	3C.2	2014-02-18	SIAT1/SIAT1	<	160	160	320	160	320	160	80	80	320
A/Genova/09/2014	3C.3	2014-02-25	SIAT1/SIAT1	<	160	160	320	320	320	160	80	80	640
A/Sassari/17/2014	3C.3	2014-03-05	Cx/SIAT1	<	160	160	320	160	160	160	40	80	320

1. < = <40; * new antiserum

Vaccine

Table 8-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-06-06 + 2014-06-10

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	160	160	160	40	80	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	80	320	320	640	320	320	320	80	160	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	1280	1280	80	160	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	320	640	640	640	640	80	160	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	1280	1280	1280	80	160	320
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	640	640	1280	640	2560	640	80	160	160
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	1280	2560	1280	160	320	320
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	160	320	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	80	160	320	160	320	160	320	320	160
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	40	40	160	160	80	160	40	80	320
TEST VIRUSES													
A/Trieste/01/2014		2013-12-28	SIAT1/SIAT1	<	160	160	320	160	160	160	80	80	320
A/Milano/13/2014		2014-01-12	SIAT1/SIAT1	<	160	160	320	320	160	320	80	80	320
A/Serbia/NS-613/2014	3C.3	2014-01-15	SIAT1	<	160	80	320	160	160	160	40	80	320
A/Serbia/NS-619/2014	3C.3	2014-01-17	SIAT1	<	160	160	320	320	320	160	80	80	320
A/Serbia/NS-630/2014		2014-01-21	SIAT1	<	320	160	640	320	320	320	80	160	640
A/Georgia/76/2014	3C.3	2014-01-22	SIAT1	<	160	160	640	320	320	320	80	160	320
A/Georgia/117/2014		2014-01-23	SIAT1	<	160	160	320	320	320	160	80	80	640
A/Georgia/108/2014		2014-01-27	SIAT1	40	160	160	640	320	320	320	80	160	640
A/Norway/513/2014		2014-01-29	SIAT1	40	320	320	640	320	640	640	160	160	640
A/Norway/504/2014		2014-02-01	SIAT1	<	320	160	640	320	320	320	80	160	640
A/Norway/507/2014	3C.3	2014-02-03	SIAT1	<	160	80	320	160	160	160	40	80	160
A/Norway/627/2014		2014-02-04	SIAT1	40	320	160	640	160	320	320	80	80	640
A/Norway/467/2014		2014-02-06	SIAT1	<	320	160	640	320	320	320	80	80	320
A/Georgia/175/2014		2014-02-06	SIAT1	40	320	320	640	640	640	640	160	320	640
A/Georgia/215/2014	3C.2	2014-02-06	SIAT1	<	80	80	320	160	160	160	40	80	160
A/Georgia/231/2014		2014-02-07	SIAT1	<	160	160	320	160	160	160	80	80	320
A/Georgia/241/2014		2014-02-10	SIAT1	40	320	320	640	320	320	320	80	160	640
A/Norway/580/2014	3C.3	2014-02-11	SIAT2	<	160	160	640	320	160	320	80	80	320
A/Serbia/NS-666/2014	3C.3	2014-02-11	SIAT1	<	160	80	160	160	160	160	40	40	160
A/Serbia/NS-669/2014	3C.3	2014-02-13	SIAT1	<	160	80	320	160	160	160	40	80	320
A/Belgium/14H0010/2014	3C.3	2014-02-13	SIAT1	<	160	80	320	160	160	160	80	80	320
A/Serbia/NS-670/2014		2014-02-13	SIAT1	40	320	320	640	640	640	320	160	160	640
A/Belgium/14S0070/2014	3C.2	2014-02-16	SIAT1	40	160	160	640	160	160	320	80	80	320
A/Georgia/328/2014	3C.3	2014-02-18	SIAT1	<	320	160	640	320	320	320	80	160	320
A/Serbia/NS-682/2014	3C.3	2014-02-21	SIAT1	<	160	80	320	160	160	160	40	80	160
A/Ukraine/6095/2014		2014-02-27	C3/SIAT1	80	320	320	1280	640	640	640	160	160	640
A/Ukraine/6096/2014		2014-02-27	C2/SIAT1	40	320	320	640	640	320	320	80	160	640
A/Ukraine/6097/2014	3C.3	2014-02-27	C2/SIAT1	40	320	160	640	320	320	320	80	160	320
A/Ukraine/6098/2014		2014-02-28	C2/SIAT1	80	640	320	1280	640	640	640	160	160	640
A/Serbia/NS-707/2014	3C.3	2014-02-28	SIAT1	80	160	320	640	640	640	320	80	160	640
A/Ukraine/6069/2014	3C.3	2014-03-01	C0/SIAT1	40	320	320	640	320	320	320	160	160	640
A/Ukraine/6070/2014		2014-03-02	C0/SIAT1	80	320	320	640	640	640	320	160	160	640
A/Ukraine/6174/2014		2014-03-04	C1/SIAT1	40	320	320	640	640	640	320	80	160	640
A/Ukraine/6141/2014	3C.3	2014-03-04	C0/SIAT1	40	320	160	640	320	320	320	80	160	320
A/Belgium/14S0265/2014	3C.3	2014-03-07	SIAT1	<	160	80	320	320	160	160	80	80	320
A/Ukraine/6138/2014	3C.3	2014-03-10	C0/SIAT2	<	160	80	320	160	160	160	40	80	320
A/Ukraine/6079/2014	3C.3	2014-03-12	C1/SIAT1	<	160	160	640	320	320	160	80	80	320
A/Ukraine/6080/2014	3C.3	2014-03-13	C1/SIAT1	<	160	160	320	160	160	160	40	80	320
A/Ukraine/6156/2014	3C.3	2014-03-24	C1/SIAT1	80	640	320	1280	640	1280	640	160	320	640
A/Ukraine/6158/2014	3C.3	2014-03-25	C1/SIAT1	<	160	80	320	160	160	160	40	80	320
A/Ukraine/6159/2014		2014-03-25	C1/SIAT1	80	320	320	640	640	640	640	160	160	640
A/Ukraine/6161/2014	3C.2	2014-03-26	C1/SIAT1	80	320	320	1280	640	640	640	160	160	640

1. < = <40

Vaccine

Table 8-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-06-17 + 2014-06-20

				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
Viruses		Collection Date	Passage History	A/Perth 16/09	A/Vic 361/11	A/Texas 50/12	A/Samara 73/13	A/Serbia NS-210/13	A/HK 146/13	NIB-85	A/Sth Afr 4655/13	A/Stock 1/13	A/Stock 6/14	*A/Nor 466/14
				F35/11	T/C F09/12	Egg F42/13	F24/13	F39/13	F40/13	F45/13	F10/14	F12/14	F14/14	F13/14
	Genetic group				3C.1	3C.1	3C.3	3C.3	3C.2	3C.3	3C.3 cl 101-60	3C.2 cl 36-18	3C.3a	3C.3a
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	160	160	160	40	80	40	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	80	320	320	640	160	320	160	160	160		
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	640	1280	640	1280	80	160	160	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	320	1280	640	640	640	160	160	640	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	2560	1280	2560	1280	2560	80	160	160	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	640	1280	1280	1280	2560	1280	160	320	160	160
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	2560	1280	2560	2560	2560	160	320	320	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	160	320	160	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E6 clone 36-18	80	80	160	320	160	320	160	320	320	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	80	320	160	160	160	40	160	320	320
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT1	<	40	40	80	40	80	80	<	80	320	320
TEST VIRUSES														
A/Bratislava/111/2014	3C.3	2014-02-06	MDCK1/SIAT1	<	160	160	640	320	320	320	80	80	320	160
A/Ukraine/6103/2014		2014-02-13	C1/SIAT1	<	160	160	320	320	160	320	80	160	320	160
A/Ukraine/6054/2014	3C.3	2014-02-14	C1/SIAT1	<	320	160	640	320	320	320	80	160	640	320
A/Ukraine/6052/2014		2014-02-17	C1/SIAT1	<	160	160	320	160	160	160	80	80	320	160
A/Ukraine/6101/2014	3C.3	2014-02-17	C1/SIAT1	<	80	80	160	160	160	160	40	80	320	160
A/Ukraine/6102/2014		2014-02-17	C1/SIAT1	<	160	80	320	160	160	160	40	80	320	160
A/Ukraine/125/2014	3C.3	2014-02-19	SIAT2/SIAT1	<	160	80	320	160	160	160	40	80	320	160
A/Ukraine/6053/2014		2014-02-20	C1/SIAT2	40	160	160	640	160	160	160	80	80	320	160
A/Odessa/259/2014		2014-02-23	SIAT2/SIAT1	<	160	160	320	160	160	160	40	80	320	160
A/Ukraine/6067/2014		2014-02-24	C1/SIAT2	40	320	160	640	320	320	320	80	160	640	320
A/Khmelnitsky/243/2014		2014-02-24	SIAT2/SIAT1	40	160	160	320	160	320	320	80	80	320	160
A/Sastin/137/2014	3C.3	2014-02-24	MDCK1/SIAT1	<	160	80	320	160	160	160	<	80	320	160
A/Ukraine/6172/2014		2014-02-27	C2/SIAT2	<	160	160	320	320	160	320	80	160	320	320
A/Norway/850/2014	3C.3	2014-02-27	SIAT3	40	320	160	640	320	320	320	80	160	640	320
A/England/256/2014	3C.3	2014-02-27	SIAT1/SIAT1	<	320	160	320	320	320	320	80	160	320	160
A/Latvia/03-013033/2014	3C.2	2014-03-06	MDCK1/SIAT1	<	40	40	160	40	80	80	40	40	80	80
A/Latvia/03-018148/2014		2014-03-08	MDCK1/SIAT1	40	320	320	640	320	640	320	160	320	640	320
A/Khmelnitsky/240/2014		2014-03-11	SIAT2/SIAT1	40	320	160	640	320	320	320	80	160	640	320
A/Khmelnitsky/246/2014		2014-03-11	SIAT2/SIAT1	40	320	160	640	320	320	640	160	160	640	320
A/Khmelnitsky/248/2014		2014-03-11	SIAT2/SIAT1	40	320	160	320	320	320	320	160	160	640	320
A/Khmelnitsky/249/2014	3C.3	2014-03-11	SIAT2/SIAT1	80	320	320	640	320	320	320	160	320	640	320
A/Khmelnitsky/250/2014		2014-03-11	SIAT2/SIAT1	40	160	160	320	160	160	320	80	160	320	160
A/Khmelnitsky/244/2014	3C.3	2014-03-12	SIAT2/SIAT1	<	160	80	320	160	160	160	40	80	320	160
A/Latvia/03-026603/2014	3C.3	2014-03-12	MDCK1/SIAT1	40	320	320	640	320	320	640	160	160	640	320
A/Khmelnitsky/245/2014		2014-03-15	SIAT2/SIAT1	<	320	160	320	320	160	320	80	160	320	160
A/Khmelnitsky/247/2014		2014-03-15	SIAT2/SIAT1	<	320	160	320	320	160	320	80	160	640	320
A/Prievidza/182/2014	3C.3	2014-03-18	MDCK1/SIAT1	<	160	160	320	320	320	320	80	160	320	320
A/England/400/2014	3C.2	2014-03-21	SIAT1/SIAT1	40	320	160	640	160	320	320	80	160	320	320
A/Ireland/21550/2014	3C.3	2014-04-04	MDCK2/SIAT1	<	80	80	160	80	80	80	<	40	160	80
A/Ireland/22350/2014	3C.3	2014-04-07	MDCK2/SIAT1	<	160	80	160	160	80	160	40	80	160	160
A/Ireland/22359/2014		2014-04-07	MDCK2/SIAT1	<	320	160	640	320	320	320	80	160	320	160
A/Ireland/22041/2014	3C.3	2014-04-08	MDCK1/SIAT1	<	80	80	160	160	160	160	<	80	160	160
A/Ireland/22878/2014	3C.3	2014-04-09	MDCK1/SIAT1	40	160	160	320	160	160	160	40	80	320	160
A/Levice/221/2014		2014-04-11	MDCK1/SIAT1	<	160	160	320	160	160	160	40	80	320	160
A/Levice/223/2014	3C.3b	2014-04-15	MDCK1/SIAT1	40	160	160	320	320	160	160	40	80	320	160

1. < = <40; * new antiserum

Vaccine

Sequences in phylogenetic trees

Table 8-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-06-24

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a
				Haemagglutination inhibition titre[1] — Post infection ferret antisera									
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	320	80	80	160	80	160	<	40	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	80	320	160	640	320	320	80	160	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	1280	80	160	160	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	1280	640	640	160	160	640	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	1280	640	80	160	160	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	160	320	320	1280	640	1280	80	160	80	80
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	320	320	80	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	80	80	160	320	160	320	320	320	80	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	40	40	160	160	80	40	80	320	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT1	<	40	<	80	40	80	<	80	320	160
TEST VIRUSES													
A/Sarajevo/13/2013		2013-01-14	SIAT1	<	320	160	640	320	320	80	160	640	320
A/Sarajevo/19/2013		2013-01-14	SIAT1	<	160	80	320	160	160	80	80	320	160
A/Slovenia/254/2014		2014-01-24	MDCK0/SIAT1	<	160	80	320	160	160	40	80	320	160
A/Slovenia/286/2014		2014-01-28	MDCK1/SIAT1	40	320	160	640	320	320	80	160	320	320
A/Slovenia/321/2014		2014-01-29	SIAT1/SIAT1	<	160	160	320	160	160	40	80	320	160
A/Ukraine/6031/2014		2014-02-04	C1/SIAT3	<	160	80	640	160	320	80	160	320	160
A/Slovenia/417/2014		2014-02-05	MDCK0/SIAT1	<	160	80	320	160	160	40	80	320	160
A/Slovenia/609/2014		2014-02-13	MDCK1/SIAT1	40	160	160	640	160	320	80	160	320	160
A/Slovenia/622/2014	3C.3	2014-02-14	MDCK1/SIAT1	<	80	80	160	160	80	40	80	160	160
A/Slovenia/694/2014		2014-02-18	SIAT0/SIAT1	<	160	160	640	320	320	80	160	320	160
A/Slovenia/697/2014		2014-02-18	SIAT0/SIAT1	<	160	160	320	160	320	80	160	320	160
A/Slovenia/706/2014		2014-02-19	MDCK0/SIAT1	80	640	320	1280	640	640	160	320	640	320
A/Slovenia/712/2014	3C.3	2014-02-19	SIAT0/SIAT1	<	160	80	320	160	320	40	160	320	160
A/Slovenia/752/2014		2014-02-21	MDCK1/SIAT1	40	320	160	640	320	320	80	160	640	320
A/Ukraine/6093/2014		2014-02-24	C3/SIAT3	40	160	160	320	160	320	80	160	320	320
A/Slovenia/823/2014		2014-02-26	MDCK0/SIAT1	<	160	160	320	160	160	80	80	320	160
A/Slovenia/842/2014	3C.3	2014-02-28	MDCK1/SIAT1	<	160	80	320	160	160	40	80	320	160
A/Slovenia/853/2014		2014-02-28	MDCK1/SIAT1	<	160	160	320	320	320	80	80	320	160
A/Slovenia/1162/2014	3C.2	2014-03-27	SIAT1/SIAT1	40	320	160	640	320	320	80	160	320	160
A/Slovenia/1213/2014	3C.3	2014-04-02	MDCK1/SIAT2	<	160	160	320	160	160	80	80	160	160
A/Ireland/23577/2014	3C.3	2014-04-12	MDCK1/SIAT2	<	160	80	320	160	160	40	80	320	160

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-06-27 + 2014-07-01 + 2014-07-04

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	A/Sth Afr 4655/13 F10/14	A/Stock 1/13 F12/14	A/Stock 6/14 F19/14	A/Stock 6/14 F20/14	A/Stock 6/14 F14/14	A/Nor 466/14 F13/14
	Genetic group				3C.1	3C.1	3C.3	3C.3	3C.2	3C.3 cl 101-60	3C.2 cl 36-18	3C.3a	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	160	160	40	80	40	80	40	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	640	320	640	640	640	160	320	640	320	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	1280	640	80	160	80	1280	160	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	640	640	1280	640	1280	160	320	320	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	1280	1280	80	160	80	1280	160	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	640	640	1280	640	2560	80	160	160	640	160	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	640	320	160	80	160	160
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	80	160	320	160	320	320	320	160	80	80	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E2 (Am1/Al1)	80	160	40	160	40	40	<	160	640	320	320	320
A/Stockholm/6/2014	3C.3a	2014-02-06	E3 (Am2/Al1)	80	160	160	160	320	160	<	40	320	320	160	160
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	40	160	80	320	320	320	80	80	320	160	320	320
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	80	80	160	160	160	40	160	320	160	640	320
TEST VIRUSES															
A/Sarajevo/22/2013		2013-01-14	SIAT2	40	160	80	320	160	160	40	80	ND	ND	160	160
A/Sarajevo/23/2013		2013-01-14	SIAT2	40	160	80	320	160	160	40	80	ND	ND	160	160
A/Sarajevo/58/2013	3A	2013-01-18	SIAT2	40	80	80	160	40	80	<	40	80	80	80	80
A/Stockholm/2/2014		2014-01-02	MDCK2/SIAT1	80	160	80	320	160	160	40	80	ND	ND	320	160
A/Slovenia/142/2014		2014-01-16	MDCK0/SIAT1	80	160	160	320	320	320	80	160	320	320	320	160
A/Czech Republic/1/2014	3C.3	unknown	MDCK3/SIAT1	80	160	160	320	320	320	80	80	320	160	320	160
A/Czech Republic/6/2014	3C.3	unknown	MDCK3/SIAT1	40	80	80	160	160	80	40	40	160	80	160	80
A/Czech Republic/10/2014		unknown	MDCK3/SIAT2	<	160	80	320	160	160	80	160	ND	ND	640	320
A/Iceland/23/2014	3C.3	2014-02-06	MDCK1/SIAT1	160	160	160	640	640	320	160	160	ND	ND	640	320
A/Austria/777909/2014		2014-02-08	SIAT1/SIAT1	40	160	40	320	160	160	40	80	ND	ND	160	160
A/Stockholm/14/2014		2014-02-12	MDCK1/SIAT1	80	320	160	640	320	320	80	160	ND	ND	320	320
A/Finland/405/2014	3C.3	2014-02-13	MDCK1/SIAT1	40	160	80	320	160	160	40	80	ND	ND	320	160
A/Iceland/34/2014		2014-02-19	MDCK1/SIAT1	80	160	80	320	160	160	80	80	ND	ND	320	160
A/Stockholm/13/2014	3C.3	2014-02-21	MDCK0/SIAT1	<	80	80	320	80	160	40	80	ND	ND	160	160
A/Stockholm/12/2014	3C.3	2014-02-24	MDCK1/SIAT1	40	160	160	320	160	160	80	160	ND	ND	320	320
A/Iceland/45/2014	3C.3	2014-02-27	MDCK1/SIAT1	80	160	160	320	160	80	80	80	ND	ND	320	160
A/Finland/411/2014	3C.3	2014-03-06	MDCK1/SIAT1	<	160	80	160	160	160	40	80	ND	ND	160	160
A/England/399/2014		2014-03-10	SIAT1/SIAT3	80	160	160	320	320	320	80	80	320	160	320	160
A/Austria/784503/2014	3C.3	2014-03-15	SIAT1/SIAT1	80	160	80	160	160	80	40	80	ND	ND	160	160
A/Vrancea/162893/2014	3C.3	2014-03-20	SIAT1/SIAT2	<	80	40	160	80	80	40	40	ND	ND	160	80
A/Botosani/162689/2014		2014-03-23	SIAT1/SIAT1	40	160	80	320	160	160	40	80	ND	ND	320	160
A/Botosani/162690/2014	3C.3	2014-03-23	SIAT1/SIAT1	80	160	80	320	160	160	40	80	ND	ND	320	160
A/Iasi/162712/2014		2014-03-24	SIAT1/SIAT1	160	320	320	1280	640	640	160	320	ND	ND	640	320
A/Calarasi/162804/2014	3C.3	2014-03-24	SIAT1/SIAT1	40	160	80	320	160	160	80	80	ND	ND	320	160
A/St. Petersburg/25/2014		2014-03-25	C1/SIAT1	80	160	160	320	320	160	80	80	320	160	320	160
A/St. Petersburg/45/2014	3C.3	2014-03-25	C1/SIAT1	80	320	160	640	320	320	80	160	320	160	640	160
A/Kaliningrad/2/2014		2014-03-25	C1/SIAT1	40	80	80	160	80	80	40	40	160	80	160	80
A/Austria/786648/2014	3C.3	2014-03-26	SIAT1/SIAT1	80	80	40	160	80	160	40	40	ND	ND	160	160
A/Mures/163025/2014	3C.3	2014-03-26	SIAT1/SIAT1	40	80	80	320	160	160	40	80	ND	ND	160	160
A/Austria/787217/2014		2014-03-27	SIAT2/SIAT1	40	160	80	320	160	160	40	80	ND	ND	160	160
A/Austria/788090/2014	3C.3	2014-04-01	SIAT2/SIAT1	40	160	80	320	160	160	40	80	ND	ND	320	160
A/Latvia/04-003950/2014		2014-04-02	MDCK2/SIAT1	80	320	320	640	320	320	80	160	320	320	320	320
A/Latvia/04-011765/2014		2014-04-04	MDCK1/SIAT1	160	320	320	640	320	320	80	160	320	320	320	160
A/Austria/790710/2014	3C.3b	2014-04-12	SIAT1/SIAT1	<	80	40	160	80	80	<	40	ND	ND	80	80

1. < = <40; ND = Not Done

Vaccine

Sequences in phylogenetic trees

Table 8-17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-07-08

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	A/Sth Afr 4655/13 F10/14	A/Stock 1/13 F12/14	A/Stock 6/14 F14/14	A/Nor 466/14 F13/14
				Haemagglutination inhibition titre[1] — Post infection ferret antisera									
	Genetic group				3C.1	3C.1	3C.3	3C.3	3C.2	3C.3 cl 101-60	3C.2 cl 36-18	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	320	80	160	80	80	160	<	80	80	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	160	640	320	320	80	160	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	640	80	160	320	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	160	640	320	640	160	160	640	160
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	1280	1280	80	160	320	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	320	320	1280	640	1280	80	160	160	80
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	160	320	640	320	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	160	160	80	160	320	160	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	40	160	160	160	80	80	640	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	80	40	80	320	160
TEST VIRUSES													
A/Gambia/G0063940/2012	3C.3	2012-10-21	SIAT1	40	160	80	320	160	160	40	80	320	160
A/Gambia/G0064536/2012		2012-10-27	SIAT1	80	160	160	320	160	160	80	80	640	160
A/Gambia/G0065236/2012	3C.3	2012-10-29	SIAT1	40	160	80	320	160	160	40	160	320	160
A/Gambia/G0065240/2012	3C.3	2012-10-29	SIAT1	80	160	160	640	320	320	80	160	640	160
A/Gambia/G0065536/2012		2012-10-30	SIAT1	80	160	80	320	160	160	40	80	320	80
A/Gambia/G0065540/2012		2012-10-30	SIAT1	80	160	160	320	320	320	80	160	320	160
A/Gambia/G0065640/2012		2012-10-30	SIAT1	40	160	80	320	160	160	80	80	640	160
A/Gambia/G0065736/2012	3C.3	2012-10-30	SIAT1	80	160	160	320	160	320	80	160	320	160
A/Gambia/G0065740/2012		2012-10-30	SIAT1	40	160	80	320	160	160	40	80	320	160
A/Gambia/G0067136/2012		2012-11-02	SIAT1	160	320	320	640	640	640	160	320	1280	320
A/Gambia/G0067140/2012		2012-11-02	SIAT1	80	160	160	320	320	320	80	160	640	160
A/Gambia/G0069636/2012	3C.3	2012-11-08	SIAT1	80	160	80	320	160	160	80	80	320	160
A/Gambia/G0069640/2012	3C.3	2012-11-08	SIAT1	80	160	160	640	320	320	80	160	640	160
A/Gambia/G0071436/2012	3C.3	2012-11-13	SIAT1	<	80	80	320	160	160	40	80	320	80
A/Gambia/G0074340/2012		2012-11-17	SIAT1	40	160	80	320	160	160	80	160	640	160
A/Gambia/G0075736/2012		2012-11-22	SIAT1	80	160	160	320	160	320	80	160	640	160
A/Sarajevo/9/2013	3C.3	2013-01-11	SIAT1	40	160	80	320	160	160	40	80	320	80
A/Austria/770588/2014		2013-12-11	SIAT1/SIAT2	80	160	160	320	320	320	80	160	640	160
A/Austria/772655/2014	3C.3	2014-01-10	SIAT2/SIAT2	80	160	160	320	160	160	80	160	640	160
A/Austria/773236/2014	3C.3	2014-01-15	SIAT1/SIAT2	80	320	160	640	320	320	80	160	640	160
A/Austria/776877/2014		2014-02-04	SIAT2/SIAT2	80	320	160	640	320	320	80	160	640	160
A/Finland/410/2014	3C.2	2014-02-26	SIAT1/SIAT3	160	160	160	320	160	160	80	160	320	160
A/Austria/781450/2014		2014-02-27	SIAT2/SIAT2	40	160	80	320	160	160	80	80	320	160
A/Iceland/62/2014	3C.3	2014-03-13	MDCK1/SIAT3	80	160	80	320	160	160	80	80	320	160
A/Austria/785558/2014	3C.3	2014-03-19	SIAT1/SIAT2	80	160	160	320	160	320	80	160	640	160
A/Olt/162753/2014	3C.2	2014-03-24	SIAT1/SIAT3	40	160	80	320	160	160	40	80	320	160
A/Bucuresti/162834/2014		2014-03-24	SIAT1/SIAT3	80	160	160	320	160	320	80	160	640	160
A/Austria/786355/2014		2014-03-25	SIAT1/SIAT2	40	80	80	160	160	160	40	80	320	80

1. < = <40

Vaccine

Table 8-18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-07-11

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] / Post infection ferret antisera: A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	80	80	80	160	160	40	80	40	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320	640	640	1280	640	320	160	320	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	640	80	160	320	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	640	640	1280	640	640	640	320	640	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	1280	1280	80	160	320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	640	640	1280	640	1280	80	160	160	80
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	640	160	320	320	320	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	80	80	160	320	160	160	320	320	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	80	320	320	160	80	80	640	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	80	40	80	640	160
TEST VIRUSES													
A/Gambia/G0069436/2012		2012-11-08	SIAT2	80	320	160	640	320	320	160	160	640	320
A/Gambia/G0070336/2012	3C.3	2012-11-12	SIAT2	80	320	160	640	320	320	160	160	640	320
A/Gambia/G0074336/2012		2012-11-17	SIAT2	80	320	160	640	320	320	160	160	640	320
A/Gambia/G0074836/2012		2012-11-19	SIAT2	80	320	160	640	320	160	80	80	640	160
A/Finland/428/2014	3C.3a	2014-02-17	MDCK1	<	80	80	320	160	160	80	160	640	320
A/Finland/428/2014	3C.3a	2014-02-17	SIAT1	<	40	40	160	160	160	40	80	640	320
A/Serbia/NS-738/2014		2014-03-07	SIAT1	80	160	160	640	320	320	80	160	640	160
A/Serbia/NS-783/2014	3C.3	2014-03-20	SIAT1	80	160	160	640	320	160	80	160	640	160
A/Hong Kong/5320/2014	3C.3	2014-03-20	SIAT1	80	320	160	640	320	320	80	160	640	320
A/Finland/437/2014	3C.3a	2014-03-24	MDCK1	<	40	40	160	160	160	40	80	640	160
A/Finland/437/2014	3C.3a	2014-03-24	SIAT1	<	80	80	320	160	160	80	160	640	320
A/Finland/438/2014	3C.3a	2014-04-03	MDCK1	<	40	40	160	160	160	40	80	640	320
A/Finland/438/2014	3C.3a	2014-04-03	SIAT1	<	40	40	160	80	160	40	80	640	320
A/Hong Kong/5578/2014	3C.3	2014-04-04	SIAT1	80	320	320	640	640	320	80	160	640	320
A/Belgium/14H0034/2014		2014-04-10	SIAT1	80	320	160	640	320	320	160	160	640	320
A/Belgium/14G0508/2014	3C.3	2014-04-16	SIAT1	80	320	160	640	320	320	80	160	640	320
A/Belgium/14G0510/2014	3C.3b	2014-04-16	SIAT1	80	160	160	320	320	160	40	80	320	160
A/Hong Kong/5695/2014	3C.3	2014-04-21	SIAT1	160	320	320	640	640	320	160	160	1280	320
A/Finland/439/2014	3C.3a	2014-04-23	MDCK1	<	80	40	160	160	160	40	80	640	320
A/Finland/439/2014	3C.3a	2014-04-23	SIAT1	<	40	40	160	160	160	40	80	640	320
A/Finland/440/2014	3C.3a	2014-04-28	MDCK1	<	80	80	320	160	160	80	160	640	320
A/Finland/440/2014	3C.3a	2014-04-28	SIAT1	<	80	80	320	160	160	80	160	640	320
A/Iceland/07150/2014		2014-05-20	SIAT1	80	160	160	640	320	320	80	160	640	160
A/Iceland/07151/2014		2014-05-20	SIAT1	80	160	160	640	320	320	80	160	640	160
A/Iceland/08202/2014	3C.3	2014-06-10	SIAT1	80	320	160	640	320	320	80	160	640	320

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-07-15 + 2014-07-22 + 2014-07-25

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	*A/Switz 9715293/13 NIBSC F13/14 3C.3a
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	160	160	160	40	80	80	<	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	160	640	320	320	80	160	640	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	640	1280	640	80	160	160	40	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	320	640	80	160	640	160	80
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	640	1280	640	80	160	160	80	40
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	320	320	640	640	1280	80	160	160	40	40
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	40	80	320	80	160	320	160	80	40	<
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	80	160	80	160	320	160	80	40	<
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	40	40	320	160	80	40	80	640	160	80
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	80	40	80	320	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	80	320	320	160	80	80	640	320	160
TEST VIRUSES														
A/Sarajevo/21/2012	6	2012-03-28	SIAT1	80	160	80	320	160	80	40	80	160	40	<
A/Sarajevo/57/2012		2012-04-26	SIAT1	80	160	80	320	160	160	40	80	320	80	40
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK3	<	40	<	80	40	40	<	<	160	40	<
A/Hong Kong/4800/2014	3C.2a	2014-02-26	MDCK1/SIAT2	<	40	<	80	40	40	<	<	160	40	<
A/Tomsk/6/2014	3C.3	2014-03-01	MDCK1/SIAT1	160	160	160	640	320	320	80	160	640	160	80
A/Hong Kong/5320/2014	3C.3	2014-03-20	MDCK2	40	160	80	320	320	160	80	80	640	160	40
A/Yaroslavl/234/2014	3C.3	2014-03-21	MDCK1/SIAT1	80	160	80	320	160	160	80	80	320	160	80
A/Moscow/206/2014	3C.2	2014-03-21	MDCK1/SIAT1	80	320	160	640	320	640	80	160	640	320	80
A/Vladimir/220/2014	3C.3	2014-03-26	MDCK1/SIAT1	160	160	160	640	160	160	80	160	640	160	40
A/Palau/6759/2014	3C.3a	2014-03-26	C2/SIAT1	40	160	160	640	320	160	80	160	640	320	80
A/Belgium/14G0496/2014	3C.3	2014-04-03	SIAT2	160	320	320	640	320	320	80	160	640	160	80
A/Hong Kong/5578/2014	3C.3	2014-04-04	MDCK2	160	320	320	1280	640	640	160	320	640	320	160
A/Mogilev/1273/2014	3C.3	2014-04-07	MDCK2/SIAT1	160	160	160	320	320	160	40	80	320	160	40
A/V. Novgorod/223/2014	3C.3	2014-04-08	MDCK1/SIAT1	160	160	160	640	160	160	40	80	320	160	40
A/Mogilev/1484/2014	3C.3	2014-04-15	MDCK2/SIAT1	80	160	160	640	320	320	80	160	640	160	80
A/Hong Kong/5739/2014	3C.2a	2014-04-29	MDCK1/SIAT2	<	40	<	80	40	40	<	<	160	40	<
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK2	<	40	40	160	80	80	40	40	320	80	<
A/Glasgow/4165/2014	3C.3a	2014-06-08	SIAT1	<	40	40	80	80	80	40	80	640	160	80

1. < = <40: * new antiserum

Vaccine

Sequences in phylogenetic trees

Table 8-20. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-07-29

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	A/Sth Afr 4655/13 F10/14	A/Stock 1/13 F12/14	A/Stock 6/14 F14/14	A/Nor 466/14 F13/14	A/Switz 9715293/13 NIBSC F13/14	*A/Poland 1702/14 NIBSC F18/14
				Haemagglutination inhibition titre[1] Post infection ferret antisera											
					3C.1	3C.1	3C.3	3C.3	3C.2	3C.3 cl 101-60	3C.2 cl 36-18	3C.3a	3C.3a	3C.3a	3C.3
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	160	160	40	80	80	<	<	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	160	640	320	320	80	160	640	160	80	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	2560	1280	80	160	320	40	80	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	1280	320	640	80	160	640	160	160	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	640	1280	1280	1280	1280	1280	80	160	320	80	80	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	640	640	1280	640	2560	160	320	160	80	80	640
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	40	80	320	80	160	320	160	160	40	<	160
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	80	160	80	160	320	160	80	40	<	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	40	80	320	160	80	80	80	640	160	160	80
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	80	40	80	320	160	80	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	40	320	160	80	40	80	640	160	160	80
A/Poland/1702/2014	3C.3	2014-04-08	SIAT2/SIAT4	40	160	80	320	160	160	40	80	320	160	80	320
TEST VIRUSES															
A/Nebraska/4/2014	3C.2a	2014-03-11	C2/SIAT3	<	40	40	160	80	40	<	40	320	160	80	<
A/Glasgow/4144/2014	3C.3a	2014-05-24	SIAT2	<	40	40	320	160	80	40	80	640	160	80	40

1. < = <40; * new antiserum

Vaccine: A/Texas/50/12 (Egg F42/13)

Sequences in phylogenetic trees: A/Nebraska/4/2014

Table 8-21. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-01

				Haemagglutination inhibition titre[1]												
				Post infection ferret antisera												
Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	*A/Nor 466/14 F23/14 3C.3a cl 32	*A/Nor 466/14 F24/14 3C.3a cl 58	A/Switz 9715293/13 NIBSC F13/14 3C.3a	*A/Switz 9715293/14 F25/14 3C.3a cl 123
REFERENCE VIRUSES																
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	320	320	80	80	160	40	40	<	40	160
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	320	1280	320	320	160	640	320	160	80	160	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	1280	2560	2560	1280	1280	640	160	320	40	160	320	80	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	640	640	320	1280	1280	1280	320	1280	320	160	160	320	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	1280	1280	1280	1280	2560	640	320	160	80	160	320	80	1280
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	640	640	160	320	320	80	40	80	40	40
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	160	320	320	80	320	160	40	<	40	40	<
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	40	320	160	40	80	640	160	80	160	320	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	<	80	640	160	80	40	160	160
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 32	40	320	320	320	320	320	80	640	160	5120	640	320	2560
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	<	40	40	40	40	<	<	320	80	320	160	80	320
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	40	320	80	40	80	640	160	ND	ND	160	ND
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4 clone 123	40	320	160	160	640	80	40	640	160	1280	320	160	2560
TEST VIRUSES																
A/South Africa/271/2014		2014-01-16	MDCK3/SIAT1	160	320	320	1280	640	160	160	1280	320	160	160	160	80
A/Kazakhstan/3661/2014	3C.3	2014-03-04	SIAT1	80	320	160	640	320	80	160	640	320	40	80	80	40
A/Kazakhstan/3663/2014	3C.3	2014-03-04	SIAT1	80	320	320	640	320	160	160	1280	320	80	80	160	80
A/Kazakhstan/3666/2014	3C.3	2014-03-11	SIAT1	160	640	320	1280	640	160	320	1280	640	160	160	320	160
A/Kazakhstan/3667/2014	3C.3	2014-03-11	SIAT1	320	320	160	640	320	80	320	640	320	40	80	80	40
A/Valjevo/2029/2014	3C.3	2014-03-13	C2/SIAT1	80	320	320	1280	640	160	320	1280	320	80	160	320	320
A/Kragujevac/2337/2014	3C.3	2014-03-25	C2/SIAT1	40	160	80	320	160	80	160	640	160	40	40	80	80
A/Kragujevac/2340/2014		2014-03-25	C2/SIAT1	80	320	320	1280	640	160	320	1280	320	80	160	320	160
A/Kragujevac/2378/2014	3C.3	2014-03-26	C2/SIAT1	160	320	320	1280	640	160	160	1280	320	160	160	160	160
A/Palau/6759/2014	3C.3a	2014-03-26	E5/E1	80	320	320	320	640	80	320	1280	320	2560	320	1280	320
A/Belgrade/2668/2014	3C.3	2014-04-04	C2/SIAT1	160	640	320	1280	640	160	320	2560	640	160	160	320	160
A/South Africa/3761/2014	3C.3	2014-05-12	MDCK2,SIAT1	80	320	160	1280	640	160	160	1280	320	160	80	160	80
A/South Africa/4090/2014	3C.3	2014-05-21	MDCK2,SIAT1	160	640	320	1280	1280	160	320	2560	640	320	160	320	160
A/South Africa/4371/2014	3C.3	2014-05-26	MDCK2,SIAT1	160	320	320	1280	640	160	320	1280	320	160	80	160	160
A/South Africa/4100/2014	3C.3	2014-05-28	MDCK1/SIAT1	80	320	320	1280	640	160	320	1280	320	160	160	80	80
A/South Africa/4370/2014		2014-05-28	MDCK2,SIAT1	80	320	160	640	320	160	320	1280	320	80	40	160	80
A/South Africa/4374/2014		2014-06-03	MDCK2,SIAT1	80	320	160	1280	320	160	320	1280	320	80	80	160	80
A/South Africa/4569/2014		2014-06-04	MDCK2,SIAT1	80	320	320	1280	320	160	320	1280	320	80	80	160	160
A/South Africa/4766/2014		2014-06-10	MDCK2,SIAT1	320	640	320	1280	640	320	320	1280	640	320	320	320	160
A/South Africa/4934/2014	3C.3	2014-06-23	MDCK2,SIAT1	<	80	40	160	80	<	80	320	80	<	<	<	<
A/South Africa/4991/2014		2014-06-24	MDCK2,SIAT1	40	320	320	640	640	160	160	1280	320	80	80	160	40
A/South Africa/5403/2014	3C.3	2014-07-02	MDCK1/SIAT1	160	640	320	1280	640	320	320	1280	640	160	160	320	160
A/South Africa/5416/2014	3C.3	2014-07-02	MDCK1/SIAT1	160	320	160	640	640	160	160	1280	320	80	80	160	80

1. < = <40: * new antisera

Vaccine

Sequences in phylogenetic trees

Table 8-22. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-05

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Nor 466/14 F24/14 3C.3a cl 58
				Haemagglutination inhibition titre[1] – Post infection ferret antisera									
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	160	320	40	160	80	40	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320	320	320	1280	640	160	320	640	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	160	320	320	80	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	320	320	1280	640	160	320	1280	320	320
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	640	640	1280	2560	160	320	160	160	320
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	160	320	640	640	640	640	320	160	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	80	80	160	320	320	320	320	160	80	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	80	320	160	80	160	640	320	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	40	80	640	320	80
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	<	40	40	80	80	<	<	320	160	160
TEST VIRUSES													
A/Kazakhstan/3393/2014		2014-01-10	SIAT1	80	160	160	320	320	80	160	320	160	40
A/Kazakhstan/3645/2014		2014-01-22	SIAT1	80	320	160	640	320	80	320	640	320	40
A/Belgrade/529/2014		2014-01-27	C1/SIAT1	160	320	320	640	640	160	320	1280	640	160
A/Sabac/630/2014		2014-01-27	C1/SIAT1	160	320	320	1280	640	160	320	1280	640	160
A/Kazakhstan/3480/2014	3C.3	2014-02-03	SIAT1	80	320	160	640	320	80	160	640	320	80
A/Kazakhstan/3698/2014		2014-02-03	SIAT1	80	320	160	640	320	160	320	640	320	80
A/Israel/Z1213/2014	3C.3	2014-02-10	SIAT1	80	320	160	640	320	320	320	640	320	80
A/Nis/1059/2014	3C.2	2014-02-10	C0/SIAT1	160	320	320	1280	640	320	320	640	320	160
A/Nis/1053/2014		2014-02-10	C1/SIAT1	160	320	320	640	320	320	320	1280	320	160
A/Nis/1055/2014		2014-02-10	C1/SIAT1	160	320	160	640	320	320	320	640	320	160
A/Nis/1056/2014		2014-02-10	C1/SIAT1	160	320	320	640	320	320	320	640	320	160
A/Nis/1057/2014		2014-02-10	C1/SIAT1	160	320	320	1280	640	320	320	1280	640	160
A/Israel/Z1222/2014		2014-02-15	SIAT1	160	320	320	640	640	160	320	640	320	160
A/Israel/Z1257/2014		2014-02-15	SIAT1	80	160	160	640	320	80	160	640	320	40
A/Israel/Z1263/2014	3C.3	2014-02-15	SIAT1	80	320	320	640	320	80	320	640	320	80
A/Israel/Z1267/2014		2014-02-15	SIAT1	80	320	320	640	320	320	320	1280	320	160
A/Israel/Z1206/2014		2014-02-16	SIAT1	80	320	160	640	320	80	160	320	320	40
A/Israel/Z1227/2014	3C.3	2014-02-17	SIAT1	40	160	80	320	160	40	80	320	160	<
A/Israel/Z1277/2014		2014-02-17	SIAT1	320	640	320	1280	640	160	320	1280	640	160
A/Sabac/1331/2014		2014-02-19	C1/SIAT1	160	320	320	1280	640	320	320	1280	320	160
A/Pozarevac/1336/2014	3C.2	2014-02-21	C0/SIAT1	160	320	320	1280	640	320	320	1280	640	160
A/Sabac/1334/2014	3C.2	2014-02-21	C1/SIAT1	160	320	320	640	640	320	320	1280	640	160
A/Kraljevo/1420/2014		2014-02-24	C1/SIAT1	160	320	320	640	640	320	320	1280	320	160
A/Belgrade/1417/2014		2014-02-25	C1/SIAT1	80	320	160	640	320	320	320	640	320	80
A/Kazakhstan/3776/2014	3C.3	2014-03-02	SIAT1	80	320	160	640	320	80	320	640	320	80
A/Lisboa/MS105/2014	3C.3	2014-04-02	SIAT1	80	160	160	320	160	80	160	320	320	40
A/Extremadura/1753/2014	3C.3	2014-04-07	SIAT1/SIAT1	80	160	160	640	160	40	80	160	160	<
A/Canarias/1686/2014		2014-04-08	SIAT1/SIAT1	160	160	320	640	320	80	160	640	320	80
A/Canarias/1687/2014	3C.3	2014-04-08	SIAT1/SIAT1	80	160	160	320	160	40	80	320	320	40
A/Canarias/1688/2014		2014-04-08	SIAT1/SIAT1	80	160	160	640	160	40	80	320	320	40
A/Canarias/1689/2014	3C.3	2014-04-08	SIAT1/SIAT1	80	160	160	320	160	40	80	320	320	40
A/Extremadura/1752/2014		2014-04-10	SIAT1/SIAT1	640	640	640	1280	640	160	320	1280	640	320
A/Extremadura/1749/2014	3C.3	2014-04-23	SIAT1/SIAT1	160	320	320	640	640	160	320	1280	640	160
A/South Africa/3622/2014	3C.3	2014-04-30	MDCK2/SIAT1	80	160	160	640	320	80	160	640	320	80
A/Madrid/SO12318/2014	3C.3b	2014-06-12	SIAT1/SIAT1	80	160	160	320	160	40	80	320	320	<

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-23. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-08

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Nor 466/14 F24/14 3C.3a cl 58
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	320	320	80	80	160	40	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	320	1280	320	320	160	640	320	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	1280	1280	1280	1280	1280	160	160	320	80	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	640	640	320	1280	1280	1280	320	1280	320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	1280	1280	1280	1280	2560	640	320	160	80	320
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	640	640	160	320	320	80	80
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	160	320	320	80	320	160	40	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT2	<	80	40	320	160	40	80	640	160	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	160	80	<	80	640	160	40
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	<	40	40	40	40	<	<	160	160	160
TEST VIRUSES													
A/Israel/Z938/2014		2014-01-24	SIAT1	80	160	160	320	320	80	160	640	160	160
A/Israel/Z941/2014	3C.3	2014-01-25	SIAT1	40	160	80	320	160	80	160	640	160	40
A/Israel/Z1035/2014	3C.3	2014-01-30	SIAT1	40	160	80	320	160	40	160	320	160	80
A/Israel/Z1048/2014	3C.2	2014-02-01	SIAT1	40	160	160	320	320	80	160	320	320	40
A/Israel/Z1047/2014		2014-02-02	SIAT1	40	160	160	640	320	80	160	640	320	80
A/Kazakhstan/3602/2014		2014-02-11	SIAT2	80	160	160	640	160	80	160	640	320	80
A/Israel/Z1225/2014	3C.3	2014-02-11	SIAT2	80	160	160	320	320	80	160	320	160	40
A/Kazakhstan/3603/2014	3C.2	2014-02-12	SIAT2	40	80	80	320	160	40	80	320	160	40
A/Kazakhstan/3605/2014		2014-02-14	SIAT2	40	80	80	320	160	40	80	320	160	40
A/Kazakhstan/3606/2014	3C.3	2014-02-16	SIAT2	40	160	80	320	160	80	160	320	160	40
A/Kazakhstan/3597/2014		2014-02-23	SIAT2	80	160	80	320	320	80	160	640	320	40
A/Lisboa/SU540/2014	3C.2	2014-02-26	SIAT2	80	80	80	320	160	40	80	80	160	<
A/Lisboa/MS102/2014	3C.3	2014-03-19	SIAT2	<	80	80	320	160	40	80	320	160	<
A/Galicia/1786/2014	3C.3	2014-07-03	SIAT1	40	80	80	320	80	40	80	320	160	<

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-24. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-12

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera: A/Perth 16/09 F35/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F36/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Nor 466/14 F24/14 3C.3a cl 58
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	320	160	160	40	80	80	40	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	1280	1280	640	160	160	640	320	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	5120	640	640	80	160	160	40	160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	1280	1280	1280	160	320	640	320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	320	320	2560	640	1280	80	320	80	80	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	40	160	320	320	320	320	160	80	40
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	160	160	160	320	320	80	40	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	80	80	160	160	40	80	640	160	80
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	40	40	80	40	<	80	320	160	40
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	<	40	40	40	80	<	<	80	80	160
TEST VIRUSES													
A/Lithuania/5009/2014	3C.3	2014-02-17	MDCK2/SIAT1	<	40	320	160	40	<	40	160	80	<
A/Lithuania/5879/2014	3C.3	2014-02-25	MDCK2/SIAT1	<	40	160	160	40	<	40	160	80	<
A/Lithuania/7266/2014		2014-03-10	MDCK2/SIAT1	40	320	640	640	320	40	80	160	160	80
A/Lithuania/7452/2014	3C.3	2014-03-12	MDCK2/SIAT1	<	80	320	160	80	<	40	160	80	<
A/Lithuania/7489/2014		2014-03-13	MDCK2/SIAT1	80	160	640	640	320	40	160	320	160	40
A/Lithuania/7802/2014	3C.3	2014-03-17	MDCK2/SIAT1	<	160	640	320	160	160	160	320	160	40
A/Lithuania/8237/2014	3C.3	2014-03-19	MDCK3/SIAT1	160	320	1280	640	640	80	80	320	320	40
A/Lithuania/8397/2014		2014-03-19	MDCK4/SIAT1	<	160	320	320	160	160	80	320	160	40
A/Lithuania/9144/2014		2014-03-22	MDCK2/SIAT1	<	80	160	160	80	<	40	160	160	<
A/Lithuania/9133/2014	3C.3	2014-03-27	MDCK2/SIAT1	<	80	640	320	160	40	80	320	160	40
A/Lithuania/9404/2014		2014-03-28	MDCK2/SIAT1	40	160	640	320	320	40	80	320	320	40
A/Lithuania/9430/2014	3C.3b	2014-03-30	MDCK2/SIAT1	<	40	160	160	80	<	40	160	80	<
A/Lithuania/10213/2014		2014-04-07	MDCK2/SIAT1	40	80	320	320	160	40	80	160	160	40
A/Lithuania/10373/2014	3C.3	2014-04-07	MDCK2/SIAT1	<	80	160	160	80	<	80	160	80	<
A/Lithuania/10543/2014	3C.3	2014-04-08	MDCK2/SIAT1	40	160	640	320	160	40	80	320	160	40
A/Lithuania/10716/2014	3C.3	2014-04-11	MDCK2/SIAT1	40	160	640	320	320	40	80	320	160	40
A/Lithuania/11727/2014		2014-04-11	MDCK2/SIAT1	40	160	1280	320	160	40	80	320	160	40
A/Lithuania/11496/2014	3C.3b	2014-04-18	MDCK2/SIAT1	40	160	320	320	160	40	80	320	160	40
A/Lithuania/11607/2014		2014-04-18	MDCK2/SIAT1	<	80	160	160	80	<	40	160	80	<
A/Dakar/09/2014	3C.3a	2014-04-23	C1/SIAT1	<	40	40	160	80	40	40	640	320	40
A/Lithuania/13347/2014	3C.3b	2014-05-08	MDCK2/SIAT1	40	80	320	320	160	<	80	320	160	<
A/Norway/1903/2014	3C.2	2014-05-20	MDCK1/SIAT1	<	80	160	160	80	<	40	160	80	<
A/Dakar/13/2014	3C.3a	2014-07-08	C1/SIAT1	<	<	40	80	40	<	40	320	80	40
A/Dakar/17/2014	3C.3a	2014-07-09	C1/SIAT1	<	40	40	160	40	<	80	320	160	40
A/Dakar/14/2014	3C.3a	2014-07-10	C1/SIAT1	<	<	40	80	40	<	40	320	160	40
A/Dakar/15/2014	3C.3a	2014-07-11	C1/SIAT1	<	40	40	160	40	40	80	320	160	80
A/Dakar/16/2014	3C.3a	2014-07-11	C1/SIAT1	<	40	40	160	40	40	80	320	160	80

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-25. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-19

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Nor 466/14 F24/14 3C.3a cl 58
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	160	160	320	40	80	40	40	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	1280	1280	640	160	160	640	320	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	640	640	80	160	80	40	160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	640	1280	640	2560	2560	160	640	640	640	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	320	640	640	1280	2560	80	160	160	80	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	40	160	320	320	320	320	160	80	40
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	40	80	160	160	160	160	80	80	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	80	40	160	160	40	80	320	320	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT1	<	80	40	160	160	40	160	320	320	160
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	<	40	40	40	80	<	<	80	160	160
TEST VIRUSES													
A/Ghana/DILI-0428/2014	3C.3a	2014-05-02	C1/SIAT1	<	<	<	80	40	<	40	320	160	<
A/Ghana/DILI-0441/2014		2014-05-08	C2/SIAT1	<	160	80	320	160	80	160	640	320	160
A/Ghana/FS-0514/2014	3C.3a	2014-05-17	C1/SIAT1	<	80	40	160	80	40	80	320	320	80
A/Ghana/DILI-0476/2014		2014-05-19	C1/SIAT1	<	40	40	160	80	40	80	320	320	80
A/Ghana/DILI-0479/2014	3C.3a	2014-05-19	C1/SIAT1	<	40	40	80	80	40	80	320	320	40
A/Ghana/DILI-0482/2014		2014-05-19	C1/SIAT1	<	40	40	160	80	40	80	320	320	80
A/Ghana/DARI-0101/2014	3C.3a	2014-05-19	C1/SIAT1	<	40	40	80	80	40	80	320	320	40
A/Ghana/DILI-0483/2014	3C.3a	2014-05-20	C1/SIAT1	<	40	<	80	80	40	80	320	320	40
A/Ghana/FS-0513/2014		2014-05-21	C1/SIAT1	<	80	40	160	80	40	80	320	320	80
A/Ghana/DARI-0104/2014	3C.3a	2014-05-27	C1/SIAT1	<	80	40	160	80	40	80	320	320	80
A/Ghana/DILI-0522/2014	3C.3a	2014-06-02	C2/SIAT1	<	160	40	160	160	80	160	640	640	80
A/Dakar/10/2014	3C.3a	2014-06-18	C1/SIAT1	<	80	40	160	80	40	80	320	320	80
A/Dakar/11/2014		2014-06-23	C2/SIAT1	<	80	40	160	160	40	80	320	320	160
A/Dakar/12/2014	3C.3a	2014-07-07	C2/SIAT1	<	40	40	160	160	40	80	320	320	80
A/Ghana/DILI-0657/2014		2014-07-22	C1/SIAT1	<	80	40	160	160	40	80	640	320	80
A/Ghana/DILI-0659/2014	3C.3a	2014-07-22	C1/SIAT1	<	80	40	160	80	40	80	320	320	80

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-26. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-22

				Haemagglutination inhibition titre[1]													
				Post infection ferret antisera													
Viruses	Genetic group	Collection Date	Passage History	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 1/13 F12/14 3C.2 cl 36-18	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Nor 466/14 F24/14 3C.3a cl 58	A/Switz 9715293/13 NIBSC 13/4 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl 123	A/Palau 6759/14 CDC F120/14 3C.3a	A/Palau 6759/14 CDC F118/14 3C.3a	A/Palau 6759/14 NIBSC F26/14 3C.3a
REFERENCE VIRUSES																	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320	1280	1280	640	160	160	640	320	160	160	320	160	80	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	1280	1280	1280	1280	80	160	320	40	160	80	640	80	160	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	640	2560	1280	160	640	1280	640	320	160	640	320	160	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	1280	2560	160	320	160	80	320	80	640	80	160	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	160	320	160	320	160	160	80	40	<	<	40	40	40
A/Stockholm/1/2013	3C.2	2013-01-13	E7 clone 36-18	40	80	160	160	320	160	80	80	40	<	<	40	40	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	80	80	320	160	80	160	1280	640	160	320	320	640	160	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	40	40	160	80	40	160	640	320	160	160	160	320	80	80
A/Norway/466/2014	3C.3a	2014-02-03	E4 clone 58	40	40	40	40	<	<	160	80	160	80	320	40	40	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	80	320	160	80	160	640	320	160	320	320	320	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E34 clone 123	160	160	160	320	<	40	320	320	320	160	1280	160	160	160
A/Palau/6759/2014	3C.3a	2014-03-26	C2/SIAT2	160	160	640	320	80	320	1280	640	320	320	320	640	160	320
A/Palau/6759/2014	3C.3a	2014-03-26	E5	320	320	640	640	80	320	1280	1280	640	640	1280	640	1280	2560
A/Palau/6759/2014	3C.3a	2014-03-26	E5	320	320	640	640	160	320	1280	1280	640	640	1280	640	1280	2560
TEST VIRUSES																	
A/Osaka-C/2003/2014	3C.3a	2014-01-10	MDCK1/MDCK1/SIAT1	40	40	160	80	40	160	640	320	160	160	160	320	160	80
A/Taiwan/442/2014	3C.2	2014-03-25	MDCK2/MDCK1/SIAT1	160	320	640	320	80	160	1280	320	160	160	160	320	160	160
A/Palau/6759/2014 (NIB HGR)	3C.3a	2014-03-26		320	320	320	320	80	160	1280	640	320	320	1280	320	1280	2560
A/Ghana/DILI-0658/2014	3C.3a	2014-07-22	C1/SIAT2	80	80	320	160	40	160	1280	320	160	160	160	320	160	160

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table8-27. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-08-27

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Switz 9715293/13 NIBSC 13/4 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl 123
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	160	160	40	80	40	<	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	320	640	320	160	640	320	160	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	80	320	80	80	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	640	160	640	320	160	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	640	1280	2560	160	160	80	80	640
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	320	320	320	160	80	40	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	80	80	320	160	80	640	320	160	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT2	<	<	40	160	80	40	640	320	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	40	80	320	160	80	640	320	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E34 clone 123	40	160	160	160	320	<	320	160	80	1280
TEST VIRUSES													
A/Albania/4869/2014	3C.3	2014-02-03	MDCK2/SIAT3	80	160	160	320	160	80	320	160	80	160
A/Albania/4979/2014	3C.3	2014-02-17	MDCK3/SIAT3	160	160	160	640	320	80	640	320	80	320
A/Hungary/480/2014		2014-02-24	MDCK3/E2/SIAT2	40	80	80	160	80	40	320	160	40	80
A/Sapporo/116/2014	3C.3	2013-03-10	MDCK1/MDCK1/SIAT1	40	40	40	160	80	<	160	160	<	80
A/Sakai/72/2014	3C.2a	2014-03-13	MDCK2/MDCK1/SIAT2	<	80	80	160	160	40	320	160	80	160
A/Haute Normandie/1087/2014	3C.2	2014-03-17	MDCK1/SIAT3	40	160	80	320	160	80	320	160	40	40
A/Paris/1124/2014	3C.3	2014-03-21	MDCK1/SIAT2	40	160	80	320	160	40	320	160	80	80
A/Pays de Loire/1260/2014	3C.3	2014-03-26	MDCK1/SIAT2	80	160	160	640	320	80	640	320	80	160
A/Pays de Loire/1262/2014	3C.3	2014-03-29	MDCK1/SIAT2	80	160	160	640	320	80	640	320	80	160
A/Bretagne/1267/2014	3C.3	2014-04-02	MDCK1/SIAT2	80	160	160	640	320	80	640	320	80	160
A/Paris/1268/2014	3C.3	2014-04-07	MDCK1/SIAT2	40	160	80	320	160	40	320	160	40	160
A/Centre/1329/2014	3C.3	2014-04-08	MDCK2/SIAT2	40	80	80	160	160	40	320	160	40	80
A/Centre/1497/2014	3C.2	2014-04-14	MDCK1/SIAT2	80	160	160	320	320	80	320	160	80	160
A/Ghana/DARI-0085/2014		2014-04-29	C1/SIAT2	<	80	80	320	160	80	640	320	160	320
A/Ghana/FS-0460/2014		2014-04-30	C2/SIAT2	<	80	80	160	160	40	640	320	160	160
A/Ghana/DILI-0512/2014		2014-05-28	C1/SIAT3	<	80	40	160	80	40	640	320	160	160

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-28. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-09-02

Haemagglutination inhibition titre[1]

Post infection ferret antisera

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 6/14 F14/14 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Switz 9715293/13 NIBSC 13/4 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl 123	A/Tasmania 11/14 AUS F3063-13D 3C.3a
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	80	160	<	40	40	<	80	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	160	320	320	640	320	160	320	160	80	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	80	160	80	40	640	40
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	160	640	320	160	320	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	320	320	640	640	1280	80	80	80	40	640	80
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	40	160	320	320	320	80	80	40	40	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT1	<	40	40	160	80	40	320	160	80	80	40
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT1	<	<	40	160	80	40	320	320	80	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	40	40	160	80	40	640	320	160	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4 clone 123	<	80	80	80	320	<	160	160	80	640	80
TEST VIRUSES														
A/Tasmania/11/2014	3C.3a	2014-03-16	E5/E1	<	320	40	40	40	<	80	80	80	160	80
A/Tasmania/11/2014	3C.3a	2014-03-16	MDCK4/SIAT1	<	40	40	160	80	40	640	320	160	80	40
A/Belgrade/2114/2014	3C.3a	2014-03-17	SIAT1	40	80	80	320	160	40	320	160	40	80	40
A/South Australia/55/2014	3C.3a	2014-06-14	E5/E1	<	80	80	40	40	<	160	160	80	160	40
A/South Australia/55/2014	3C.3a	2014-06-14	MDCK1/SIAT1	<	40	40	160	80	80	640	320	160	160	80
A/Newcastle/22/2014	3C.3b	2014-06-16	E4/E1	<	<	40	80	80	160	80	80	<	<	<
A/Newcastle/22/2014	3C.3b	2014-06-16	MDCK1/SIAT1	80	80	80	320	160	40	320	160	40	80	40
A/Norway/2006/2014	3C.3a	2014-06-17	SIAT1	40	160	80	320	160	80	640	160	80	80	80
A/Norway/2034/2014	3C.3a	2014-06-23	SIAT1	40	80	80	320	160	40	320	160	40	80	40

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-29. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-09-05 + 2014-09-09

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 F09/12 T/C 3C.1	A/Texas 50/12 F42/13 Egg 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 6/14 F14/14 T/C 3C.3a	A/Stock 6/14 F20/14 Egg 3C.3a	A/Nor 466/14 F13/14 3C.3a	A/Switz 9715293/13 F13/4 NIBSC 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl 123	A/Palau 6759/14 F29/14 HGR 3C.3a	A/Palau 6759/14 F41/14 HGR NIBSC 3C.3a	A/Palau 6759/14 F27/14 Egg 3C.3a	A/Palau 6759/14 F28/14 T/C 3C.3a	A/Sydney 71/14 F3066-130 Aus 3C.2a
				Haemagglutination inhibition titre[1] — Post infection ferret antisera															
REFERENCE VIRUSES																			
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	160	160	40	40	80	40	<	80	<	<	<	<	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	160	320	320	640	320	160	320	160	160	80	160	40	<	160	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	80	160	640	80	80	640	80	80	160	80	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	320	320	1280	1280	160	640	640	320	160	640	80	80	160	320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	640	1280	2560	160	80	640	80	40	640	80	160	160	80	160
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	40	80	320	160	320	80	40	80	40	<	<	<	40	80	<
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT1	<	40	40	160	80	40	320	160	320	160	80	40	<	80	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E4	80	160	160	80	160	<	160	320	160	160	640	160	160	160	160	160
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT1	<	<	40	80	80	40	320	160	320	160	160	<	<	80	80	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	40	40	160	80	40	320	160	320	160	160	40	40	80	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	320	<	320	320	160	80	1280	160	160	160	80	160
TEST VIRUSES																			
NIB-87 (A/Palau/6759/2014)	3C.3a	2014-03-26	HGR NIB	80	320	320	320	320	80	640	1280	640	640	1280	2560	2560	5120	640	ND
A/Palau/6759/2014	3C.3a	2014-03-26	E5/E1	<	160	160	80	80	<	160	320	160	80	320	160	80	160	80	ND
A/Palau/6759/2014	3C.3a	2014-03-26	C2/SIAT3	<	80	40	320	160	40	640	160	320	160	160	40	<	160	160	ND
A/Sydney/71/2014	3C.2a	2014-02-09	MDCK3/SIAT3	<	80	80	160	80	40	320	80	160	80	80	ND	ND	ND	ND	160
A/Lithuania/9947/2014		2014-04-02	SIAT2	40	80	80	160	160	<	160	80	160	<	80	ND	ND	ND	ND	<
A/St Petersburg/80/2014	3C.2a	2014-04-14	C1/SIAT3	<	80	80	160	160	40	320	160	160	80	80	ND	ND	ND	ND	160
A/Norway/1648/2014		2014-04-25	SIAT3	80	80	160	320	160	<	320	80	160	40	80	ND	ND	ND	ND	<
A/Cameroon/4046/2014	3C.3a	2014-06-13	SIAT1	<	40	40	160	80	40	640	160	320	160	160	40	<	160	160	ND
A/Cameroon/4045/2014	3C.3a	2014-06-13	SIAT1	<	40	40	160	80	40	640	160	320	160	80	40	<	80	160	ND
A/Cameroon/4427/2014	3C.3a	2014-06-30	SIAT1	<	40	40	160	160	40	640	160	320	160	160	40	40	160	160	ND
A/Cameroon/4680/2014	3C.3a	2014-07-09	SIAT1	<	40	40	160	80	40	640	160	320	160	80	40	<	160	160	ND

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT)

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1]									
				Post-infection ferret antisera									
				A/Vic 361/11 T/C F09/12	A/Texas 50/12 E F42/13	A/Stock 6/14 T/C F14/14	A/Switz 9715923/13 T/C NIBSC F13/14	A/Nor 466/14 T/C F13/14	A/Switz 9715923/13 E Cl123 F25/14	A/Nor 466/14 E Cl32 F23/14	A/Nor 466/14 E Cl58 F24/14	A/Palau 6759/14 E CDC F120/14	A/Palau 6759/14 T/C CDC F118/14
	Genetic group			3C.1	3C.1	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a
Substitutions in genetic group 3C.3a egg-adapted viruses									I140R, G186V	G186E, D225G, N246S (-CHO)	G186E, D225G	186G>V, 219S>Y, 225G>D, 246H>N*	
REFERENCE VIRUSES													
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320 (255)	320 (349)	320 (260)	80 (100)	160 (160)	320 (200)	80 (92)	80 (114)	160 (130)	160 (149)
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	1280 (1204)	1280 (1882)	320 (254)	80 (77)	80 (60)	320 (356)	160 (123)	160 (159)	160 (125)	80 (74)
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT3	40 (37)	40 (29)	80 (78)	40 (49)	80 (62)	80 (62)	40 (51)	40 (56)	40 (50)	80 (71)
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT3/MDCK1	40 (38)	40 (30)	160 (138)	40 (56)	80 (93)	80 (67)	40 (57)	40 (59)	40 (54)	80 (92)
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT2/SIAT3/MDCK2	40 (37)	40 (28)	80 (109)	80 (60)	80 (75)	80 (67)	40 (59)	80 (67)	40 (55)	80 (104)
A/Switzerland/9715293/2014	3C.3a	2013-12-06	SIAT1/SIAT2	40 (45)	40 (33)	160 (203)	160 (138)	160 (120)	80 (79)	40 (51)	40 (56)	40 (40)	80 (94)
A/Norway/466/2014	3C.3a	2014-02-03	SIAT2/SIAT3	80 (62)	80 (62)	320 (250)	160 (123)	160 (165)	80 (104)	80 (72)	80 (90)	80 (75)	160 (129)
A/Switzerland/9715293/2013 Cl123	3C.3a	2013-12-06	E4	80 (72)	80 (66)	160 (194)	80 (99)	80 (83)	160 (129)	80 (62)	40 (47)	40 (23)	40 (40)
A/Norway/466/2014 Cl32	3C.3a	2014-02-03	E4	320 (292)	320 (362)	640 (498)	320 (308)	320 (307)	1280 (974)	2560 (3348)	640 (606)	640 (755)	160 (198)
A/Norway/466/2014 Cl58	3C.3a	2014-02-03	E4	160 (172)	160 (183)	640 (550)	320 (268)	160 (202)	160 (227)	320 (353)	160 (169)	80 (89)	80 (68)
A/Palau/6759/2014	3C.3a	2014-03-26	E5	80 (106)	80 (111)	320 (279)	160 (126)	160 (204)	640 (516)	1280 (1018)	160 (188)	320 (362)	160 (193)
A/Palau/6759/2014	3C.3a	2014-03-26	C2/SIAT1	40 (28)	40 (22)	160 (148)	80 (80)	40 (50)	40 (34)	40 (25)	40 (25)	40 (22)	80 (65)
TEST VIRUSES													
A/Hong Kong/5695/2014	3C.3	2014-04-21	SIAT1	320 (300)	320 (321)	640 (496)	320 (250)	160 (128)	160 (238)	80 (71)	160 (189)	160 (153)	160 (149)
A/Hong Kong/5578/2014	3C.3	2014-04-04	SIAT1	320 (245)	320 (363)	320 (319)	320 (239)	80 (116)	160 (140)	80 (65)	80 (72)	80 (71)	80 (74)
A/Hong Kong/5320/2014	3C.3	2014-03-20	SIAT1	320 (295)	640 (651)	640 (781)	320 (296)	160 (143)	160 (213)	80 (82)	160 (131)	80 (112)	160 (214)
A/Nebraska/4/2014	3C.2a	2014-03-11	C2/SIAT1	80 (68)	40 (55)	160 (160)	160 (126)	80 (84)	80 (60)	40 (38)	40 (41)	40 (40)	40 (43)
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK2	80 (68)	80 (60)	320 (369)	160 (168)	80 (77)	80 (107)	80 (65)	80 (71)	80 (61)	160 (113)
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK2	40 (38)	40 (36)	160 (162)	80 (70)	80 (77)	80 (75)	40 (27)	40 (43)	40 (37)	40 (58)
A/Finland/440/2014	3C.3a	2014-04-28	SIAT1	40 (33)	40 (30)	160 (189)	160 (134)	80 (81)	80 (72)	40 (38)	40 (56)	40 (45)	80 (107)
A/Finland/439/2014	3C.3a	2014-04-23	SIAT1	40 (32)	40 (35)	320 (394)	320 (240)	160 (225)	80 (73)	40 (43)	80 (66)	40 (45)	80 (102)
A/Finland/438/2014	3C.3a	2014-04-03	SIAT1	40 (34)	40 (35)	320 (412)	320 (267)	320 (260)	80 (76)	40 (52)	80 (65)	40 (48)	80 (118)
A/Finland/437/2014	3C.3a	2014-03-24	SIAT1	80 (63)	40 (54)	320 (260)	160 (200)	160 (162)	80 (100)	80 (70)	80 (84)	80 (67)	160 (152)
A/Finland/428/2014	3C.3a	2014-02-17	SIAT1	80 (67)	40 (36)	320 (462)	320 (243)	320 (291)	80 (101)	40 (57)	80 (95)	80 (62)	160 (147)
					Vaccine								

[1] Readings show the antiserum doubling dilution value as read by eye while the values in parentheses show the computer calculated values, corresponding to 50% reduction

* The substitution pattern shown corresponds to sequencing of an E5/E1 virus stock

Figure 10. Phylogenetic comparison of influenza A(H3N2) HA genes

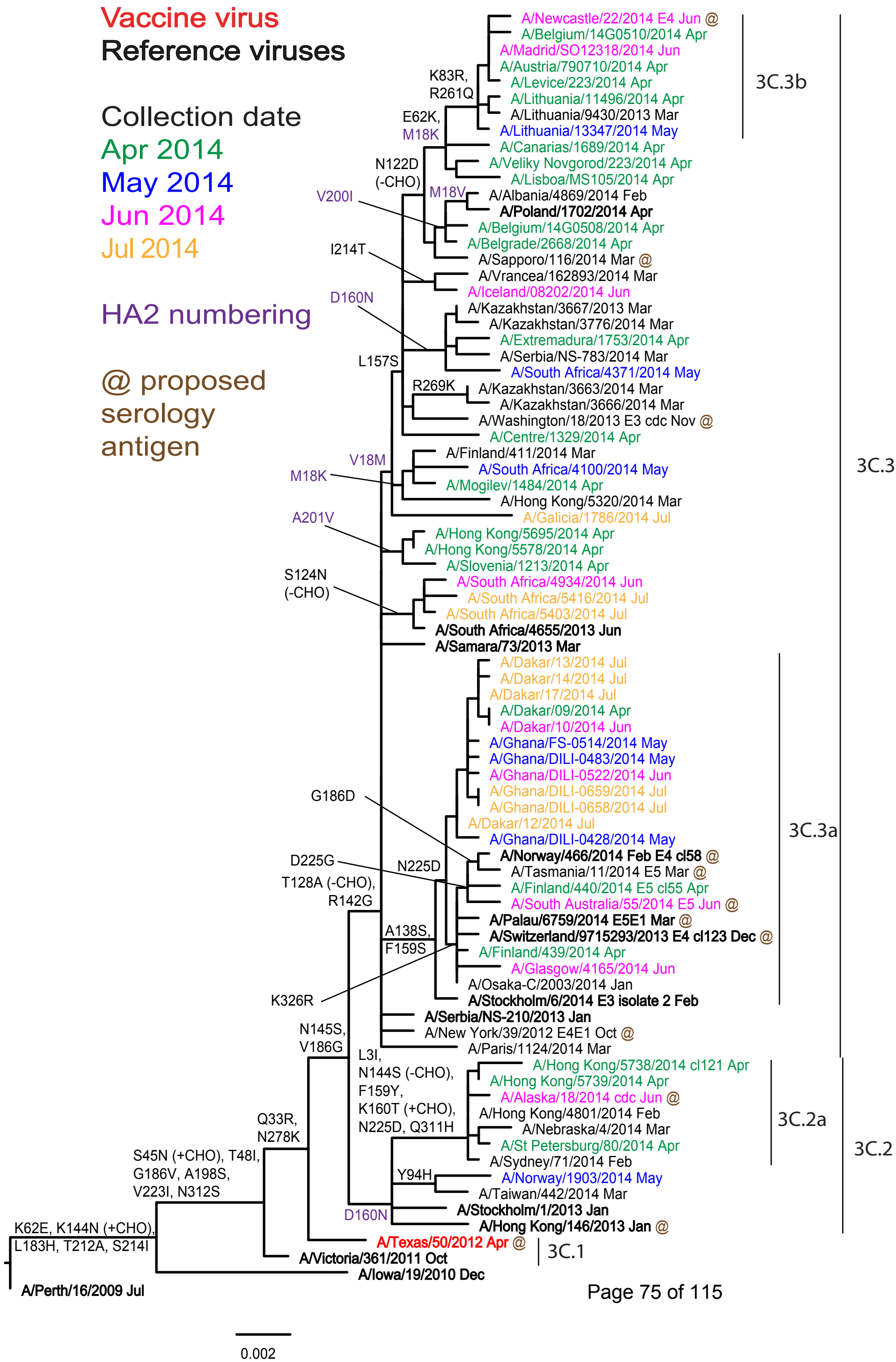


Figure 11. HA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                           10        20        30        40        50        60        70        80        90       100
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                   QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDQIEVTNATELVQSSSTGEICDSPHQILDGKNCTLIDALLGDPQCDGFQNKKWDLFVERSKAYSNCYPY
A/Iowa/19/2010                    ....................................................N........E...............................H...... 6
A/Victoria/361/2011               ............................................N..I.............E...................................... 3C.1
A/Texas/50/2012                   ................................R...........N..I.............E...................................... 3C.1
A/Hong Kong/146/2013              ................................R...........N..I.............E...................................... 3C.2
A/Stockholm/1/2013                ................................R...........N..I.............E...................................... 3C.2
A/Norway/1903/2014                ................................R...........N..I.............E...............................H...... 3C.2
A/St Petersburg/80/2014           ..I.............................R...........N..I.............E...................................... 3C.2a
A/Alaska/18/2014 cdc              ..I.............................R...........N..I.............E...................................... 3C.2a
A/Hong Kong/5738/2014 cl121       ..I.............................R...........N..I.............E...................................... 3C.2a
A/Serbia/NS-210/2013              ................................R...........N..I.............E...................................... 3C.3
A/Samara/73/2013                  ................................R...........N..I.............E...................................... 3C.3
A/South Africa/4655/2013          ................................R...........N..I.............E...................................... 3C.3
A/South Africa/5416/2014          ................................R...........N..I.............E...................................... 3C.3
A/Galicia/1786/2014               ......E.........................R...........N..I.............E............H......................... 3C.3
A/South Africa/4100/2014          ................................R...........N..I.............E...................................... 3C.3
A/Iceland/08202/2014              ................................R...........N..I.............E...................................... 3C.3
A/Poland/1702/2014                ................................R...........N..I.............E...................................... 3C.3
A/Lithuania/13347/2014            ................................R...........N..I..................................R................. 3C.3b
A/Newcastle/22/2014 E4            ................................R...........N..I..................................R................. 3C.3b
A/Madrid/SO12318/2014             ................................R...........N..I..................................R................. 3C.3b
A/Stockholm/6/2014 isolate 2      ................................R...........N..I.............E...................................... 3C.3a
A/Finland/440/2014 E5 cl55        ................................R...........N..I.............E...................................... 3C.3a
A/Norway/466/2014 E4 cl58         ................................R...........N..I.............E...................................... 3C.3a
A/Switzerland/9715293/2013 E4     ................................R...........N..I.............E...................................... 3C.3a
A/Palau/6759/2014 E5E1            ..................S.............R...........N..I.............E...................................... 3C.3a
A/Tasmania/11/2014 E5             K...............................R...........N..I.............E...................................... 3C.3a
A/South Australia/55/2014 E5      ................................R...........N..I.............E...................................... 3C.3a
A/Ghana/DILI-0522/2014            ................................R...........N..I.............E...................................... 3C.3a
A/Dakar/10/2014                   ................................R...........N..I.............E...................................... 3C.3a

                                          110       120       130       140       150       160       170       180       190       200
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                   DVPDYASLRSLVASSGTLEFNNESFNWTGVTQNGTSSACIRRSKNSFFSRLNWLTHLNFKYPALNVTMPNNEQFDKLYIWGVLHPGTDKDQIFLYAQASG
A/Iowa/19/2010                    ...........................................N......................................H..X.........T..A.
A/Victoria/361/2011               ...........................................N...........R..........................H..V...........S..
A/Texas/50/2012                   ...........................N...............N......................................H..V...........P..
A/Hong Kong/146/2013              .......................................K...NS.....................................H..W...........S..
A/Stockholm/1/2013                .....................................S.....NS.....................................H..........P...S..
A/Norway/1903/2014                ...........................................NS.....................................H..............S..
A/St Petersburg/80/2014           ...........................................SS.............YT......................H..............S..
A/Alaska/18/2014 cdc              ...........................................SS.............YT......................H..............S..
A/Hong Kong/5738/2014 cl121       ...........................................SS.............Y.......................H..........P...S..
A/Serbia/NS-210/2013              ...........................A.............G.NS.....................................H..V...........S..
A/Samara/73/2013                  ...........................A...........K.G.NS.....................................H..............S..
A/South Africa/4655/2013          .......................N...A.............G.NS.....................................H..............S..
A/South Africa/5416/2014          .......................N...A.............G.NS.....................................H..............S..
A/Galicia/1786/2014               ...........................A.............G.NS.....................................H..............S..
A/South Africa/4100/2014          ...........................A.............G.NS.....................................H..............S..
A/Iceland/08202/2014              ...........................A.............G.NS...........S.........................H..............S..
A/Poland/1702/2014                .....................D.....A.............G.NS...........S.........................H..............S..
A/Lithuania/13347/2014            .....................D.....A.............G.NS...........S.........................H..............S..
A/Newcastle/22/2014 E4            .....................D.....A.........S...G.NS...........S.........................H..........P...S..
A/Madrid/SO12318/2014             .....................D.....A.............G.NS...........S.........................H..............S..
A/Stockholm/6/2014 isolate 2      ...........................A.........S...G.NS.............S.......................H..V...........S..
A/Finland/440/2014 E5 cl55        ...........................A.........S...G.NS.............S.......................F..............S..
A/Norway/466/2014 E4 cl58         ...........................A.........S...G.NS.............S.......................H..E...........S..
A/Switzerland/9715293/2013 E4     ...........................A.........S.R.G.NS.............S.......................H..V...........S..
A/Palau/6759/2014 E5E1            ...........................A.........S...G.NS.............S.......................H..X...........S..
A/Tasmania/11/2014 E5             ...........................A.........S...G.NS.............S.......................H..D...G.......S..
A/South Australia/55/2014 E5      ...........................A.........S...G.N..............S.......................H..............S..
A/Ghana/DILI-0522/2014            ...........................A.........S...G.NS.............S.......................H..............S..
A/Dakar/10/2014                   ...........................A.........S...G.NS.............S.......................H..............S..

                                          210       220       230       240       250       260       270       280       290       300
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                   RITVSTKRSQQTVSPNIGSRPRVRNIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCNSECITPNGSIPNDKPFQNVNRI
A/Iowa/19/2010                    ...........A.I...............V.................................................A....................
A/Victoria/361/2011               ...........A.I........I.............................................................................
A/Texas/50/2012                   ...........A.I....F...I.......................................................K......................
A/Hong Kong/146/2013              ...........A.I........I.......................................................K......................
A/Stockholm/1/2013                ..I........A.I........I..................V...................................K..........S...........
A/Norway/1903/2014                ...........A.I........I.......................................................K......................
A/St Petersburg/80/2014           ...........A.I........I.D.....................................................K......................
A/Alaska/18/2014 cdc              ...........A.I........I.D.....................................................K......................
A/Hong Kong/5738/2014 cl121       ..I........A.I........I.D.....................................................K......................
A/Serbia/NS-210/2013              ...........A.I....F...I.......................................................K......................
A/Samara/73/2013                  ...........A.I........I.......................................................K......................
A/South Africa/4655/2013          ...........A.I........I.......................................................K......................
A/South Africa/5416/2014          ...........A.I........I.......................................................K......................
A/Galicia/1786/2014               ...........A.I........I.......................................................K......................
A/South Africa/4100/2014          ...........A.I........I.....................................Q................K......................
A/Iceland/08202/2014              ...........A.T........I.......................................................K......................
A/Poland/1702/2014                ...........A.I........I.......................................................K......................
A/Lithuania/13347/2014            ...........A.I........I.....................................Q................K......................
A/Newcastle/22/2014 E4            ...........A.I........I.....................................Q................K......................
A/Madrid/SO12318/2014             ...........A.I........I.....................................Q................K......................
A/Stockholm/6/2014 isolate 2      ...........A.I........I.......................................................K......................
A/Finland/440/2014 E5 cl55        ...........A.I........I.G.....................................................K......................
A/Norway/466/2014 E4 cl58         ...........A.I........I.G.....................................................K......................
A/Switzerland/9715293/2013 E4     ...........A.I........I.D.....................................................K......................
A/Palau/6759/2014 E5E1            ...........A.I....X...I.X....................X...............................K......................
A/Tasmania/11/2014 E5             ...........A.I........I.G.....................................................K......................
A/South Australia/55/2014 E5      ......R....A.I........I.G.....................................................K......................
A/Ghana/DILI-0522/2014            ...........A.I........I.D.....................................................K......................
A/Dakar/10/2014                   ...........A.I........I.D.....................................................K......................
```

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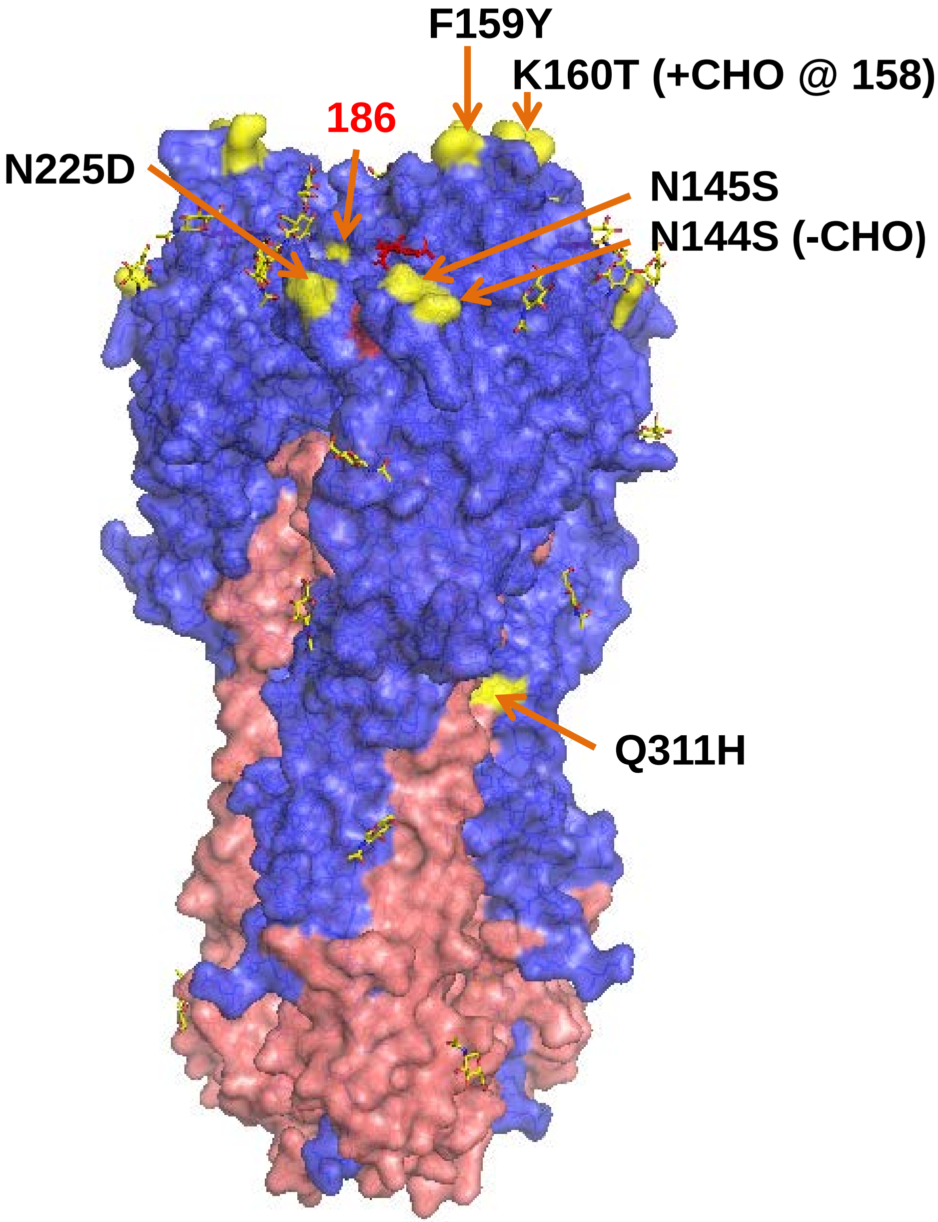
                                       310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                 TYGACPRYVKQNTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS
A/Iowa/19/2010                  ....................................................................................................
A/Victoria/361/2011             ...........S........................................................................................
A/Texas/50/2012                 ...........S........................................................................................
A/Hong Kong/146/2013            ....................................................................................................
A/Stockholm/1/2013              ...........S........................................................................................
A/Norway/1903/2014              ...........S........................................................................................
A/St Petersburg/80/2014         ..........HS........................................................................................
A/Alaska/18/2014 cdc            ..........HS........................................................................................
A/Hong Kong/5738/2014 cl121     ..........H.........................................................................................
A/Serbia/NS-210/2013            ...........S........................................................................................
A/Samara/73/2013                ...........S........................................................................................
A/South Africa/4655/2013        ...........S........................................................................................
A/South Africa/5416/2014        ...........S........................................................................................
A/Galicia/1786/2014             ...........S..................................M.....................................................
A/South Africa/4100/2014        ...........S..................................K.....................................................
A/Iceland/08202/2014            ...........S..................................K.....................................................
A/Poland/1702/2014              ...........S........................................................................................
A/Lithuania/13347/2014          ...........S..................................K.....................................................
A/Newcastle/22/2014 E4          ...........S..................................K.....................................................
A/Madrid/SO12318/2014           ...........S..................................K.....................................................
A/Stockholm/6/2014 isolate 2    ...........S........................................................................................
A/Finland/440/2014 E5 cl55      ...........S.............R..........................................................................
A/Norway/466/2014 E4 cl58       ...........S.............R..........................................................................
A/Switzerland/9715293/2013 E4   ...........S.............R..........................................................................
A/Palau/6759/2014 E5E1          ...........S.............R..........................................................................
A/Tasmania/11/2014 E5           ...........S.............R..........................................................................
A/South Australia/55/2014 E5    ...........S.............R..........................................................................
A/Ghana/DILI-0522/2014          ...........S........................................................................................
A/Dakar/10/2014                 ...........S........................................................................................

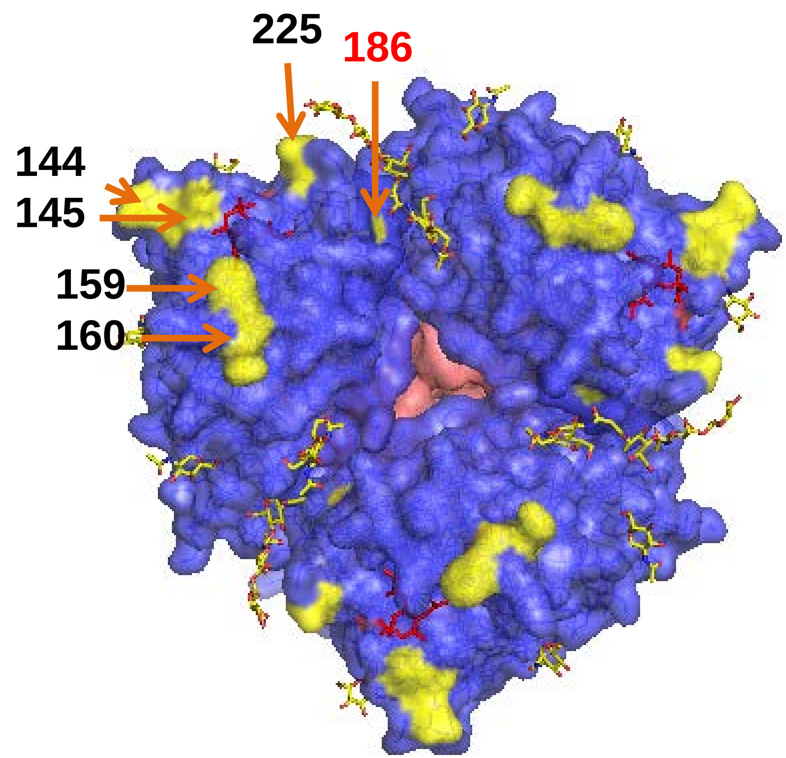
                                       410       420       430       440       450       460       470       480       490       500
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                 EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF
A/Iowa/19/2010                  ....................................................................................................
A/Victoria/361/2011             ....................................................................................................
A/Texas/50/2012                 ....................................................................................................
A/Hong Kong/146/2013            ........................................................................................N...........
A/Stockholm/1/2013              ........................................................................................N...........
A/Norway/1903/2014              ........................................................................................N...........
A/St Petersburg/80/2014         ........................................................................................N...........
A/Alaska/18/2014 cdc            ........................................................................................N...........
A/Hong Kong/5738/2014 cl121     ........................................................................................N...........
A/Serbia/NS-210/2013            ....................................................................................................
A/Samara/73/2013                ....................................................................................................
A/South Africa/4655/2013        .....V..............................................................................................
A/South Africa/5416/2014        ....................................................................................................
A/Galicia/1786/2014             ........................................................................................N...........
A/South Africa/4100/2014        ....................................................................................................
A/Iceland/08202/2014            ....................................................................................................
A/Poland/1702/2014              ....................................................................................................
A/Lithuania/13347/2014          ....................................................................................................
A/Newcastle/22/2014 E4          ....................................................................................................
A/Madrid/SO12318/2014           ....................................................................................................
A/Stockholm/6/2014 isolate 2    ....................................................................................................
A/Finland/440/2014 E5 cl55      ....................................................................................................
A/Norway/466/2014 E4 cl58       ....................................................................................................
A/Switzerland/9715293/2013 E4   ....................................................................................................
A/Palau/6759/2014 E5E1          ....................................................................................................
A/Tasmania/11/2014 E5           ....................................................................................................
A/South Australia/55/2014 E5    ....................................................................................................
A/Ghana/DILI-0522/2014          ........................................L...........................................................
A/Dakar/10/2014                 ....................................................................................................

                                       510       520       530       540       550
                                ....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                 QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI
A/Iowa/19/2010                  ..................................................
A/Victoria/361/2011             ..................................................
A/Texas/50/2012                 ..................................................
A/Hong Kong/146/2013            ..................................................
A/Stockholm/1/2013              ..................................................
A/Norway/1903/2014              ..................................................
A/St Petersburg/80/2014         ..................................................
A/Alaska/18/2014 cdc            ..................................................
A/Hong Kong/5738/2014 cl121     ..................................................
A/Serbia/NS-210/2013            ..................................................
A/Samara/73/2013                ..................................................
A/South Africa/4655/2013        ..................................................
A/South Africa/5416/2014        ..................................................
A/Galicia/1786/2014             ..R..................T............................
A/South Africa/4100/2014        ..................................................
A/Iceland/08202/2014            ..................................................
A/Poland/1702/2014              ............................I.....................
A/Lithuania/13347/2014          ..................................................
A/Newcastle/22/2014 E4          ..................................................
A/Madrid/SO12318/2014           ..................................................
A/Stockholm/6/2014 isolate 2    ..................................................
A/Finland/440/2014 E5 cl55      ..................................................
A/Norway/466/2014 E4 cl58       ..................................................
A/Switzerland/9715293/2013 E4   ..................................................
A/Palau/6759/2014 E5E1          ..................................................
A/Tasmania/11/2014 E5           ..................................................
A/South Australia/55/2014 E5    ..................................................
A/Ghana/DILI-0522/2014          ..................................................
A/Dakar/10/2014                 ..................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 12A. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.2a





View from top, looking down the 3-fold axis

Figure 12B. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.3a

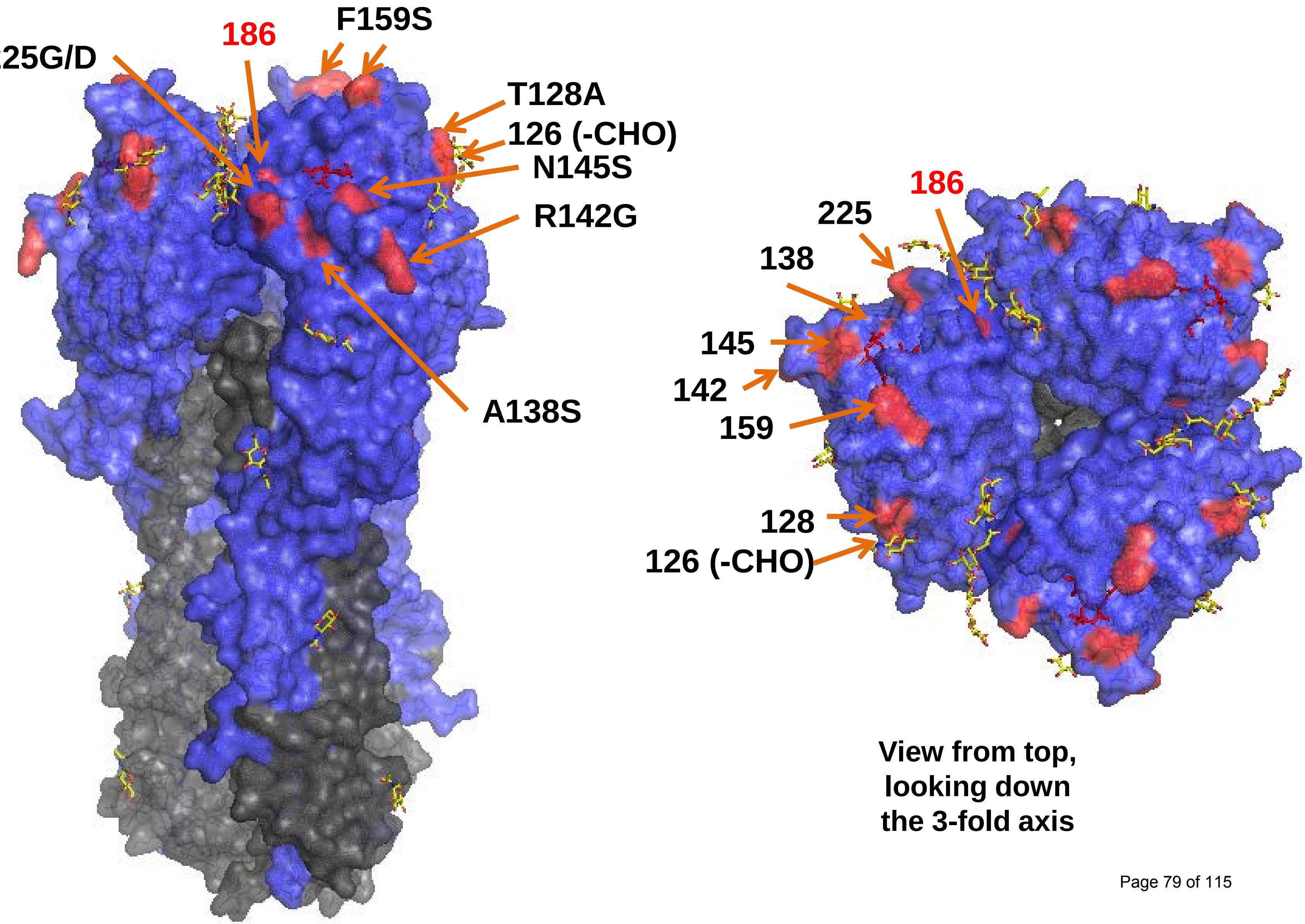
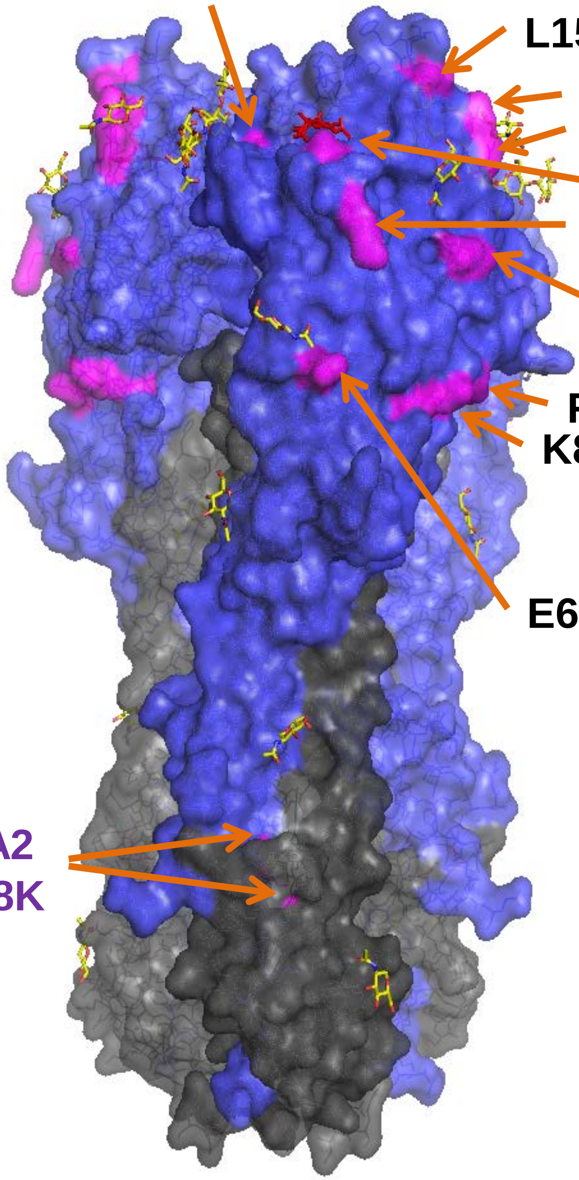
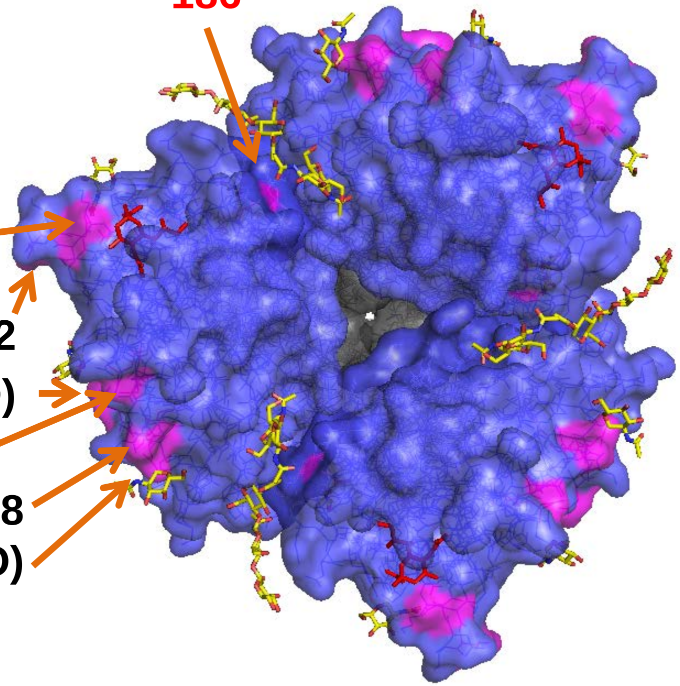


Figure 12C. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.3b





View from top, looking down the 3-fold axis

Figure 13. Phylogenetic comparison of influenza A(H3N2) NA genes

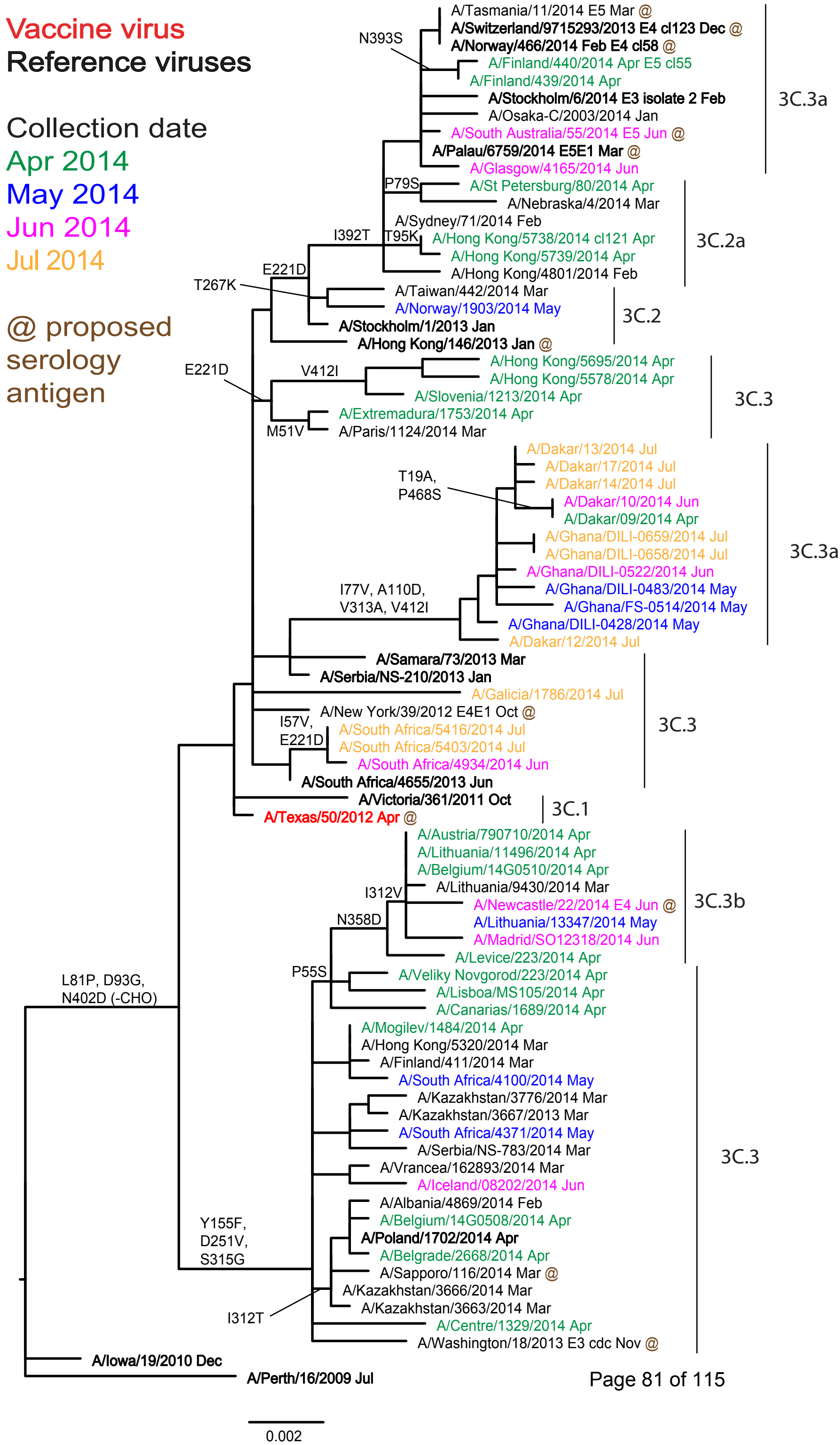


Figure 14. NA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

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                                           10        20        30        40        50        60        70        80        90       100
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                    MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKLAEYRNWSKPQCDITGFAPF
A/Iowa/19/2010                     .................................................................................................... 6
A/Victoria/361/2011                ................................................................................P...........G....... 3C.1
A/Texas/50/2012                    ................................................................................P...........G....... 3C.1
A/Hong Kong/146/2013               ................................................................................P...........G....... 3C.2
A/Stockholm/1/2013                 ................................................................................P...........G....... 3C.2
A/Norway/1903/2014                 ................................................................................P.K.........G....... 3C.2
A/St Petersburg/80/2014            ..............................................................................S.P...........G....... 3C.2a
A/Hong Kong/5738/2014 cl121        ................................................................................P...........G.K..... 3C.2a
A/Serbia/NS-210/2013               ................................................................................P...........G....... 3C.3
A/Samara/73/2013                   ...............................................................K................P...........G....... 3C.3
A/South Africa/4655/2013           ................................................................................P...........G....... 3C.3
A/South Africa/5416/2014           ........................................................V.......................P...........G....... 3C.3
A/Galicia/1786/2014                ................................................................................P...........G....... 3C.3
A/South Africa/4100/2014           ..........................................Y.....................................P...........G....... 3C.3
A/Iceland/08202/2014               ................................................................................P...........G....... 3C.3
A/Poland/1702/2014                 ................................................................................P...........G....... 3C.3
A/Lithuania/13347/2014             ......................................................S.........................P...........G....... 3C.3b
A/Newcastle/22/2014 E4             ......................................................S.........................P...........G....... 3C.3b
A/Madrid/SO12318/2014              ......................................................S.........................P...........G....... 3C.3b
A/Stockholm/6/2014 isolate 2       ................................................................................P...........G....... 3C.3a
A/Finland/440/2014 E5 cl55         .........................................................M......................P...........G....... 3C.3a
A/Norway/466/2014 E4 cl58          ................................................................................P...........G....... 3C.3a
A/Switzerland/9715293/2013 E4      ................................................................................P...........G....... 3C.3a
A/Palau/6759/2014 E5E1             ................................................................................P...........G....... 3C.3a
A/Tasmania/11/2014 E5              ................................................................................P...........G....... 3C.3a
A/South Australia/55/2014 E5       ........................................K.......................................P...........G....... 3C.3a
A/Ghana/DILI-0522/2014             ............................................................................V...P...........G....... 3C.3a
A/Dakar/10/2014                    ..................A.........................................................V...P...........G....... 3C.3a

                                          110       120       130       140       150       160       170       180       190       200
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                    SKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTLNNVHSNNTVRDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Iowa/19/2010                     ....................................................................................................
A/Victoria/361/2011                ........................................................X...........................................
A/Texas/50/2012                    .................................................H..................................................
A/Hong Kong/146/2013               ....................................................................................................
A/Stockholm/1/2013                 ....................................................................................................
A/Norway/1903/2014                 ..................................................X.................................................
A/St Petersburg/80/2014            ..................................................X.................................................
A/Hong Kong/5738/2014 cl121        ....................................................................................................
A/Serbia/NS-210/2013               ..............................H.....................................................................
A/Samara/73/2013                   ....................................................................................................
A/South Africa/4655/2013           ....................................................................................................
A/South Africa/5416/2014           ....................................................................................................
A/Galicia/1786/2014                ........................................S...........................................................
A/South Africa/4100/2014           ......................................................F.............................................
A/Iceland/08202/2014               ......................................................F.............................................
A/Poland/1702/2014                 ......................................................F.............................................
A/Lithuania/13347/2014             ..................................................X...F.............................................
A/Newcastle/22/2014 E4             ......................................................F.............................................
A/Madrid/SO12318/2014              ......................................................F.............................................
A/Stockholm/6/2014 isolate 2       ....................................................................................................
A/Finland/440/2014 E5 cl55         ....................................................................................................
A/Norway/466/2014 E4 cl58          ....................................................................................................
A/Switzerland/9715293/2013 E4      ....................................................................................................
A/Palau/6759/2014 E5E1             ....................................................................................................
A/Tasmania/11/2014 E5              ....................................................................................................
A/South Australia/55/2014 E5       ....................................................................................................
A/Ghana/DILI-0522/2014             .........D..........................................................................................
A/Dakar/10/2014                    .........D..........................................................................................

                                          210       220       230       240       250       260       270       280       290       300
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                    ATASFIYNGRLVDSVVSWSKEILRTQESECVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNR
A/Iowa/19/2010                     ...................Q................................................................................
A/Victoria/361/2011                .........................................................K..........................................
A/Texas/50/2012                    ....................................................................................................
A/Hong Kong/146/2013               ....................................................................................................
A/Stockholm/1/2013                 ....................D...............................................................................
A/Norway/1903/2014                 ....................D.............................................K.................................
A/St Petersburg/80/2014            ....................N...............................................................................
A/Hong Kong/5738/2014 cl121        ....................D...............................................................................
A/Serbia/NS-210/2013               ....................................................................................................
A/Samara/73/2013                   ....................................................................................................
A/South Africa/4655/2013           ....................................................................................................
A/South Africa/5416/2014           ....................D...............................................................................
A/Galicia/1786/2014                .....................V..............................................................................
A/South Africa/4100/2014           ..................................................V.................................................
A/Iceland/08202/2014               ..................................................V.................................................
A/Poland/1702/2014                 ..................................................V.................................................
A/Lithuania/13347/2014             ..................................................V.................................................
A/Newcastle/22/2014 E4             ..................................................V.................................................
A/Madrid/SO12318/2014              ..................................................V.................................................
A/Stockholm/6/2014 isolate 2       ....................D...............................................................................
A/Finland/440/2014 E5 cl55         ....................D...............................................................................
A/Norway/466/2014 E4 cl58          ....................D...............................................................................
A/Switzerland/9715293/2013 E4      ....................D...............................................................................
A/Palau/6759/2014 E5E1             ....................D...............................................................................
A/Tasmania/11/2014 E5              ....................D...............................................................................
A/South Australia/55/2014 E5       ....................D...............................................................................
A/Ghana/DILI-0522/2014             ....................................................................................................
A/Dakar/10/2014                    ....................................................................................................
```

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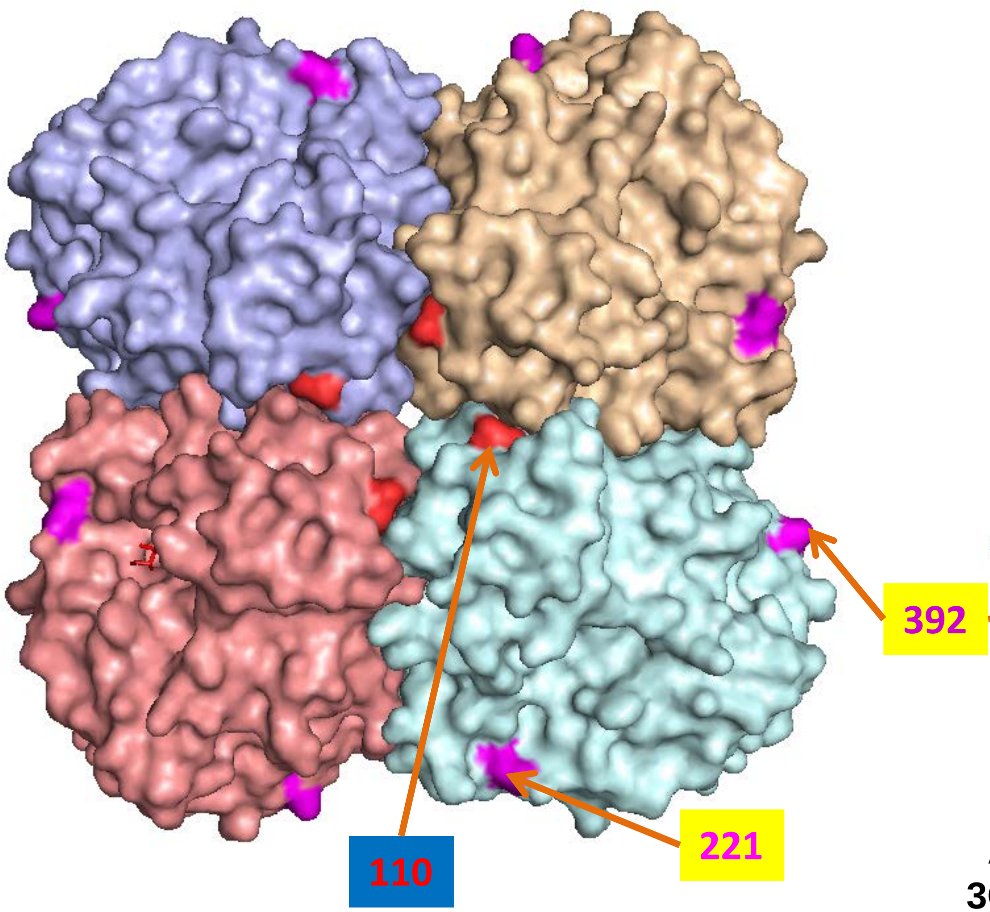
                                       310       320       330       340       350       360       370       380       390       400
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRTISEKSRLGYETFKVIEGWSNPKSKLQINRQVIVDR
A/Iowa/19/2010                 ..................................................................N.T......................M........
A/Victoria/361/2011            ............................T.....................................N.T...............................
A/Texas/50/2012                ..................................................................N.T...............................
A/Hong Kong/146/2013           ..................................................................N.T...............................
A/Stockholm/1/2013             ..................................................................N.T...............................
A/Norway/1903/2014             ..................................................................N.T...............................
A/St Petersburg/80/2014        ..................................................................N.T......................T........
A/Hong Kong/5738/2014 cl121    ..................................................................N.T......................T........
A/Serbia/NS-210/2013           ..................................................................N.T...............................
A/Samara/73/2013               ..................................................................N.T...............................
A/South Africa/4655/2013       ..................................................................N.T...............................
A/South Africa/5416/2014       ..................................................................N.T...............................
A/Galicia/1786/2014            ..................................G...N...........................N.T...............................
A/South Africa/4100/2014       ..............G...................................................N.T............................I..
A/Iceland/08202/2014           ..............G...................................................N.T................S..............
A/Poland/1702/2014             ...........T..G...................................................N.T...............................
A/Lithuania/13347/2014         ...........V..G..........................................D........N.T...............................
A/Newcastle/22/2014 E4         ...........V..G.............T............................D........N.T...............................
A/Madrid/SO12318/2014          ...........V..G..........................................D........N.T...............................
A/Stockholm/6/2014 isolate 2   ..................................................................N.T......................T........
A/Finland/440/2014 E5 cl55     ..................................................................NDT......................TS.......
A/Norway/466/2014 E4 cl58      ..................................................................N.T......................T........
A/Switzerland/9715293/2013 E4  ..................................................................N.T......................T........
A/Palau/6759/2014 E5E1         ..................................................................N.T......................T........
A/Tasmania/11/2014 E5          ..................................................................N.T......................T........
A/South Australia/55/2014 E5   ..................................................................N.T......................T........
A/Ghana/DILI-0522/2014         ............A.....................................................N.T...............................
A/Dakar/10/2014                ............A.....................................................N.T...............................

                                       410       420       430       440       450       460
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Perth/16/2009                GNRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADINLMPI
A/Iowa/19/2010                 ...............................................................L.....
A/Victoria/361/2011            .D.............................................................L.....
A/Texas/50/2012                .D.............................................................L.....
A/Hong Kong/146/2013           .D.............................................................L.....
A/Stockholm/1/2013             .D.............................................................L.....
A/Norway/1903/2014             .D.............................................................L.....
A/St Petersburg/80/2014        .D.............................................................L.....
A/Hong Kong/5738/2014 cl121    .D.............................................................L.....
A/Serbia/NS-210/2013           .D.............................................................L.....
A/Samara/73/2013               .D........................T....................................L.....
A/South Africa/4655/2013       .D.............................................................L.....
A/South Africa/5416/2014       .D.............................................................L.....
A/Galicia/1786/2014            .D.............................................................L.....
A/South Africa/4100/2014       .D.............................................................L.....
A/Iceland/08202/2014           .D.............................................................L.....
A/Poland/1702/2014             .D.............................................................L.....
A/Lithuania/13347/2014         .D.............................................................L.F...
A/Newcastle/22/2014 E4         .D.............................................................L.....
A/Madrid/SO12318/2014          .D.............................................................L.....
A/Stockholm/6/2014 isolate 2   .D.............................................................L.....
A/Finland/440/2014 E5 cl55     .D.............................................................L.....
A/Norway/466/2014 E4 cl58      .D.............................................................L.....
A/Switzerland/9715293/2013 E4  .D.............................................................L.....
A/Palau/6759/2014 E5E1         .D.............................................................L.....
A/Tasmania/11/2014 E5          .D.............................................................L.....
A/South Australia/55/2014 E5   .D.............................................................L.....
A/Ghana/DILI-0522/2014         .D.........I...................................................L.....
A/Dakar/10/2014                .D.........I...................................................L...S.
```

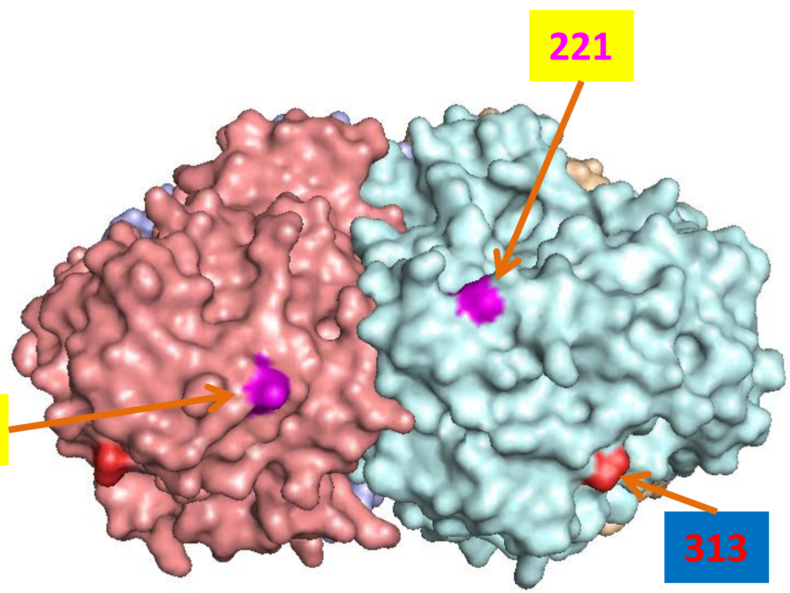
Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 15. NA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroups 3C.2a, 3C.3a & 3C.3a (West Africa)

3C.2a	3C.3a	3C.3a (WA)
		I77V
		A110D
E221N/D	E221D	E221
		V313A
I392T	I392T	I392
		V412I



View from top, looking down the 4-fold axis



Amino acid positions defining the 3C.2a and 3C.3a (magenta) subgroups and differentiating the 3C.3a viruses from West Africa (red).

Influenza B viruses.

Influenza B viruses represented just over 15% of samples received with collection dates after 2014-01-31 (**Table 2**). B/Yamagata lineage viruses predominated over those of the B/Victoria lineage at a ratio of approximately 2:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from six countries in Europe, three in Africa (Cameroon, Ghana and South Africa), Argentina and Hong Kong SAR. The collection details and summary results are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-3**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are those with collection dates after 2014-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees presented here are highlighted. Over 80% of cell-propagated test viruses showed ≥8-fold reductions in HI titres compared with the titres with the homologous virus, with post-infection ferret antisera raised against the recommended vaccine virus for quadrivalent vaccines, B/Brisbane/60/2008. Similar proportions of the test viruses were poorly recognised by post-infection ferret antisera raised against the egg-propagated reference viruses B/Malta/636714/2011 and B/South Australia/81/2012, while no test virus was recognised by the antiserum raised against B/Johannesburg/3964/2012 at a titre within 4-fold of the titre for the homologous virus. In contrast, much higher proportions of the test viruses showed reactivity within 4-fold of the titres with the corresponding homologous virus with antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells. As shown in **Tables 11-1 to 11-3**, these antisera were raised against B/Paris/1762/2009, B/Hong Kong/514/2009, B/Odessa/3886/2010 and B/Formosa/V2367/2012. These viruses are considered to be surrogate cell-propagated antigens representing the egg-propagated prototype strain. Antisera raised against B/Hong Kong/514/2009, B/Paris/1762/2009 and B/Odessa/3886/2010 recognised between 95% and ~80% of test viruses at titres within 4-fold of the titres with the homologous viruses. The antiserum raised against B/Formosa/V2367/2012 recognised fewer test viruses with just over 50% being recognised by the antiserum at a titre within 4-fold of the titre with the homologous virus.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 16** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 17**. The HA genes of the great majority of recently collected viruses sent to NIMR fall into the B/Brisbane/60/2008 genetic clade in subgroup 1A. **Figure 18** shows a phylogenetic tree for the NA gene and **Figure 19** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa/Indian Ocean (Cameroon, South Africa and La Réunion), the Middle East (Israel), Argentina and Hong Kong SAR. The collection details and results are summarised in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-8.** Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 13-1** and **13-3 to 13-8** are viruses with collection dates after 2014-01-31. The clade into which the HA falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. Post-infection ferret antiserum raised against the current, egg-propagated, vaccine virus B/Massachusetts/02/2012 recognised approximately 65% of viruses collected after 2014-01-31 at titres within 4-fold of the titre with the homologous virus. A ferret antiserum raised against a cell-propagated cultivar of B/Massachusetts/02/2012 recognised ~60% of viruses collected since 2014-01-31 at titres within 4-fold of its titre with the homologous virus. Antisera raised against two other viruses with HA genes belonging to the B/Massachusetts/02/2012 clade (Clade 2), B/Estonia/55669/2011 and B/Hong Kong/3577/2012, recognised approximately 45% and 60% of the test viruses, respectively, at titres within 4-fold of the titres of the antisera with their homologous viruses. An antiserum raised against the previous vaccine virus B/Wisconsin/1/2010 recognised 90% of test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus and ~66% were recognised similarly by antiserum raised against cell-propagated B/Novosibirsk/01/2012, a virus belonging to the B/Wisconsin/1/2010 clade, Clade 3. Almost all (98%) viruses were recognised by an antiserum raised against an egg-propagated cultivar of A/Stockholm/12/2011, a Clade 3 virus, at titres within 4-fold of the titre for the homologous virus; this antiserum does not differentiate between viruses of Clade 2 and Clade 3. Similarly, an antiserum from the WHO CC in Melbourne, raised against B/Phuket/3073/2013, recognised all 4 test and the reference viruses analysed (**Table 13-8**) at titres within 4-fold of the titre for the homologous virus; this antiserum also did not seem to distinguish between viruses of the two clades. Despite the results with these two antisera, viruses of the two clades are considered antigenically distinguishable with some antisera.

Figure 20 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 21**. **Figure 22** shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 23** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. The HA and NA genes of recently collected viruses fell into the B/Massachusetts/02/2012 clade (Clade 2) and the B/Wisconsin/1/2010 clade (Clade 3) with viruses in Clade 3 predominating over those in Clade 2 at a ratio of ~2:1.

Antiviral Analysis.

No evidence for reduced sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was seen for influenza B viruses collected after 2014-01-31 **(Table 14).**

Conclusion.

Viruses of the B/Yamagata lineage have predominated. For viruses of the B/Victoria lineage most were antigenically similar to viruses closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell into two antigenically distinguishable groups with viruses of the B/Wisconsin/1/2010 clade (Clade 3) predominating. Approximately 66% of recently circulating test viruses (collected after 2014-01-31) were recognised by post-infection ferret antiserum raised against egg-propagated B/Massachusetts/02/2012, the current vaccine virus, at titres within 4-fold of the titre with the homologous virus.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2014-01-31

MONTH	B Victoria lineage				
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]
FEBRUARY					
Cameroon	1	1	1		
Germany	2	2	1		1
Russia	1	1	1		
South Africa	1	1		1	
United Kingdom	1	1			1
MARCH					
Cameroon	4	4	3	1	
Iceland	2	2		1	1
Norway	1	1	1		
Russia	1	1	1		
United Kingdom	1	1			1
APRIL					
Belgium	1	1		1	
Cameroon	1	1	1		
Ghana	1	1		1	
Iceland	1	1		1	
Russia	2	2		2	
MAY					
Ghana	2	2	2		
Hong Kong	3	3			3
JUNE					
Cameroon	2	2		2	
Ghana	2	2	2		
JULY					
Cameroon	3	3		1	2
	30	30	13	10	7
10 Countries			43.4%	33.3%	23.3%

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 11-1. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post infection ferret antisera									
				B/Bris[1,4] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F26/13	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F9/13	B/Odessa[2] 3886/10 F19/11	B/Malta[4] 636714/11 F29/13	B/Jhb[2] 3964/12 F01/13	B/For[3] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
				1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	320	80	<	10	<	80	160	80	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	320	40	40	40	320	320	160	1280
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK1	2560	10	80	40	80	80	40	20	40	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	20	160	80	160	160	160	80	80	320
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	320	40	80	80	160	160	160	640
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	40	40	40	320	160	160	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	20	160	80	640	1280	640	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	20	160	80	80	40	160	80	160	640
B/South Australia/81/2012	1A	2012-11-28	E4/E1	5120	160	1280	80	160	<	320	640	320	1280
TEST VIRUSES													
B/Aichi/62/2013	1A	2012-12-11	MDCK1/MDCK2	5120	20	160	80	80	80	80	40	80	160
B/Dakar/21/2013	1A	2013-09-26	MDCK2	5120	10	80	40	40	40	40	10	80	160
B/Cote d'Ivoire/2021/2013	1A	2013-11-22	SIAT2	2560	20	80	40	80	80	40	20	80	160
B/Jiangsu-Tianning/1795/2013	1B	2013-11-28	E3/E1	1280	80	640	40	40	40	320	320	160	1280
B/Togo/LNG415/2013	1A	2013-11-29	MDCK3	5120	20	160	40	40	80	80	40	80	160
B/Togo/LNG419/2013	1B	2013-12-03	MDCK2	2560	20	160	80	80	80	80	40	80	160
B/Cote d'Ivoire/2091/2013	1A	2013-12-03	MDCK2	5120	40	160	40	80	80	40	20	80	80
B/Togo/LNG424/2013	1A	2013-12-09	MDCK2	2560	80	160	20	40	20	160	320	160	640
B/Cameroon/7386/2013	1A	2013-12-19	MDCK2	5120	10	160	40	80	160	40	10	40	80
B/Tokat/143/2014	1A	2014-01-08	SIAT1/MDCK1	1280	10	80	20	40	10	10	20	40	40
B/Cankırı/226/2014	1A	2014-01-11	SIAT1/SIAT1	2560	10	80	20	20	10	10	20	20	40
B/Berlin/1/2014	1A	2014-01-15	C2/MDCK1	1280	10	80	20	20	10	10	20	40	40
B/Cankiri/348/2014	1A	2014-01-17	MDCK2	2560	20	80	40	40	<	20	<	20	80
B/Sachsen/1/2014	1A	2014-02-03	C2/MDCK2	1280	10	20	<	40	<	10	10	20	40
B/Brandenburg/2/2014	1A	2014-02-10	C2/MDCK2	5120	10	160	40	80	80	20	20	80	80
B/England/199/2014	1A	2014-02-14	MDCK1/MDCK1	2560	20	80	80	80	<	10	20	<	80
B/England/373/2014	1A	2014-03-12	SIAT1/MDCK1	2560	20	40	40	80	<	10	20	<	40

1. < = <40; 2. < = <10; 3. < = <20; 4. hyperimmune sheep serum.

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-2. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre / Post infection ferret antisera: B/Bris[1,4] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F9/13	B/Odessa[2] 3886/10 F19/11	B/Malta[2] 636714/11 F29/13	B/Jhb[3] 3964/12 F01/13	B/For[2] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
				1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	1280	80	20	20	<	80	320	80	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	160	320	80	80	40	320	320	160	1280
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	2560	10	40	160	80	80	20	40	40	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	10	40	160	80	160	80	80	80	160
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	5120	10	40	160	80	160	40	40	80	80
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	160	320	80	80	80	640	320	320	1280
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	320	320	160	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	80	320	80	80	40	160	320	320	640
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	80	320	40	40	40	320	320	320	640
TEST VIRUSES													
B/Gambia/G0066836/2012		2012-11-01	MDCK1	2560	10	20	160	80	80	40	40	40	80
B/Gambia/G0068640/2012	1A	2012-11-06	MDCK2	1280	10	20	80	40	40	40	40	80	80
B/Gambia/G0069936/2012		2012-11-09	MDCK1	2560	<	20	80	40	40	20	40	40	80
B/Gambia/G0076538/2012		2012-11-23	MDCK1	2560	<	40	80	80	80	40	40	80	80
B/Gambia/G0142736/2013		unknown	MDCK1	5120	10	80	160	80	80	80	80	80	160
B/Gambia/G0137636/2013	1A	2013-10-04	MDCK1	1280	10	80	20	40	40	40	80	80	160
B/Gambia/G0139236/2013		2013-10-08	MDCK1	2560	10	80	40	40	40	40	80	40	160
B/Gambia/G0139336/2013	1A	2013-10-08	MDCK1	1280	<	40	40	40	40	20	40	40	80
B/Gambia/G0141336/2013	1A	2013-10-18	MDCK1	1280	10	80	40	40	40	40	80	40	160
B/Gambia/G0141340/2013		2013-10-18	MDCK1	1280	10	80	40	40	40	40	80	80	160
B/Gambia/G0141836/2013	1A	2013-10-19	MDCK1	2560	10	160	40	40	40	40	80	80	160
B/Moscow/2/2014	1A	2014-02-19	MDCK1/MDCK1	2560	20	40	160	160	80	80	80	80	160
B/South Africa/662/2014	1A	2014-02-27	MDCK1/MDCK1	2560	20	40	160	80	40	80	80	80	160
B/Iceland/63/2014	1A	2014-03-15	MDCK1/MDCK1	1280	10	40	20	40	20	10	20	40	40
B/Iceland/66/2014	1A	2014-03-18	MDCK1/MDCK1	1280	10	40	40	40	40	20	40	40	80
B/Moscow/12/2014	1A	2014-03-28	MDCK1/MDCK1	2560	10	40	80	80	80	20	40	40	80
B/Belgium/14G0509/2014	1A	2014-04-16	MDCK1	2560	20	40	160	80	40	80	80	80	160
B/Khabarovsk/14/2014	1A	2014-04-25	C3/MDCK2	2560	40	80	40	80	40	80	80	80	160
B/Tomsk/1/2014	1A	2014-04-26	MDCK1/MDCK1	2560	<	10	80	80	40	10	<	40	40
B/Iceland/77/2014	1A	2014-04-29	MDCK1/MDCK1	1280	10	80	40	40	40	20	40	40	80
B/Hong Kong/3221/2014	1A	2014-05-08	MDCK1/MDCK1	5120	20	160	<[a]	160	<	<[a]	160	160	320
B/Hong Kong/3224/2014	1A	2014-05-09	MDCK1/MDCK1	5120	20	160	80	160	<	<[a]	80	160	320
B/Hong Kong/3298/2014	1A	2014-05-21	MDCK1/MDCK1	2560	20	160	<[a]	80	<	<[a]	80	160	160

1. < = <40; 2. < = <10; 3. < = <20; 4. hyperimmune sheep serum; a. < = <40.

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post infection ferret sera									
				B/Bris[1,3] 60/08 Sh 522 1A	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12 1A	B/Paris[2] 1762/09 F07/11 1A	B/HK[2] 514/09 F9/13 1B	B/Odessa[2] 3886/10 F19/11 1B	B/Malta[2] 636714/11 F29/13 1A	B/Jhb[2] 3964/12 F01/13 1A	B/For[2] V2367/12 F04/13 1A	B/Sth Aus[2] 81/12 F41/13 1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	80	<	20	<	80	160	80	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	160	320	80	80	40	640	640	320	1280
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	2560	10	20	80	80	80	40	40	40	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	10	80	160	160	160	160	160	80	320
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	2560	<	40	80	160	160	40	80	80	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	160	40	40	20	320	320	160	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	80	160	80	1280	1280	640	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	80	320	80	80	40	320	320	320	640
B/South Australia/81/2012		2012-11-28	E4/E1	1280	160	320	80	80	40	320	320	320	1280
TEST VIRUSES													
B/Cameroon/743/2014		2014-02-05	MDCK1	2560	10	40	80	80	160	20	40	80	160
B/Norway/970/2014	1A	2014-03-07	MDCK1	5120	10	40	80	80	160	40	40	80	160
B/Kumamoto/46/2014	1A	2014-03-14	MDCK1/MDCK1/MDCK1	5120	20	80	160	160	80	80	160	160	320
B/Cameroon/2080/2014		2014-03-24	MDCK2/MDCK1	5120	10	40	40	80	40	10	<	40	40
B/Cameroon/2053/2014		2014-03-26	MDCK2/MDCK1	5120	<	40	160	80	160	40	80	80	160
B/Cameroon/2052/2014	1A	2014-03-26	MDCK2/MDCK1	5120	<	20	80	80	160	40	10	40	80
B/Cameroon/2315/2014	1A	2014-03-31	MDCK2/MDCK1	5120	20	40	160	160	160	160	160	160	160
B/Cameroon/2293/2014	1A	2014-04-03	MDCK2/MDCK1	5120	<	20	80	80	160	40	10	40	80
B/Ghana/DILI-0434/2014	1A	2014-04-28	C1/MDCK1	2560	<	20	160	80	40	80	80	80	80
B/Ghana/DILI-0487/2014	1A	2014-05-20	C1/MDCK1	2560	20	20	160	80	80	40	40	80	160
B/Ghana/DILI-0506/2014	1A	2014-05-27	C1/MDCK1	2560	10	40	80	80	80	80	40	80	160
B/Ghana/DILI-0531/2014	1A	2014-06-03	MDCK1	5120	10	20	80	80	80	80	80	80	160
B/Ghana/DILI-0572/2014	1A	2014-06-16	C1/MDCK1	2560	10	20	80	80	80	80	80	80	160
B/Cameroon/4641/2014		2014-06-25	MDCK1/MDCK1	5120	10	20	20	80	40	40	10	40	40
B/Cameroon/4314/2014	1A	2014-06-25	MDCK1/MDCK1	5120	10	20	20	80	40	40	10	40	40
B/Cameroon/4736/2014	1A	2014-07-04	MDCK1/MDCK1	5120	10	20	20	80	20	20	10	40	40
B/Cameroon/4737/2014		2014-07-05	MDCK1/MDCK1	5120	10	20	20	80	80	20	10	40	40
B/Cameroon/4681/2014	1A	2014-07-09	MDCK1/MDCK1	2560	<	20	20	80	40	20	10	40	20

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Figure 16. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

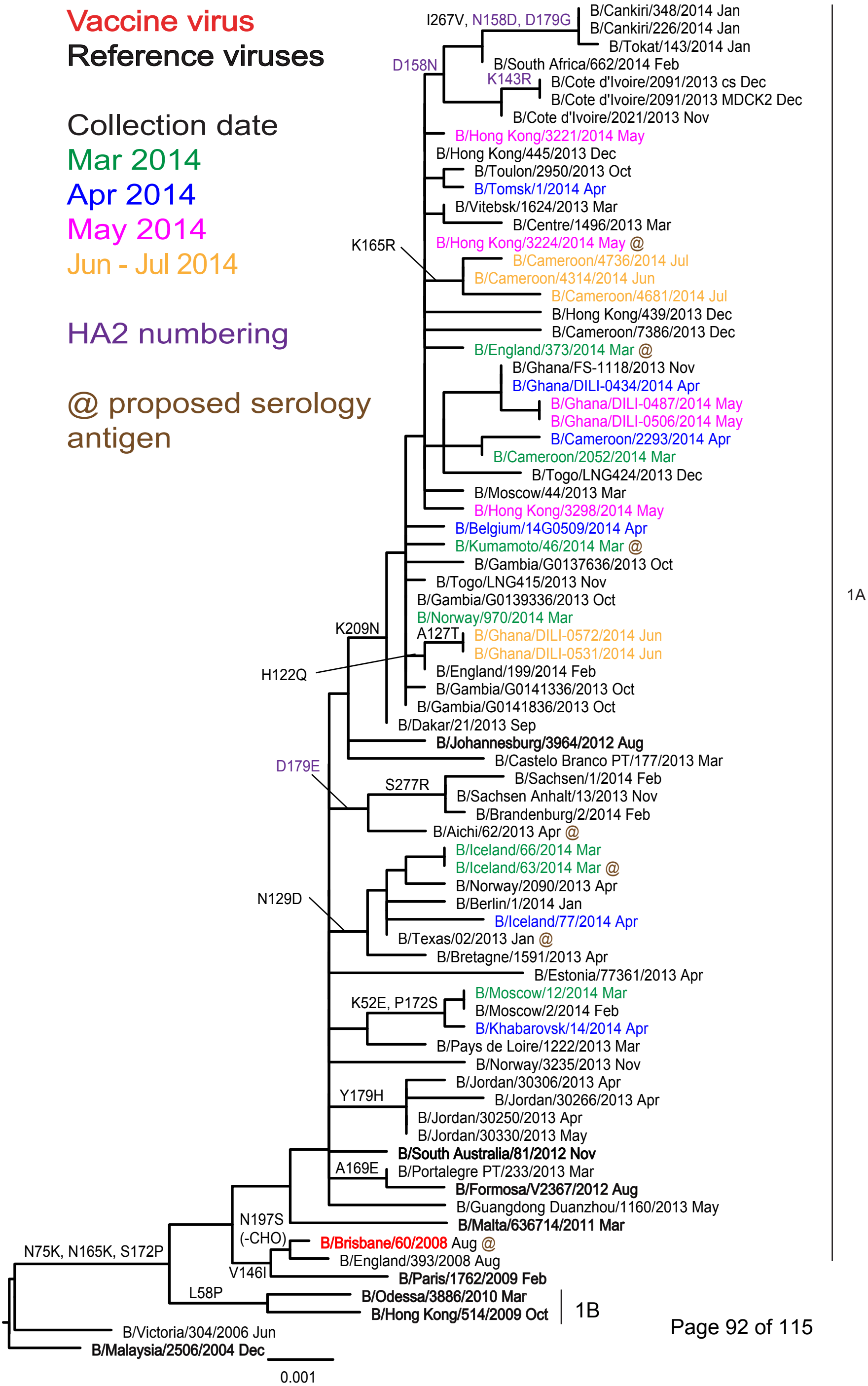


Figure 17. HA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                   10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGNIPSARVSILHEVRPVTSGCFPIMHD
B/Hong Kong/514/2009       .........................................................P................K......................... 1B
B/Odessa/3886/2010         .........................................................P................K......................... 1B
B/Brisbane/60/2008         ..........................................................................K......................... 1A
B/Paris/1762/2009          ..........................................................................K......................... 1A
B/Malta/636714/2011        ..........................................................................K......................... 1A
B/Formosa/V2367/2012       .................................A........................................K......................... 1A
B/South Australia/81/2012  ..........................................................................K......................... 1A
B/Johannesburg/3964/2012   ..............................................A...........................K......................... 1A
B/Aichi/62/2013            ....................................X.....................................K............K............ 1A
B/Cote d'Ivoire/2091/2013  ..........................................................................K......................... 1A
B/Texas/02/2013            ..........................................................................K......................... 1A
B/Khabarovsk/14/2014       ...................................................E......................K......................... 1A
B/Moscow/12/2014           ...................................................E......................K......................... 1A
B/Iceland/77/2014          ..........................................................................K......................... 1A
B/Iceland/63/2014          ..........................................................................K......................... 1A
B/Sachsen/1/2014           ..........................................................................K......................... 1A
B/England/199/2014         ..........................................................................K......................... 1A
B/Ghana/DILI-0572/2014     ..........................................................................K......................... 1A
B/Norway/970/2014          ..........................................................................K......................... 1A
B/Kumamoto/46/2014         ..........................................................................K......................... 1A
B/Belgium/14G0509/2014     ..........................................................................K......................... 1A
B/Cameroon/2293/2014       ...................................................................S......K......................... 1A
B/Ghana/DILI-0506/2014     ..........................................................................K......................... 1A
B/England/373/2014         ..........................................................................K......................... 1A
B/Cameroon/4736/2014       ..........................................................................K......................... 1A
B/Hong Kong/3224/2014      ..........................................................................K......................... 1A
B/Tomsk/1/2014             ..........................................................................K......................... 1A
B/South Africa/662/2014    ..........................................................................K......................... 1A
B/Tokat/143/2014           ..........................................................................K......................... 1A

                                  110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNVTNGNGFFATMAWAVPKNDNNKTATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ
B/Hong Kong/514/2009       ................................................................K......P..........................T.
B/Odessa/3886/2010         ................................................................K......P..........................T.
B/Brisbane/60/2008         .............................................I..................K......P........................S.T.
B/Paris/1762/2009          .............................................I..................K......P.................I........T.
B/Malta/636714/2011        ................................................................K......P........................S.T.
B/Formosa/V2367/2012       ................................................................K...E..P..........................T.
B/South Australia/81/2012  ................................................................K......P..........................A.
B/Johannesburg/3964/2012   ................................................................K......P........................X.T.
B/Aichi/62/2013            ................................................................K......P..........................T.
B/Cote d'Ivoire/2091/2013  ................................................................K......P..........................T.
B/Texas/02/2013            ............................D...................................K......P........................X.X.
B/Khabarovsk/14/2014       .................K..............................................K.................................T.
B/Moscow/12/2014           ................................................................K.................................T.
B/Iceland/77/2014          ............................D...................................K......P..........................T.
B/Iceland/63/2014          ............................D...................................K......P..........................T.
B/Sachsen/1/2014           ...................................E..........................K.K......P..........................T.
B/England/199/2014         .....................Q..........................................K......P..........................T.
B/Ghana/DILI-0572/2014     .....................Q....T.....................................K......P..........................T.
B/Norway/970/2014          ................................................................K......P..........................T.
B/Kumamoto/46/2014         ................................................................K......P..........................T.
B/Belgium/14G0509/2014     ................................................................K......P..........................T.
B/Cameroon/2293/2014       ................................................................K......P..........................T.
B/Ghana/DILI-0506/2014     ................................................................K......P..........................T.
B/England/373/2014         ................................................................K......P.................I........T.
B/Cameroon/4736/2014       ................................................................R......P..........................T.
B/Hong Kong/3224/2014      ................................................................K......P..........................T.
B/Tomsk/1/2014             ................................................................K......P..........................T.
B/South Africa/662/2014    ................................................................K......P..........................T.
B/Tokat/143/2014           ................................................................K......P..........................T.

                                  210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Hong Kong/514/2009       ..............................................................................K.....................
B/Odessa/3886/2010         ....................................................................................................
B/Brisbane/60/2008         ....................................................................................................
B/Paris/1762/2009          ........................I.............................P.............................................
B/Malta/636714/2011        ....................................................................................................
B/Formosa/V2367/2012       ....................................................................................................
B/South Australia/81/2012  ....................................................................................................
B/Johannesburg/3964/2012   ....................................................................................................
B/Aichi/62/2013            ........................................X...........................................................
B/Cote d'Ivoire/2091/2013  ........N...........................................................................................
B/Texas/02/2013            ....................................................................................................
B/Khabarovsk/14/2014       ....................................................................................................
B/Moscow/12/2014           ....................................................................................................
B/Iceland/77/2014          ....................................................................................................
B/Iceland/63/2014          ....................................................................................................
B/Sachsen/1/2014           ............................................................................R.......................
B/England/199/2014         ........N...........................................................................................
B/Ghana/DILI-0572/2014     ........N...........................................................................................
B/Norway/970/2014          ........N...........................................................................................
B/Kumamoto/46/2014         ........N...........................................................................................
B/Belgium/14G0509/2014     ........N...........................................................................................
B/Cameroon/2293/2014       ........N...........................................................................................
B/Ghana/DILI-0506/2014     ........N...........................................................................................
B/England/373/2014         ........N...........................................................................................
B/Cameroon/4736/2014       ........N...........................................................................................
B/Hong Kong/3224/2014      ........N...........................................................................................
B/Tomsk/1/2014             ........N...........................................................................................
B/South Africa/662/2014    ........N...........................................................................................
B/Tokat/143/2014           ........N.........................................................V.................................
```

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                                       310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Paris/1762/2009             ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Aichi/62/2013               ....................................................................................................
B/Cote d'Ivoire/2091/2013     ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Khabarovsk/14/2014          ....................................................................................................
B/Moscow/12/2014              ....................................................................................................
B/Iceland/77/2014             ....................................................................................................
B/Iceland/63/2014             ....................................................................................................
B/Sachsen/1/2014              ....................................................................................................
B/England/199/2014            ....................................................................................................
B/Ghana/DILI-0572/2014        ....................................................................................................
B/Norway/970/2014             ....................................................................................................
B/Kumamoto/46/2014            ....................................................................................................
B/Belgium/14G0509/2014        ..............................................................................I.....................
B/Cameroon/2293/2014          ....................................................................................................
B/Ghana/DILI-0506/2014        ....................................................................................................
B/England/373/2014            ....................................................................................................
B/Cameroon/4736/2014          ....................................................................................................
B/Hong Kong/3224/2014         ....................................................................................................
B/Tomsk/1/2014                ....................................................................................................
B/South Africa/662/2014       ....................................................................................................
B/Tokat/143/2014              ....................................................................................................

                                       410       420       430       440       450       460       470       480       490       500
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Paris/1762/2009             ............................D.......................................................................
B/Malta/636714/2011           ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Aichi/62/2013               ....................................................................................................
B/Cote d'Ivoire/2091/2013     .........................................................................................R..........
B/Texas/02/2013               ....................................................................................................
B/Khabarovsk/14/2014          ....................................................................................................
B/Moscow/12/2014              ....................................................................................................
B/Iceland/77/2014             .............................................................................I......................
B/Iceland/63/2014             ....................................................................................................
B/Sachsen/1/2014              ....................................................................................................
B/England/199/2014            ....................................................................................................
B/Ghana/DILI-0572/2014        ....................................................................................................
B/Norway/970/2014             ....................................................................................................
B/Kumamoto/46/2014            ....................................................................................................
B/Belgium/14G0509/2014        ....................................................................................................
B/Cameroon/2293/2014          ....................................................................................................
B/Ghana/DILI-0506/2014        ....................................................................................................
B/England/373/2014            ....................................................................................................
B/Cameroon/4736/2014          ....................................................................................................
B/Hong Kong/3224/2014         ....................................................................................................
B/Tomsk/1/2014                ....................................................................................................
B/South Africa/662/2014       ....................................................................................................
B/Tokat/143/2014              ....................................................................................................

                                       510       520       530       540       550       560       570
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Hong Kong/514/2009          ......................................................................
B/Odessa/3886/2010            ......................................................................
B/Brisbane/60/2008            ......................................................................
B/Paris/1762/2009             ......................................................................
B/Malta/636714/2011           ......................................................................
B/Formosa/V2367/2012          ......................................................................
B/South Australia/81/2012     ......................................................................
B/Johannesburg/3964/2012      ......................................................................
B/Aichi/62/2013               .........................E............................................
B/Cote d'Ivoire/2091/2013     ....N.................................................................
B/Texas/02/2013               ......................................................................
B/Khabarovsk/14/2014          ......................................................................
B/Moscow/12/2014              ......................................................................
B/Iceland/77/2014             ......................................................................
B/Iceland/63/2014             ......................................................................
B/Sachsen/1/2014              .........................E............................................
B/England/199/2014            ......................................................................
B/Ghana/DILI-0572/2014        ......................................................................
B/Norway/970/2014             ......................................................................
B/Kumamoto/46/2014            ......................................................................
B/Belgium/14G0509/2014        ......................................................................
B/Cameroon/2293/2014          ......................................................................
B/Ghana/DILI-0506/2014        ......................................................................
B/England/373/2014            ......................................................................
B/Cameroon/4736/2014          ......................................................................
B/Hong Kong/3224/2014         ......................................................................
B/Tomsk/1/2014                ......................................................................
B/South Africa/662/2014       ....N.................................................................
B/Tokat/143/2014              .........................G............................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 18. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

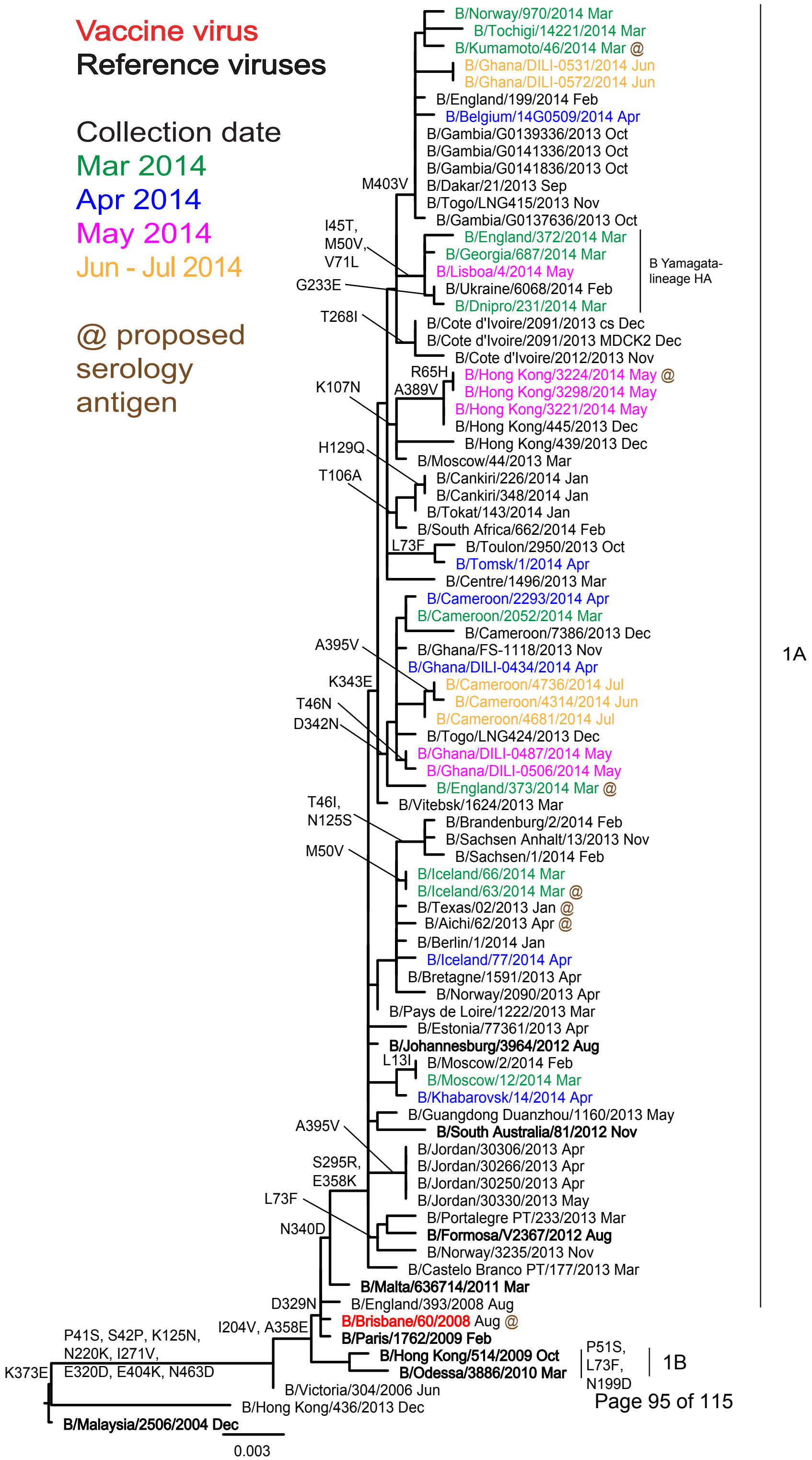

Figure 19. NA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

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                                   10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFPSTEITAPTMPLDCANASNVQAVNRSATKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Hong Kong/514/2009       ........................................SP........S.....................S........................... 1B
B/Odessa/3886/2010         ..........................L.............SP........S.....................F........................... 1B
B/Brisbane/60/2008         ........................................SP.......................................................... 1A
B/Paris/1762/2009          ........................................SP.......T.................................................. 1A
B/Malta/636714/2011        ........................................SP......A................................................... 1A
B/Formosa/V2367/2012       ........................................SP..............................F........................... 1A
B/South Australia/81/2012  ........................................SP........................................................N. 1A
B/Johannesburg/3964/2012   ........................................SP.......................................................... 1A
B/Aichi/62/2013            ........................................SP.......................................................... 1A
B/Cote d'Ivoire/2091/2013  ........................................SP.......................................................... 1A
B/Texas/2/2013             ........................................SP.......................................................... 1A
B/Khabarovsk/14/2014       ........................................SP.......................................................... 1A
B/Moscow/12/2014           ............I...........................SP.......................................................... 1A
B/Iceland/77/2014          ........................................SP.......................................................... 1A
B/Iceland/63/2014          ........................................SP.......V.................................................. 1A
B/Sachsen/1/2014           ........................................SP...I...................................................... 1A
B/England/199/2014         ........................................SP.......................................................... 1A
B/Ghana/DILI-0572/2014     ........................................SP.......................................................... 1A
B/Norway/970/2014          ........................................SP.......................................................... 1A
B/Kumamoto/46/2014         ........................................SP.......................................................... 1A
B/Belgium/14G0509/2014     ........................................SP.......................................................... 1A
B/Cameroon/2293/2014       ........................................SP.......................................................... 1A
B/Ghana/DILI-0506/2014     ........................................SP...N...................................................... 1A
B/England/373/2014         ........................................SP.......................................................... 1A
B/Cameroon/4736/2014       ........................................SP.......................................................... 1A
B/Hong Kong/3224/2014      ........................................SP......................H................................... 1A
B/Tomsk/1/2014             ........................................SP..............................F........................... 1A
B/South Africa/662/2014    ........................................SP.......................................................... 1A
B/Tokat/143/2014           ........................................SP.......................................................... 1A

                                  110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNNA
B/Hong Kong/514/2009       ........................N.........................................................................D.
B/Odessa/3886/2010         ........................N.........................................................................D.
B/Brisbane/60/2008         ........................N...........................................................................
B/Paris/1762/2009          ........................N...........................................................................
B/Malta/636714/2011        ........................N...........................................................................
B/Formosa/V2367/2012       ........................N...........................................................................
B/South Australia/81/2012  ........................N...........................................................................
B/Johannesburg/3964/2012   ........................N...........................................................................
B/Aichi/62/2013            ........................N...........................................................................
B/Cote d'Ivoire/2091/2013  ........................N...........................................................................
B/Texas/2/2013             ........................N...........................................................................
B/Khabarovsk/14/2014       ........................N...........................................................................
B/Moscow/12/2014           ........................N...........................................................................
B/Iceland/77/2014          ........................N...........................................................................
B/Iceland/63/2014          ........................N...........................................................................
B/Sachsen/1/2014           ........................S...........................................................................
B/England/199/2014         ........................N...........................................................................
B/Ghana/DILI-0572/2014     ........................N...........................................................................
B/Norway/970/2014          ........................N...........................................................................
B/Kumamoto/46/2014         ........................N...........................................................................
B/Belgium/14G0509/2014     ........................N...........................................................................
B/Cameroon/2293/2014       ........................N...........................................................................
B/Ghana/DILI-0506/2014     ........................N...........................................................................
B/England/373/2014         ........................N...........................................................................
B/Cameroon/4736/2014       ........................N...........................................................................
B/Hong Kong/3224/2014      ......N.................N...........................................................................
B/Tomsk/1/2014             ........................N...........................................................................
B/South Africa/662/2014    .....A..................N...........................................................................
B/Tokat/143/2014           .....A..................N...........................................................................

                                  210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004       LLKIKYGEAYTDTYHSYANNILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRIKHTEECTCGFASNKTIECACRDNSYTAKR
B/Hong Kong/514/2009       ...V...............K..................................................V.............................
B/Odessa/3886/2010         ...V...............K..................................................V.............................
B/Brisbane/60/2008         ...V...............K..................................................V.............................
B/Paris/1762/2009          ...V...............K..................................................V.............................
B/Malta/636714/2011        ...V...............K..................................................V.............................
B/Formosa/V2367/2012       ...V...............K..................................................V.......................R.....
B/South Australia/81/2012  ...V...............K..................................................V.......................R.....
B/Johannesburg/3964/2012   ...V...............K..................................................V.......................R.....
B/Aichi/62/2013            ...V...............K..................................................V.......................R.....
B/Cote d'Ivoire/2091/2013  ...V...............K...............................................I..V.......................R.....
B/Texas/2/2013             ...V...............K..................................................V.......................R.....
B/Khabarovsk/14/2014       ...V...............K..................................................V.......................R.....
B/Moscow/12/2014           ...V...............K..................................................V.......................R.....
B/Iceland/77/2014          ...V...............K..................................................V.......................R.....
B/Iceland/63/2014          ...V...............K..................................................V.......................R.....
B/Sachsen/1/2014           ...V...............K..................................................V.......................R.....
B/England/199/2014         ...V...............K..................................................V.......................R.....
B/Ghana/DILI-0572/2014     ...V...............K..................................................V.......................R.....
B/Norway/970/2014          ...V...............K..................................................V.......................R.....
B/Kumamoto/46/2014         ...V...............K..................................................V.......................R.....
B/Belgium/14G0509/2014     ...V...............K..................................................V.......................R.....
B/Cameroon/2293/2014       ...V...............K..................................................V.......................R.....
B/Ghana/DILI-0506/2014     ...V...............K..................................................V.......................R.....
B/England/373/2014         ...V...............K..................................................V.......................R.....
B/Cameroon/4736/2014       ...V...............K..................................................V.......................R.....
B/Hong Kong/3224/2014      ...V...............K..................................................V.......................R.....
B/Tomsk/1/2014             ...V...............K..................................................V.......................R.....
B/South Africa/662/2014    ...V...............K..................................................V.......................R.....
B/Tokat/143/2014           ...V...............K..................................................V.......................R.....
```

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                                 310       320       330       340       350       360       370       380       390       400
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         PFVKLNVETDTAEIRLMCTETYLDTPRPDDGSITGPCESNGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWADSDALAFSGVM
B/Hong Kong/514/2009         ...................D.....................................E..............E...........................
B/Odessa/3886/2010           ...................D.....................................E..............E...........................
B/Brisbane/60/2008           ...................D........N............................E..............E...........................
B/Paris/1762/2009            ...................D........N............................E..............E...........................
B/Malta/636714/2011          ...................D........N..........D.................E..............E...........................
B/Formosa/V2367/2012         ...................D........N..........D.................K..............E...........................
B/South Australia/81/2012    ...................D........N..........D.................K..............E...........................
B/Johannesburg/3964/2012     ...................D........N..........D.................K..............E...........................
B/Aichi/62/2013              ...................D........N..........D.................K..............E...........................
B/Cote d'Ivoire/2091/2013    ...................D........N..........D..E..............K..............E...........................
B/Texas/2/2013               ...................D........N..........D.................K..............E...........................
B/Khabarovsk/14/2014         ...................D........N..........D.................K..............E...........................
B/Moscow/12/2014             ...................D........N..........D.................K..............E...........................
B/Iceland/77/2014            ...................D........N..........D.................K..............E...........................
B/Iceland/63/2014            ...................D........N..........D.................K..............E...........................
B/Sachsen/1/2014             ...................D........N..........D.................K..............E...........................
B/England/199/2014           ...................D........N..........D..E..............K..............E.......................G...
B/Ghana/DILI-0572/2014       ...................D........N..........D..E..............K..............E...........................
B/Norway/970/2014            ...................D........N.............E..............K..............E...........................
B/Kumamoto/46/2014           ...................D........N..........D..E....V.........K..............E...........................
B/Belgium/14G0509/2014       ...................D........N..........D..E..............K..............E...........................
B/Cameroon/2293/2014         ...................D........N..........D.NE..............K..............E...........................
B/Ghana/DILI-0506/2014       ...................D........N..........D.NE..............K..............E...........................
B/England/373/2014           ...................D........N..........D.NE..............K..............E...........................
B/Cameroon/4736/2014         ...................D........N..........D.NE..............K..............E.....................V.....
B/Hong Kong/3224/2014        ...................D........N..........D..E..............K..............E...............V...........
B/Tomsk/1/2014               ...................D........N..........D..E..............K..............E...........................
B/South Africa/662/2014      ...................D........N..........D..E..............K..............E.....................V.....
B/Tokat/143/2014             ...................D........N..........D..E..............K..............E...........................
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                                 410       420       430       440       450       460
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Malaysia/2506/2004         VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Hong Kong/514/2009         ...K..........................................................D...
B/Odessa/3886/2010           ...K..........................................................D...
B/Brisbane/60/2008           ...K..........................................................D...
B/Paris/1762/2009            ...K..........................................................D...
B/Malta/636714/2011          ...K..........................................................D...
B/Formosa/V2367/2012         ...K..........................................................D...
B/South Australia/81/2012    ...K..........................................................D...
B/Johannesburg/3964/2012     I..K..........................................................D...
B/Aichi/62/2013              ...K..........................................................D...
B/Cote d'Ivoire/2091/2013    ...K..........................................................D...
B/Texas/2/2013               ...K..........................................................D...
B/Khabarovsk/14/2014         ...K..........................................................D...
B/Moscow/12/2014             ...K..........................................................D...
B/Iceland/77/2014            ...K..........................................................D...
B/Iceland/63/2014            ...K..........................................................D...
B/Sachsen/1/2014             ...K..........................................................D...
B/England/199/2014           ..VK..........................................................D...
B/Ghana/DILI-0572/2014       ..VK..........................................................D...
B/Norway/970/2014            ..VK..........................................................D...
B/Kumamoto/46/2014           ..VK..........................................................D...
B/Belgium/14G0509/2014       ..VK.......................................................A..D...
B/Cameroon/2293/2014         ...K..........................................................D...
B/Ghana/DILI-0506/2014       ...K..........................................................D...
B/England/373/2014           ...K..............................................................
B/Cameroon/4736/2014         ...K..........................................................D...
B/Hong Kong/3224/2014        ...K..........................................................D...
B/Tomsk/1/2014               ...K..........................................................D...
B/South Africa/662/2014      ...K..........................................................D...
B/Tokat/143/2014             ...K..........................................................D...
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2014-01-31

MONTH Country	B Yamagata lineage Number received	 Number propagated[1]	 ≤2 fold[2]	 4 fold[2]	 ≥8 fold[2]
FEBRUARY					
Cameroon	1	1			1
Georgia	3	3		1	2
Germany	2	2		2	
Israel	2	2		2	
Italy	2	2		1	1
Portugal	1	1	1		
Sweden	2	2			2
Ukraine	2	2		1	1
MARCH					
Austria	3	3		1	2
Cameroon	2	2		1	1
France	1	1		1	
Georgia	1	1			1
Israel	4	4		3	1
Latvia	1	1		1	
Romania	1	1		1	
Russia	2	2	2		
Slovakia	1	1		1	
Spain	1	1		1	
Switzerland	2	2		1	1
Ukraine	2	2	1	1	
United Kingdom	1	1		1	
APRIL					
Russia	3	in process	2		
MAY					
Belarus	2	2	2		
France	1	1		1	
Iceland	1	1			1
Norway	2	2		1	1
Reunion	2	2	1		1
Portugal	1	1		1	
JUNE					
Hong Kong	3	3	2	1	
Norway	1	1		1	
Reunion	2	2		1	1
South Africa	1	1			1
JULY					
Spain	2	2			2
	58	55	11	26	20
23 Countries			19.3%	45.6%	35.1%

1. Propagated to sufficient titre to perform HI assay
2. Reduction in HI titre with ferret antiserum raised against B/Massachusetts/02/2012 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F15/13
				Haemagglution Inhibition Titre – Post infection ferret antisera									
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	640	160	80	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	640	160	80	320	640	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	640	160	160	160	320	<	40	40	320	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	80	80	160	10	40	40	160	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	80	80	80	640	80	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	160	160	320	320	160	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	80	80	80	80	640	80	640	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	320	640	160	640	160	40	1280	640	640
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	320	160	320	320	80	320	640	320
TEST VIRUSES													
B/Osaka/18/2013		2012-12-07	MDCK1/MDCK1/MDCK1	640	160	80	160	320	40	80	80	160	80
B/Henan-Nanzhaoling/1406/2013		2013-11-02	C3/MDCK1	1280	160	160	160	320	40	160	160	160	160
B/Shandong-Mudan/1374/2013		2013-11-03	C5/MDCK1	2560	160	160	320	320	320	320	320	320	640
B/Cameroon/6302/2013	2	2013-11-09	MDCK3	2560	160	160	80	80	640	40	640	320	640
B/Chongqing-Yuzhong/11616/2013		2013-11-12	E3/E1	1280	160	640	40	160	80	10	160	320	160
B/Cameroon/6633/2013		2013-11-18	MDCK3	2560	160	160	80	160	640	40	1280	640	640
B/Cameroon/6453/2013	2	2013-11-19	MDCK2	5120	160	640	160	40	320	320	2560	1280	2560
B/Cameroon/6794/2013	2	2013-12-04	MDCK2	1280	80	320	40	<	160	80	1280	640	1280
B/Cameroon/6999/2013		2013-12-05	MDCK4	5120	320	640	320	1280	640	640	1280	1280	1280
B/Cameroon/7011/2013		2013-12-10	MDCK2	1280	80	320	40	<	160	80	2560	640	1280
B/Cameroon/7009/2013	2	2013-12-11	MDCK2	2560	320	320	40	<	160	80	2560	1280	1280
B/Trabzon/30/2014	2	2014-01-03	SIAT1/MDCK1	5120	320	320	320	640	640	640	1280	320	1280
B/Catalonia/2151450NS/2014	3	2014-01-21	C0/MDCK1	2560	160	160	320	640	160	160	160	160	320
B/Berlin/2/2014	2	2014-02-06	C2/MDCK1	5120	160	160	160	320	160	80	640	160	640
B/Bayern/1/2014	3	2014-02-10	C2/MDCK1	1280	160	80	160	320	40	80	80	160	80

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre — Post infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	640	160	80	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	640	160	80	320	640	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	320	320	320	640	<	40	40	640	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	5120	320	160	160	320	<	40	40	320	80
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	160	80	160	1280	80	1280	320	1280
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	320	320	640	640	320	640	640	320	1280
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	5120	320	160	160	320	1280	160	1280	320	1280
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	1280	320	1280	160	40	640	1280	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	1280	1280	640	640	640	160	1280	1280	1280
TEST VIRUSES													
B/Cameroon/3865/2013	2	2013-07-11	MDCK2/MDCK1	1280	20	80	40	160	640	40	640	160	320
B/Madagascar/03261/2013	2	2013-09-02	MDCK2	2560	160	80	40	80	320	40	640	320	640
B/Madagascar/03256/2013	2	2013-09-03	MDCK2	2560	160	80	40	80	640	40	640	320	640
B/Cote d'Ivoire/1875/2013	3	2013-11-04	MDCK2	2560	160	80	160	320	40	160	160	320	160
B/Cameroon/6585/2013	2	2013-11-25	MDCK1/MDCK1	5120	80	160	80	160	1280	80	1280	320	1280
B/Cameroon/6788/2013	2	2013-11-28	MDCK2/MDCK1	2560	160	80	40	80	640	40	640	160	640
B/Croatia/2547/2013	3	2013-12-11	MDCK2	1280	160	80	160	320	20	80	80	320	80
B/Cameroon/7102/2013		2013-12-13	MDCK1/MDCK1	2560	160	80	40	80	640	40	640	160	640
B/Cameroon/7103/2013		2013-12-13	MDCK1/MDCK1	2560	160	80	40	80	640	40	640	160	640
B/Cameroon/7095/2013	2	2013-12-13	MDCK2/MDCK1	2560	160	80	40	320	640	40	640	160	640
B/Cameroon/7143/2013	2	2013-12-16	MDCK2/MDCK1	5120	160	160	80	160	640	40	640	320	640
B/Cameroon/7430/2013	2	2013-12-30	MDCK1/MDCK1	2560	160	80	40	160	640	40	640	320	640
B/Cameroon/7433/2013		2013-12-30	MDCK1/MDCK1	2560	160	80	40	80	320	40	640	160	640
B/Cameroon/7435/2013	2	2013-12-30	MDCK1/MDCK1	2560	160	80	40	80	640	40	640	160	320
B/Cameroon/7429/2013		2013-12-30	MDCK1/MDCK1	5120	160	160	80	160	1280	160	1280	320	640
B/Croatia/5/2014	3	2014-01-02	MDCK2	640	80	80	80	160	10	80	80	160	80
B/Malatya/301/2014	3	2014-01-16	MDCK2	2560	320	160	320	320	80	160	160	320	160

1. < = <40;   2. < = <10;   3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre — Post infection ferret antisera									
				B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	320	160	20	160	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	160	320	160	20	320	1280	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	640	160	160	80	160	<	20	40	160	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	40	160	<	20	40	160	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	640	80	40	20	20	640	40	640	80	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	640	80	80	160	160	80	160	160	80	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	80	80	80	160	320	80	640	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	80	320	80	20	160	640	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	160	320	320	40	640	320	640
TEST VIRUSES													
B/Zambia/13-086/2013	2	2013-08-15	Cx/MDCK1	2560	160	160	160	320	640	40	1280	320	80
B/Zambia/13-089/2013	2	2013-08-19	Cx/MDCK1	2560	80	160	80	160	640	40	640	160	80
B/Zambia/13-090/2013		2013-08-23	Cx/MDCK1	2560	160	160	160	320	640	40	640	320	160
B/Zambia/13-327/2013	2	2013-09-12	Cx/MDCK2	1280	80	80	40	160	640	40	640	160	160
B/Zambia/13-348/2013	2	2013-09-14	Cx/MDCK1	1280	80	160	40	160	640	20	640	160	80
B/Zambia/13-177/2013	2	2013-09-26	Cx/SIAT1	640	80	80	20	40	320	20	320	80	320
B/Cameroon/5643/2013	2	2013-10-09	MDCK3	5120	160	320	160	320	640	40	1280	320	160
B/Cameroon/6168/2013	2	2013-10-31	MDCK3	2560	160	160	80	160	640	40	640	320	80
B/Firenze/2/2014	3	2014-01-14	MDCK2/MDCK1	640	40	40	40	40	20	40	80	80	80
B/Switzerland/9943612/2014	3	2014-01-22	MDCK2	640	80	80	80	160	40	80	160	160	160
B/Georgia/203/2014	3	2014-02-10	MDCK3	1280	160	80	160	160	80	160	160	160	160
B/Dnipro/177/2014	3	2014-02-12	SIAT2/MDCK1	640	80	80	80	80	20	80	80	80	80
B/Sassari/9/2014	3	2014-02-13	Cx/MDCK1	1280	80	80	160	160	80	160	160	80	320
B/Georgia/435/2014	3	2014-02-24	MDCK2	320	40	40	40	40	10	40	40	80	40
B/Sassari/10/2014	3	2014-02-25	Cx/SIAT1	2560	160	160	320	320	320	320	640	160	320
B/Georgia/465/2014	3	2014-02-26	MDCK3	320	40	40	40	40	10	40	40	80	40
B/Switzerland/10295792/2014	3	2014-03-27	MDCK2	320	80	40	80	80	20	40	80	160	40
B/Georgia/687/2014	3	2014-03-28	MDCK2	320	40	20	40	40	10	40	40	80	40
B/Switzerland/10294006/2014	3	2014-03-28	MDCK2	320	40	40	80	80	10	40	40	80	40

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre – Post infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	*B/Mass[2] 02/12 Egg F16/14	B/Mass[2] 02/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2	2
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	320	160	20	160	1280	ND	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	160	320	160	20	320	1280	ND	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	640	160	160	80	160	<	20	40	160	ND	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	40	160	<	20	40	160	ND	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	640	80	40	20	20	640	40	640	80	ND	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	640	80	80	160	160	160	320	320	160	ND	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	80	80	160	320	320	320	320	160	ND	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	320	640	80	40	40	640	1280	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	160	320	320	40	640	320	ND	640
TEST VIRUSES														
B/Armenia/469/2013	2	2013-03-21	MDCK1	2560	80	80	40	40	640	20	640	160	ND	320
B/England/49/2014	2	2014-01-21	SIAT1/MDCK1	2560	640	640	160	640	160	80	320	640	1280	320
B/Finland/392/2014	2	2014-01-26	MDCK1/MDCK1	1280	160	160	80	80	320	20	160	320	320	640
B/Stockholm/4/2014	2	2014-02-17	MDCK1MDCK1	1280	160	80	40	80	640	40	640	ND	160	320
B/Stockholm/5/2014	3	2014-02-21	MDCK0/MDCK1	640	80	40	80	160	40	80	80	ND	80	80
B/Ukraine/6068/2014	3	2014-02-26	C0/MDCK1	640	80	80	80	80	40	80	40	160	80	80
B/Dnipro/231/2014	3	2014-03-03	SIAT2/MDCK1	2560	160	80	160	320	320	640	320	320	640	640
B/Latvia/03-022701/2014	3	2014-03-11	MDCK 1/MDCK1	640	80	80	160	160	40	80	40	160	160	160
B/Kharkov/210/2014	3	2014-03-12	SIAT2/MDCK1	640	160	80	80	80	40	80	40	160	80	80
B/England/372/2014	3	2014-03-13	SIAT1/MDCK1	1280	80	80	160	160	40	80	40	160	160	160
B/Bratislava/175/2014	3	2014-03-13	MDCK1/MDCK1	640	80	80	160	160	20	80	40	160	80	80
B/Bucuresti/1772-1324/2014	3	2014-03-20	MDCK1/MDCK1	2560	160	160	320	320	320	320	640	ND	320	640
B/St Petersburg/76/2014	3	2014-04-08	C2/MDCK1	5120	320	320	320	640	320	640	640	ND	640	640
B/Iceland/78/2014	3	2014-05-28	MDCK0/MDCK1	320	40	40	160	160	20	40	40	80	ND	40
												Vaccine	Vaccine	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; * new antiserum

Sequences in phylogenetic trees

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13
				Haemagglution Inhibition Titre									
				Post infection ferret antisera									
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	160	640	80	40	640	640	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	1280	160	40	1280	1280	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	640	320	160	160	640	<	40	160	320	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	320	<	40	160	320	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	640	160	80	40	40	640	40	2560	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	1280	320	160	320	640	160	320	1280	320	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	640	80	40	40	80	320	40	2560	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	160	640	80	20	640	640	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	640	640	80	1280	640	640
TEST VIRUSES													
B/Armenia/311/2013	2	2013-03-11	MDCK1	1280	320	160	40	160	640	40	640	160	640
B/Lisboa/3/2014	2	2014-02-27	MDCK1	1280	160	160	160	320	640	160	640	320	640
B/Austria/783231/2014	3	2014-03-10	SIAT1/MDCK1	2560	320	160	320	640	160	320	320	160	320
B/Austria/784502/2014	2	2014-03-15	SIAT1/MDCK1	5120	640	320	160	640	1280	160	1280	320	1280
B/Austria/784609/2014	3	2014-03-17	SIAT1/MDCK1	2560	320	320	640	640	320	640	640	320	640
B/Moscow/10/2014	3	2014-03-21	MDCK1/MDCK1	1280	160	160	160	320	40	160	640	320	160
B/Moscow/9/2014	3	2014-03-24	MDCK1/MDCK1	640	160	80	160	320	40	80	640	320	80
B/Moscow/13/2014	3	2014-04-01	MDCK1/MDCK1	1280	320	80	160	320	80	160	640	320	160
B/Gomel/1634/2014	2	2014-05-09	MDCK2/MDCK1	1280	160	80	40	80	320	40	640	320	320
B/Gomel/1635/2014	2	2014-05-09	MDCK2/MDCK1	1280	160	80	40	80	640	40	640	320	320
B/Lisboa/4/2014	3	2014-05-19	MDCK1	320	80	40	160	160	20	80	80	160	80
B/Hong Kong/3417/2014	3	2014-06-04	MDCK1/MDCK1	1280	320	80	80	320	40	80	160	320	80
B/Hong Kong/3419/2014	3	2014-06-04	MDCK1/MDCK1	2560	320	160	640	640	320	320	640	320	320
B/Hong Kong/3423/2014	3	2014-06-05	MDCK1/MDCK1	640	320	80	80	160	40	80	80	160	80

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-6. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre / Post infection ferret sera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13
Genetic Group				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	640	320	640	160	80	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	160	20	640	1280	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	320	160	320	640	<	20	40	320	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	160	80	320	<	20	40	320	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	80	80	160	1280	80	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	320	160	320	640	320	640	640	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	160	80	80	160	640	80	640	320	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	640	160	640	160	20	320	1280	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	320	320	320	640	80	640	640	640
TEST VIRUSES													
B/South Africa/68/2014	3	2014-01-10	MDCK1/MDCK1	1280	160	80	160	320	40	80	80	160	80
B/Israel/Z1321/2014	2	2014-02-18	MDCK1	2560	160	80	80	160	640	40	640	320	640
B/Israel/Z1323/2014	2	2014-02-18	MDCK1	1280	160	80	80	80	640	20	640	320	320
B/Israel/Z1411/2014	2	2014-03-02	MDCK1	2560	160	80	80	160	640	40	640	320	640
B/Israel/Z1508/2014	2	2014-03-02	MDCK1	1280	160	80	80	80	640	40	640	320	320
B/Israel/Z1510/2014	2	2014-03-07	MDCK1	1280	80	80	80	80	80	40	160	320	160
B/Israel/Z1532/2014	2	2014-03-09	MDCK1	1280	160	40	40	80	80	40	80	160	80
B/Extremadura/1755/2014	3	2014-03-17	SIAT1/MDCK1	5120	320	320	320	640	320	640	640	320	640
B/South Africa/4380/2014	3	2014-06-06	MDCK2/MDCK1	1280	160	40	160	160	20	80	80	160	80
B/Canarias/1801/2014	3	2014-07-02	Cx/MDCK1	640	80	80	160	160	40	160	80	160	80
B/Canarias/1802/2014	3	2014-07-02	Cx/MDCK1	1280	80	80	160	160	40	80	160	160	80

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-7. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre – Post infection ferret sera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	320	640	160	40	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	160	40	320	1280	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	320	160	320	640	<	40	40	320	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	160	160	320	<	40	40	320	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	320	80	80	160	1280	80	1280	320	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	320	320	640	640	320	640	640	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	320	160	160	320	1280	80	1280	320	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	1280	320	1280	160	40	320	1280	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	1280	640	320	640	640	80	640	640	640
TEST VIRUSES													
B/Hungary/379/2014	2	2014-01-17	MDCK2/E2/MDCK1	2560	320	320	160	160	320	20	320	640	320
B/Hungary/378/2014	2	2014-01-20	MDCK2/E2/MDCK1	2560	320	160	160	320	640	40	640	320	320
B/Miyagi/21/2014	2	2013-03-10	MDCK1/MDCK1/MDCK1	5120	320	320	320	640	1280	80	1280	320	1280
B/Tochiga/14221/2014	3	2013-03-17	MDCK1/MDCK1/MDCK1	2560	320	160	320	640	160	320	640	320	320
B/Paris/1105/2014	3	2014-03-20	MDCK1/MDCK1	1280	320	160	320	640	40	160	320	320	160
B/Centre/1663/2014	3	2014-05-12	MDCK1/MDCK1	1280	160	80	320	320	40	160	320	320	80
B/La Reunion/2062/14	2	2014-05-25	MDCK2/MDCK1	5120	320	160	160	320	1280	160	1280	640	640
B/La Reunion/2047/14	3	2014-05-30	MDCK2/MDCK1	1280	160	80	160	320	20	80	80	160	80
B/La Reunion/2068/14	3	2014-06-09	MDCK2/MDCK1	1280	160	80	320	320	20	80	80	160	80
B/La Reunion/2072/14	2	2014-06-21	MDCK2/MDCK1	2560	160	160	160	320	640	80	1280	320	640

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 13-8. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 AUS F3064-21D Egg
				Haemagglution Inhibition Titre – Post infection ferret sera										
Genetic Group				1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	320	640	160	40	320	640	160	640
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	1280	320	80	640	1280	320	1280
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	320	160	320	640	<	40	40	320	40	640
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	320	<	40	40	320	40	320
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	80	80	80	640	80	1280	320	640	160
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	1280	160	160	320	320	160	320	320	320	320	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	160	80	160	160	640	80	640	320	640	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	1280	320	1280	160	40	320	1280	320	1280
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	1280	640	320	640	640	80	640	1280	640	1280
B/Phuket/3073/2013	3	2013-11-21	E4/E1	1280	160	160	160	320	<	40	40	320	40	640
TEST VIRUSES														
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK1	1280	160	80	20	320	80	80	160	320	160	640
B/Cameroon/598/2014		2014-01-29	MDCK1	1280	80	80	40	80	320	20	320	160	320	ND
B/Cameroon/1044/2014		2014-02-19	MDCK2	1280	80	40	40	40	640	40	640	160	320	ND
B/Cameroon/1640/2014	2	2014-03-10	MDCK1/MDCK1	2560	320	160	10	160	640	40	640	320	640	320
B/Cameroon/2082/2014	2	2014-03-20	MDCK1/MDCK1	2560	160	80	10	160	640	80	640	160	640	160
B/Norway/1877/2014	3	2014-05-21	MDCK1	1280	160	80	20	320	80	80	160	320	160	320
B/Norway/2045/2014		2014-05-28	MDCK2	1280	160	80	160	320	80	80	160	160	160	ND
B/Norway/2011/2014	3	2014-06-19	MDCK1	640	160	80	20	320	40	80	160	320	80	320
B/Brisbane/9/2014	3	D/M unknown	E4/E1	640	160	80	160	320	<	20	20	160	40	320

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Vaccine

Figure 20. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

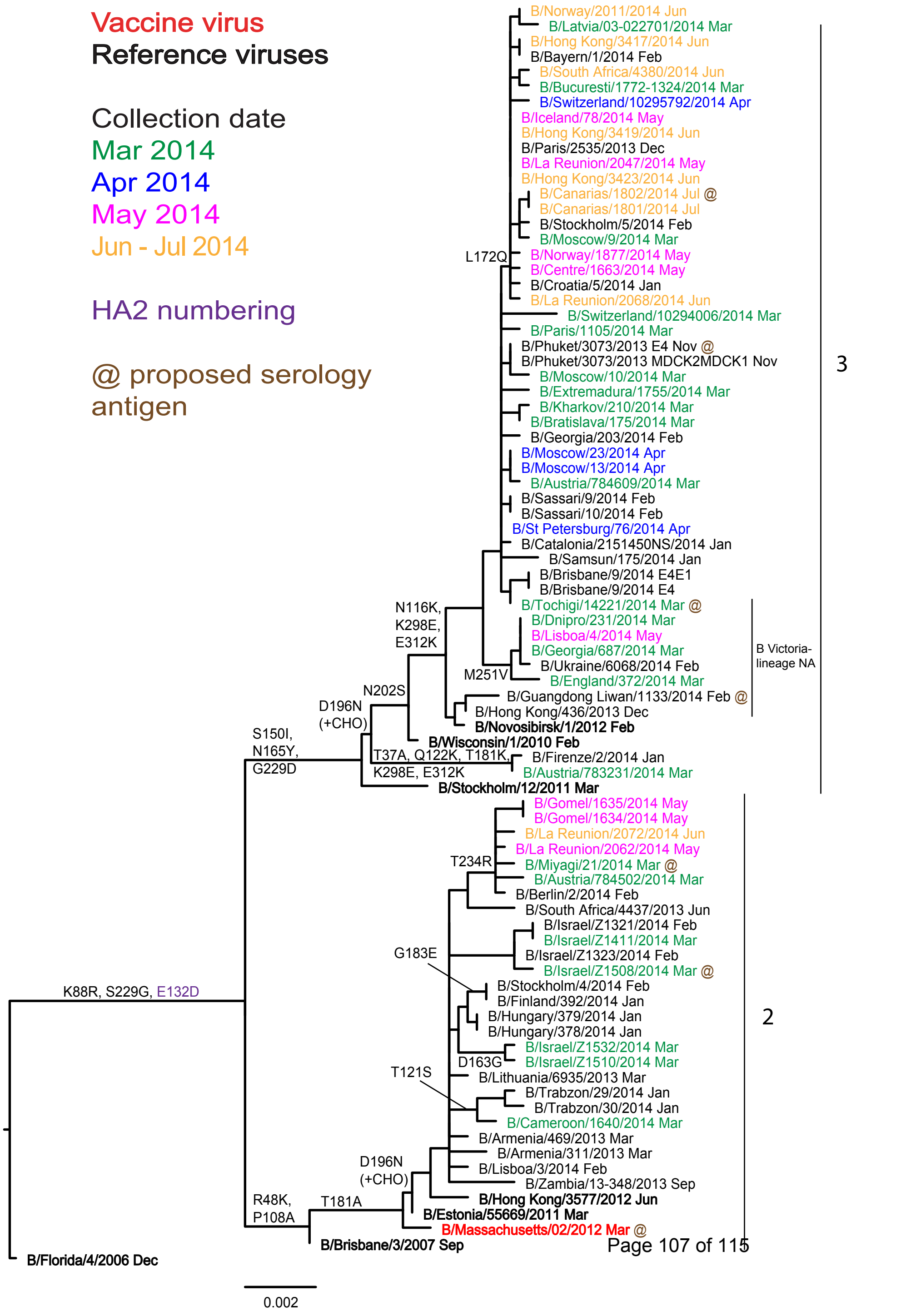

Figure 21. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIMHD
B/Brisbane/3/2007             ...............................................K.......................................R............ 2
B/Estonia/55669/2011          ...............................................K.......................................R............ 2
B/Massachusetts/02/2012       ...............................................K.......................................R............ 2
B/Hong Kong/3577/2012         ...............................................K.......................................R............ 2
B/Cameroon/1640/2014          ...............................................K.......................................R............ 2
B/Israel/Z1532/2014           ...............................................K.......................................R............ 2
B/Israel/Z1508/2014           ...............................................K.......................................R............ 2
B/Austria/784502/2014         ...............................................K.......................................R............ 2
B/Miyagi/21/2014              ...............................................K.......................................R............ 2
B/La Reunion/2072/2014        ...............................................K.......................................R............ 2
B/Gomel/1635/2014             ...............................................K.......................................R............ 2
B/South Africa/4437/2013      ...............................................K.......................................R............ 2r
B/Wisconsin/1/2010            .......................................................................................R............ 3
B/Stockholm/12/2011           ............................A..........................................................R............ 3
B/Novosibirsk/1/2012          .......................................................................................R............ 3
B/Phuket/3073/2013            .......................................................................................R............ 3
B/Austria/783231/2014         ....................................A..................................................R............ 3
B/Brisbane/9/2014             .......................................................................................R............ 3
B/St Petersburg/76/2014       .......................................................................................R............ 3
B/Moscow/23/2014              .......................................................................................R............ 3
B/Centre/1663/2014            .......................................................................................R............ 3
B/Canarias/1802/2014          .......................................................................................R............ 3
B/Iceland/78/2014             .......................................................................................R............ 3
B/Switzerland/10295792/2014   .......................................................................................R............ 3
B/Hong Kong/3417/2014         .......................................................................................R............ 3
B/Lisboa/4/2014               .......................................................................................R............ 3V
B/England/372/2014            .......................................................................................R............ 3V
B/Tochigi/14221/2014          .......................................................................................R............ 3V
B/Guangdong Liwan/1133/2014   .......................................................................................R............ 3V

                                   110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              RTKIRQLPNLLRGYENIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKSGFFATMAWAVPKDNNKNATNPLTVEVPYICTEGEDQITVWGFHSDDKTQM
B/Brisbane/3/2007             .......A............................................................................................
B/Estonia/55669/2011          .......A........................................................................A..............N....
B/Massachusetts/02/2012       .......A........................................................................A...................
B/Hong Kong/3577/2012         .......A........................................................................A..............N....
B/Cameroon/1640/2014          .......A............S...........................................................A..............N....
B/Israel/Z1532/2014           .......A......................................................G.................A..............N....
B/Israel/Z1508/2014           .......A........................................................................A..............N....
B/Austria/784502/2014         .......A........................................................................A..............N....
B/Miyagi/21/2014              .......A................V.......................................................A..............N....
B/La Reunion/2072/2014        .......A........................................................................A..............N....
B/Gomel/1635/2014             .......A........................................................................A..............N....
B/South Africa/4437/2013      .......A.................N......................................................A..............N....
B/Wisconsin/1/2010            .................................................I..............Y..............................N....
B/Stockholm/12/2011           .................................................I..............Y......Q.............................
B/Novosibirsk/1/2012          ...............K.................................I..............Y..............................N....
B/Phuket/3073/2013            ...............K.................................I..............Y..............................X.X..
B/Austria/783231/2014         .....................K...........................I..............Y...............K..............N....
B/Brisbane/9/2014             ...............K.................................I..............Y..............................X....
B/St Petersburg/76/2014       ...............K.................................I..............Y..............................N....
B/Moscow/23/2014              ...............K.................................I..............Y..............................N....
B/Centre/1663/2014            ...............K.................................I..............Y......Q.......................N....
B/Canarias/1802/2014          ...............K.................................I..............Y......Q.......................N....
B/Iceland/78/2014             ...............K.................................I..............Y......Q.......................N....
B/Switzerland/10295792/2014   ...............K.................................I..............Y......Q.......................N....
B/Hong Kong/3417/2014         ...............K.................................I..............Y......Q.......................N....
B/Lisboa/4/2014               ...............K.................................I..............Y..............................N....
B/England/372/2014            ...............K.......A.........................I..............Y.................R............N....
B/Tochigi/14221/2014          ...............K.................................I..............Y..............................N....
B/Guangdong Liwan/1133/2014   ...............K.................................I..............Y..............................K....

                                   210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006              KNLYGDSNPQKFTSSANGVTTHYVSQIGSFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG
B/Brisbane/3/2007             ............................G.......................................................................
B/Estonia/55669/2011          ............................G.......................................................................
B/Massachusetts/02/2012       ............................G.......................................................................
B/Hong Kong/3577/2012         ............................G.......................................................................
B/Cameroon/1640/2014          ............................G.......................................................................
B/Israel/Z1532/2014           ............................G.......................................................................
B/Israel/Z1508/2014           ............................G.......................................................................
B/Austria/784502/2014         ............................G....R....................E.............................................
B/Miyagi/21/2014              ............................G....R..................................................................
B/La Reunion/2072/2014        ............................G....R..................................................................
B/Gomel/1635/2014             ............................G....R..................................................................
B/South Africa/4437/2013      ............................G.......................................................................
B/Wisconsin/1/2010            .S..........................D.......................................................................
B/Stockholm/12/2011           ............................D.....................V.................................................
B/Novosibirsk/1/2012          .S..........................D.....................................................................E..
B/Phuket/3073/2013 E4         .S..........................D.....................................................................E..
B/Austria/783231/2014         ............................D.....................................................................E..
B/Brisbane/9/2014 E4E1        .S..........................D.....................................................................E..
B/St Petersburg/76/2014       .S..........................D.....................................................................E..
B/Moscow/23/2014              .S..........................D.....................................................................E..
B/Centre/1663/2014            .S..........................D.....................................................................E..
B/Canarias/1802/2014          .S..........................D.....................................................................E..
B/Iceland/78/2014             .S..........................D.....................................................................E..
B/Switzerland/10295792/2014   .S..........................D.....................V...............................................E..
B/Hong Kong/3417/2014         .S..........................D.....................................................................E..
B/Lisboa/4/2014               .S..........................D.....................V...............................................E..
B/England/372/2014            .S..........................D.....................V...............................................E..
B/Tochigi/14221/2014          .S..........................D.....................................................................E..
B/Guangdong Liwan/1133/2014   .S..........................G.....................................................................E..
```

	310 320 330 340 350 360 370 380 390 400
	....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006	GLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Brisbane/3/2007	....
B/Estonia/55669/2011	....
B/Massachusetts/02/2012	....
B/Hong Kong/3577/2012	....
B/Cameroon/1640/2014	....
B/Israel/Z1532/2014	....
B/Israel/Z1508/2014	....
B/Austria/784502/2014	....
B/Miyagi/21/2014	....
B/La Reunion/2072/2014	....
B/Gomel/1635/2014	....
B/South Africa/4437/2013	....
B/Wisconsin/1/2010	....
B/Stockholm/12/2011	....
B/Novosibirsk/1/2012	...........K..
B/Phuket/3073/2013 E4	...........K..
B/Austria/783231/2014	...........K..
B/Brisbane/9/2014 E4E1	...........K..
B/St Petersburg/76/2014	...........K..
B/Moscow/23/2014	...........K..
B/Centre/1663/2014	...........K..
B/Canarias/1802/2014	...........K..
B/Iceland/78/2014	...........K..
B/Switzerland/10295792/2014	...........K..
B/Hong Kong/3417/2014	...........K..
B/Lisboa/4/2014	...........K..
B/England/372/2014	...........K..
B/Tochigi/14221/2014	...........K..
B/Guangdong Liwan/1133/2014	...........K..

	410 420 430 440 450 460 470 480 490 500
	....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006	LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIAA
B/Brisbane/3/2007	....D......................
B/Estonia/55669/2011	....D......................
B/Massachusetts/02/2012	....D......................
B/Hong Kong/3577/2012	....D......................
B/Cameroon/1640/2014	....D......................
B/Israel/Z1532/2014	....D......................
B/Israel/Z1508/2014	....D......................
B/Austria/784502/2014	....D......................
B/Miyagi/21/2014	....D......................
B/La Reunion/2072/2014	....D......................
B/Gomel/1635/2014	....D......................
B/South Africa/4437/2013	....D......................
B/Wisconsin/1/2010	....D......................
B/Stockholm/12/2011	....D......................
B/Novosibirsk/1/2012	....D......................
B/Phuket/3073/2013 E4	....D......................
B/Austria/783231/2014	....D......................
B/Brisbane/9/2014 E4E1	....D......................
B/St Petersburg/76/2014	....D......................
B/Moscow/23/2014	....D......................
B/Centre/1663/2014	....D......................
B/Canarias/1802/2014	....D......................
B/Iceland/78/2014	....D......................
B/Switzerland/10295792/2014	....D......................
B/Hong Kong/3417/2014	....D......................
B/Lisboa/4/2014	....D......................
B/England/372/2014	....D......................
B/Tochigi/14221/2014	....D......................
B/Guangdong Liwan/1133/2014	....D..................K...

	510 520 530 540 550 560
	....|....|....|....|....|....|....|....|....|....|....|....|....|....
B/Florida/4/2006	GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Brisbane/3/2007	....
B/Estonia/55669/2011	....
B/Massachusetts/02/2012	....
B/Hong Kong/3577/2012	....
B/Cameroon/1640/2014	....
B/Israel/Z1532/2014	....
B/Israel/Z1508/2014	....
B/Austria/784502/2014	........................N..
B/Miyagi/21/2014	....
B/La Reunion/2072/2014	....
B/Gomel/1635/2014	....
B/South Africa/4437/2013	....
B/Wisconsin/1/2010	....
B/Stockholm/12/2011	....
B/Novosibirsk/1/2012	....
B/Phuket/3073/2013 E4	....
B/Austria/783231/2014	....
B/Brisbane/9/2014 E4E1	....
B/St Petersburg/76/2014	....
B/Moscow/23/2014	....
B/Centre/1663/2014	....
B/Canarias/1802/2014	....
B/Iceland/78/2014	....
B/Switzerland/10295792/2014	....
B/Hong Kong/3417/2014	....
B/Lisboa/4/2014	....
B/England/372/2014	....
B/Tochigi/14221/2014	....
B/Guangdong Liwan/1133/2014	....

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 22. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

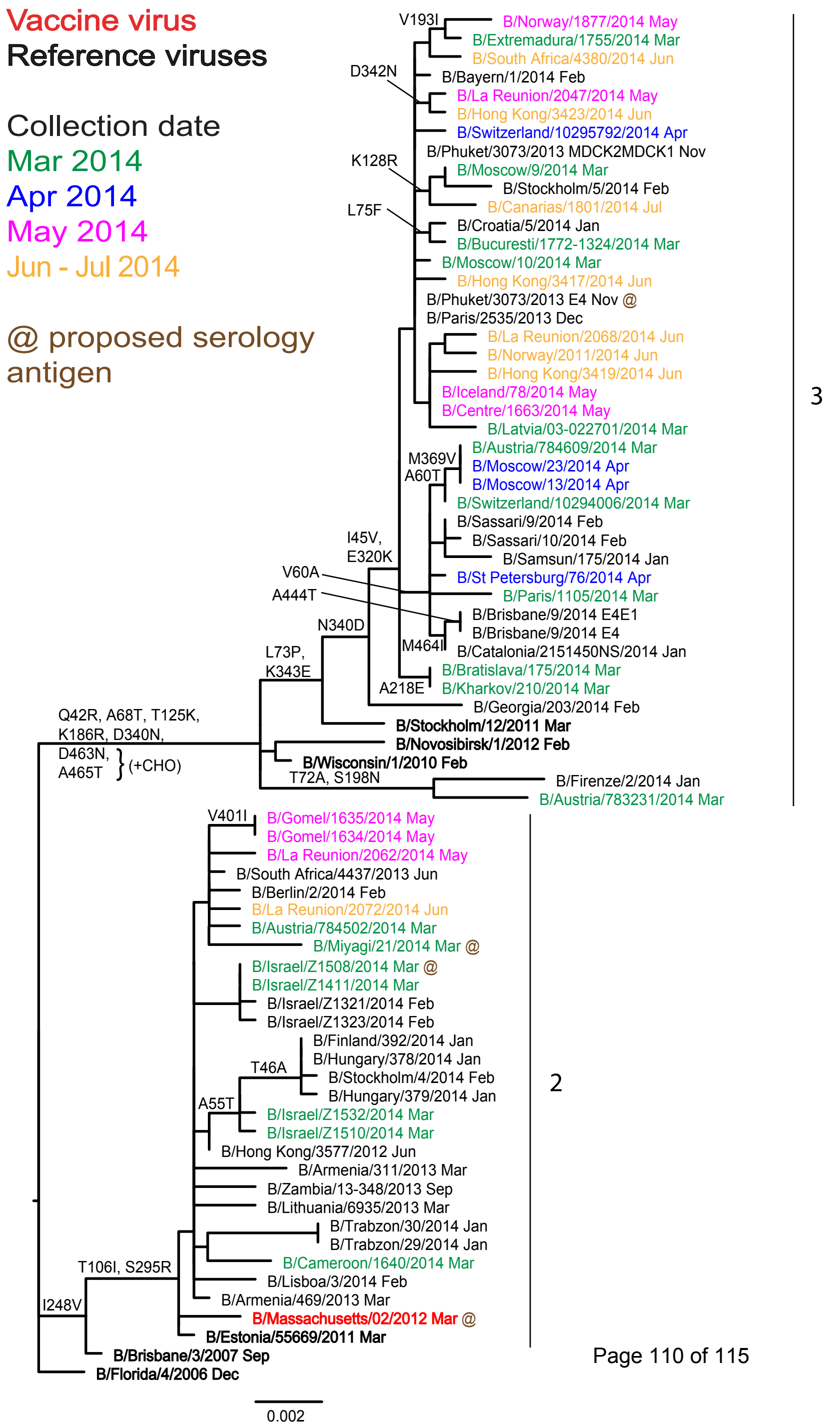

Figure 23. NA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

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                                     10        20        30        40        50        60        70        80        90       100
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Brisbane/3/2007            .......K............................................................................................ 2
B/Estonia/55669/2011         .................................................................................................... 2
B/Massachusetts/02/2012      ................................................................H................................... 2
B/Hong Kong/3577/2012        .................................................................................................... 2
B/Cameroon/1640/2014         .................................................................................................... 2
B/Israel/Z1532/2014          ......................................................T............................................. 2
B/Israel/Z1508/2014          .................................................................................................... 2
B/Austria/784502/2014        .............................................................................................R...... 2
B/Miyagi/21/2014             ................................................................H.........................I......... 2
B/La Reunion/2072/2014       .................................................................................................... 2
B/Gomel/1635/2014            .................................................................................................... 2
B/South Africa/4380/2014     .........................................R..V......................T....P........................... 2r
B/Wisconsin/1/2010           .........................................R.........................T................................ 3
B/Stockholm/12/2011          .........................................R.........................T....P........................... 3
B/Novosibirsk/1/2012         .........................................R.........................T................................ 3
B/Phuket/3073/2013           .........................................R..V......................T....P........................... 3
B/Austria/783231/2014        .........................................R.........................T...A............................ 3
B/Brisbane/9/2014            .........................................R..V..............A.......T....P........................... 3
B/St Petersburg/76/2014      .........................................R..V..............A.......T....P........................... 3
B/Moscow/23/2014             .........................................R..V..............T.......T....P........................... 3
B/Centre/1663/2014           .........................................R..V......................T....P........................... 3
B/Canarias/1801/2014         .........................................R..V......................T....P........................... 3
B/Iceland/78/2014            .........................................R..V......................T....P........................... 3
B/Switzerland/10295792/2014  .........................................R..V......................T....P........................... 3
B/Hong Kong/3417/2014        .........................................R..V......................T....P........................... 3
B/Lisboa/4/2014              .........................................P..T...TV.................T..L............................. 3V
B/England/372/2014           .........................................P..T...TV.................T..L............................. 3V
B/Tochigi/14221/2014         .........................................P......T..................T................................ 3V
A/Guangdong Liwan/1133/2014  .................................L......PS......T..................T................................ 3V

                                    110       120       130       140       150       160       170       180       190       200
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             HRFGETKGNSAPLIIREPFIACGPTECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDSNA
B/Brisbane/3/2007            ....................................................................................................
B/Estonia/55669/2011         .....I..............................................................................................
B/Massachusetts/02/2012      .....I..............................................................................................
B/Hong Kong/3577/2012        .....I..............................................................................................
B/Cameroon/1640/2014         .....I.....................R........................................................................
B/Israel/Z1532/2014          .....I..............................................................................................
B/Israel/Z1508/2014          .....I..............................................................................................
B/Austria/784502/2014        .....I..............................................................................................
B/Miyagi/21/2014             .....I..............................................................................................
B/La Reunion/2072/2014       .....I..............................................................................................
B/Gomel/1635/2014            .....I..............................................................................................
B/South Africa/4380/2014     ........................K............................................................R..............
B/Wisconsin/1/2010           ........................K............................................................R..............
B/Stockholm/12/2011          ........................K............................................................R..............
B/Novosibirsk/1/2012         ........................K............................................................R..............
B/Phuket/3073/2013           ........................K............................................................R..............
B/Austria/783231/2014        ........................K............................................................R...........N..
B/Brisbane/9/2014            ........................K............................................................R..............
B/St Petersburg/76/2014      ........................R............................................................R..............
B/Moscow/23/2014             ........................K............................................................R..............
B/Centre/1663/2014           ........................K............................................................R..............
B/Canarias/1801/2014         ........................K..R.........................................................R..............
B/Iceland/78/2014            ........................K............................................................R..............
B/Switzerland/10295792/2014  ........................K............................................................R..............
B/Hong Kong/3417/2014        ........................K............................................................R..............
B/Lisboa/4/2014              ........................N......................G.................................................N..
B/England/372/2014           ........................N......................G.................................................N..
B/Tochigi/14221/2014         ........................N......................G.................................................N..
A/Guangdong Liwan/1133/2014  ........................K......................G.................................................N..

                                    210       220       230       240       250       260       270       280       290       300
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Brisbane/3/2007            ...............................................V....................................................
B/Estonia/55669/2011         ...............................................V..............................................R.....
B/Massachusetts/02/2012      ...............................................V..............................................R.....
B/Hong Kong/3577/2012        ...............................................V..............................................R.....
B/Cameroon/1640/2014         ...............................................V..............................................R.....
B/Israel/Z1532/2014          ...............................................V..............................................R.....
B/Israel/Z1508/2014          ...............................................V..............................................R.....
B/Austria/784502/2014        ...............................................V..............................................R.....
B/Miyagi/21/2014             ...............................................V..............................................R.....
B/La Reunion/2072/2014       ...............................................V..............................................R.....
B/Gomel/1635/2014            ...............................................V..............................................R.....
B/South Africa/4380/2014     ....................................................................................................
B/Wisconsin/1/2010           ....................................................................................................
B/Stockholm/12/2011          ....................................................................................................
B/Novosibirsk/1/2012         ....................................................................................................
B/Phuket/3073/2013           ....................................................................................................
B/Austria/783231/2014        ..............................................................................................N.....
B/Brisbane/9/2014            ....................................................................................................
B/St Petersburg/76/2014      ....................................................................................................
B/Moscow/23/2014             ....................................................................................................
B/Centre/1663/2014           ....................................................................................................
B/Canarias/1801/2014         ....................................................................................................
B/Iceland/78/2014            ....................................................................................................
B/Switzerland/10295792/2014  ....................................................................................................
B/Hong Kong/3417/2014        ....................................................................................................
B/Lisboa/4/2014              ...V..............NK..............N........S...V..............................................R.....
B/England/372/2014           ...V..............NK..............N........S...V..............................................R.....
B/Tochigi/14221/2014         ...V..............NK..............N........S...V..............................................R.....
A/Guangdong Liwan/1133/2014  ..................N...............N....V...S...V.............M........TQ............................
```

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                                        310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                PFVKLNVETDTAEIRLMCTETYLDTPRPNDGSITGPCESDGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Brisbane/3/2007               ....................................................................................................
B/Estonia/55669/2011            ....................................................................................................
B/Massachusetts/02/2012         ....................................................................................................
B/Hong Kong/3577/2012           ....................................................................................................
B/Cameroon/1640/2014            ....................................................................................................
B/Israel/Z1532/2014             ....................................................................................................
B/Israel/Z1508/2014             ....................................................................................................
B/Austria/784502/2014           ....................................................................................................
B/Miyagi/21/2014                ....................................................................................................
B/La Reunion/2072/2014          .................................................................................R..................
B/Gomel/1635/2014               ..........................................................................................X.........
B/South Africa/4380/2014        ...................K......................E.........................................................
B/Wisconsin/1/2010              .......................................N............................................................
B/Stockholm/12/2011             .......................................N..E.........................................................
B/Novosibirsk/1/2012            .......................................N............................................................
B/Phuket/3073/2013              ...................K......................E.........................................................
B/Austria/783231/2014           .......................................N....N.......................................................
B/Brisbane/9/2014               ...................K......................E.........................................................
B/St Petersburg/76/2014         ...................K......................E.........................................................
B/Moscow/23/2014                ...................K......................E.........................V...............................
B/Centre/1663/2014              ...................K......................E.........................................................
B/Canarias/1801/2014            ............D......K......................E.........................................................
B/Iceland/78/2014               ...................K......................E.........................................................
B/Switzerland/10295792/2014     ...................K......................E.............T...........................................
B/Hong Kong/3417/2014           ...................K......................E.........................................................
B/Lisboa/4/2014                 ...................D......................E..............K..............E...............A..D...F....
B/England/372/2014              ...................D......................E..............K..............E...............A..D...F....
B/Tochigi/14221/2014            ........N..........D......................E........L.....K..............E...............A..D..VF....
A/Guangdong Liwan/1133/2014     ...................K...................N.G..............................E.K........N....A..D...F....

                                        410       420       430       440       450       460
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Florida/4/2006                VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVDMAL
B/Brisbane/3/2007               ..................................................................
B/Estonia/55669/2011            ..................................................................
B/Massachusetts/02/2012         ..................................................................
B/Hong Kong/3577/2012           ..................................................................
B/Cameroon/1640/2014            ..T...............................................................
B/Israel/Z1532/2014             ..................................................................
B/Israel/Z1508/2014             ..................................................................
B/Austria/784502/2014           ..................................................................
B/Miyagi/21/2014                ..................................................................
B/La Reunion/2072/2014          ..................................................................
B/Gomel/1635/2014               I.................................................................
B/South Africa/4380/2014        ..............................................................N.T.
B/Wisconsin/1/2010              ..............................................................N.T.
B/Stockholm/12/2011             ..............................................................N.T.
B/Novosibirsk/1/2012            ..............................................................N.T.
B/Phuket/3073/2013              ..............................................................N.T.
B/Austria/783231/2014           ..............................................................N.T.
B/Brisbane/9/2014               ...........................................T..................NIT.
B/St Petersburg/76/2014         ..............................................................N.T.
B/Moscow/23/2014                ..............................................................N.T.
B/Centre/1663/2014              ..............................................................N.T.
B/Canarias/1801/2014            ..............................................................N.T.
B/Iceland/78/2014               ..............................................................N.T.
B/Switzerland/10295792/2014     ..............................................................N.T.
B/Hong Kong/3417/2014           ...........................................V..................N.T.
B/Lisboa/4/2014                 ...K...............................E..............................
B/England/372/2014              ...K...............................E..........................N...
B/Tochigi/14221/2014            ..VK...............................E..............................
A/Guangdong Liwan/1133/2014     I..................................E..........................N.T.
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Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 24. Phylogenetic comparison of influenza A(H1N1)pdm09 M genes

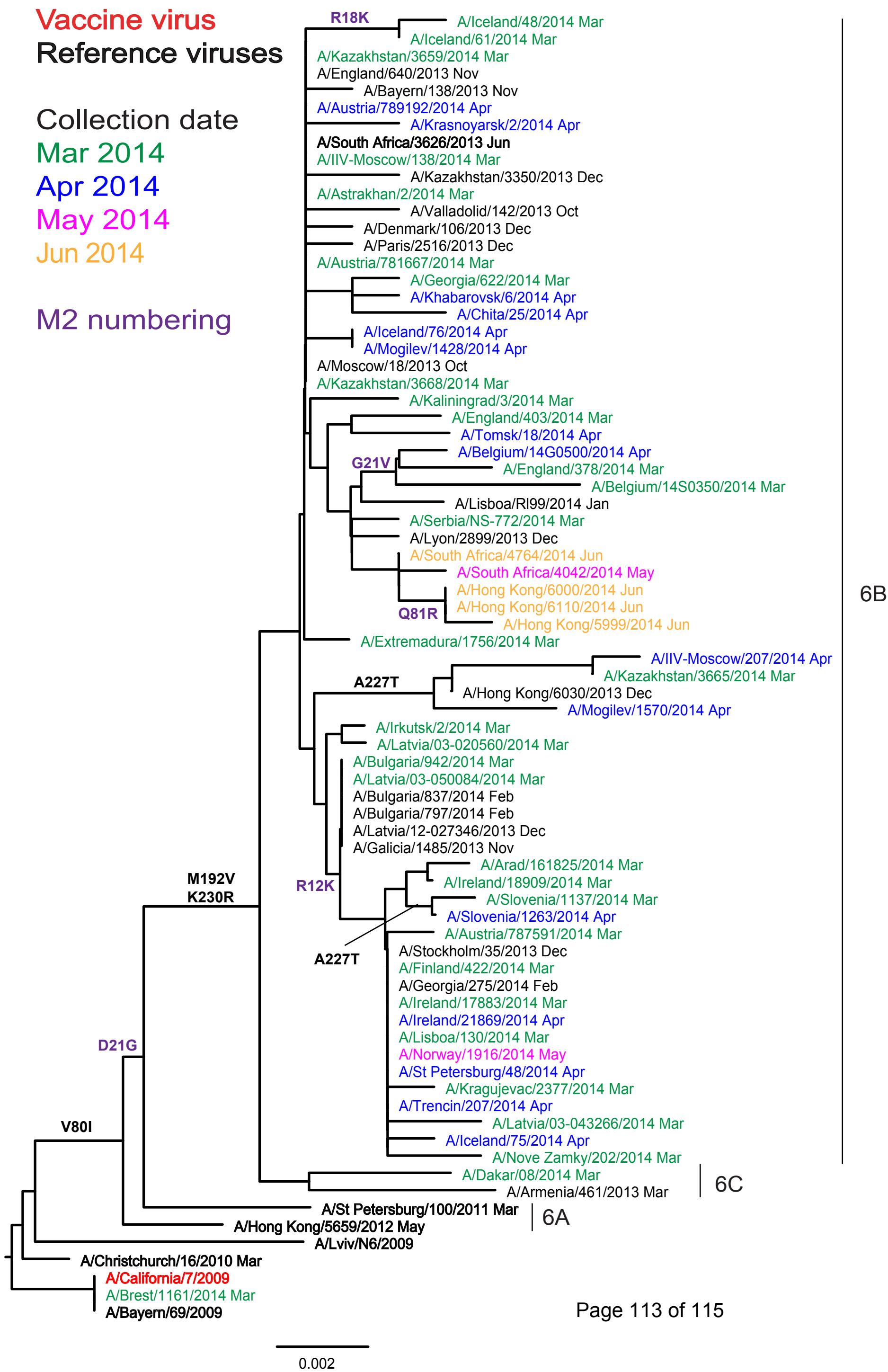

Figure 25. Phylogenetic comparison of influenza A(H3N2) M genes

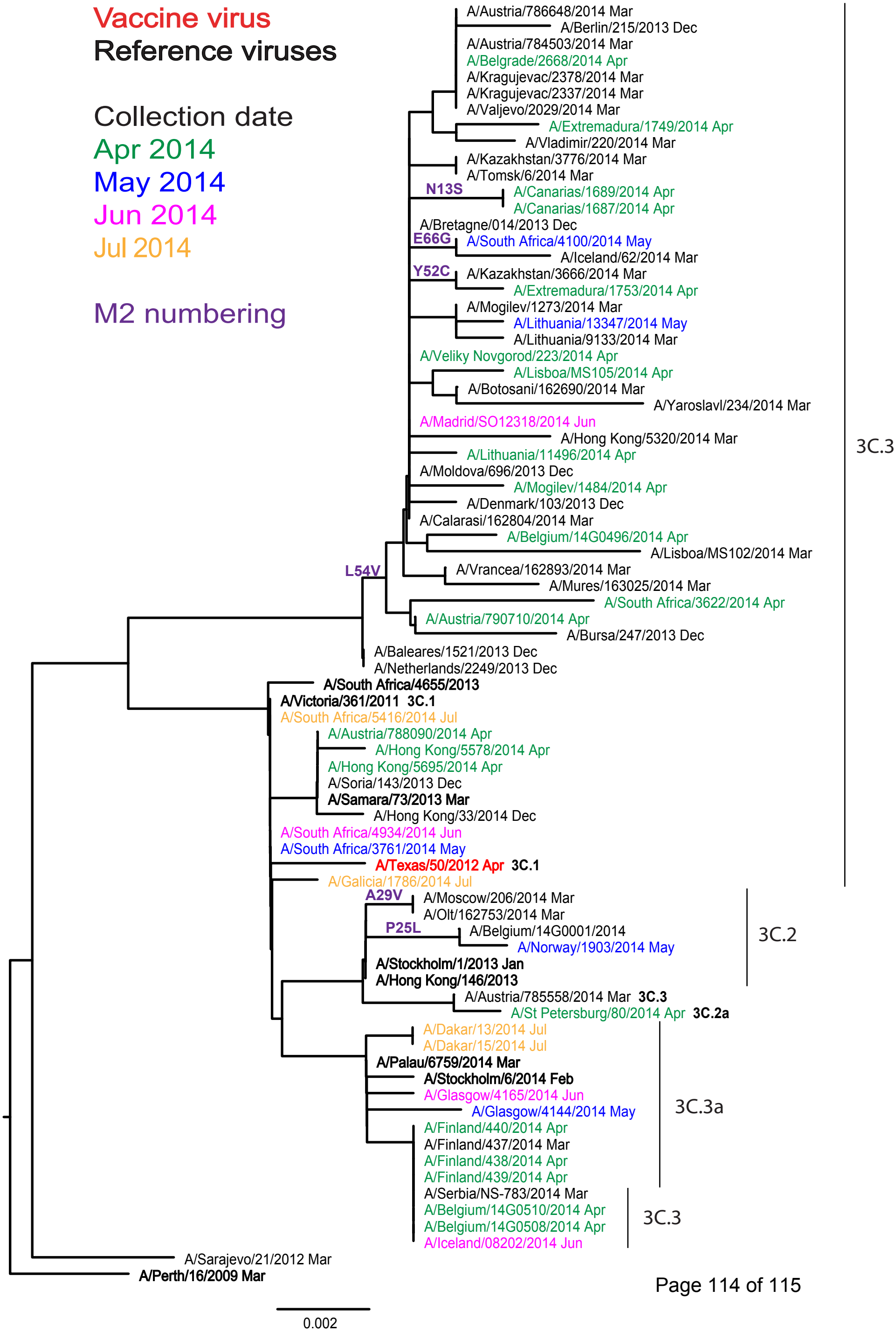

Table 14. Antiviral (NAI) testing of isolates with collection dates in 2013 and 2014

Month of Collection	Number of viruses tested						
	A(H1N1)pdm09		A(H3N2)		Flu B		Total
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2013							
January	202	2[a]	101		109		414
February	122		55		143		320
March	93		72		162		327
April	23		23		85		131
May	24		2		17		43
June	9		10		5		24
July	2		8		2		12
August	8		8		3		19
September	7		14		7		28
October	14		28		15		57
November	42		48		19		109
December	80	1[a]	182	2[d,e]	33		298
Total	626	3	551	2	600	0	1782
2014							
January	128	1[b] + 1	177	1[f]	14		322
February	108	1[c]	128	3[g,h,i]	18		258
March	53		87	1[j]	22		163
April	23		46		7		76
May	3		9		13		25
June	8		8		11		27
July	2	1[k]	3		5		11
Since 2014-01-31	197	2 (1.0%)	281	4 (1.4%)	76	0	560
Total	325	4	458	5	90	0	882

Viruses were subjected to phenotypic testing against oseltamivir and zanamivir

a A(H1N1)pdm09 virus (A/Austria/713625/2013) carries H275Y	Fold difference = os 939 zan 6	
a A(H1N1)pdm09 virus (A/IIV-Moscow/34/2013) carries H275Y	Fold difference = os 472 zan 1	
a A(H1N1)pdm09 virus (A/Catalonia/6634S/2013) carries H275Y	Fold difference = os 504 zan 2	
b A(H1N1)pdm09 virus (A/Belgium/14G0006/2014) carries I223R	Fold difference = os 40 zan 4	
c A(H1N1)pdm09 virus (A/Cyprus/F79/2014) carries S247N	Fold difference = os 7 zan 4	Outlier
d A(H3N2) virus (A/England/651/2013) carries E119V	Fold difference = os 7 zan 3	Outlier
e A(H3N2) virus (A/Norway/53/2014) carries N329K	Fold difference = os 10 zan 11	
f A(H3N2) virus (A/Serbia/NS-613/2014) carries Q391H	Fold difference = os 8 zan 9	Outlier
g A(H3N2) virus (A/Milano/84/2014) carries V215I and S331R	Fold difference = os 10 zan 8	
h A(H3N2) virus (A/Ukraine/86/2014) carries S331R	Fold difference = os 10 zan 8	
i A(H3N2) virus (A/Stockholm/14/2014) carries I194V, S331R and E381G	Fold difference = os 26 zan 9	
j A(H3N2) virus (A/Calarasi/162804/2014) carries S331R and D339N	Fold difference = os 17 zan 1	
k A(H1N1)pdm09 virus (A/Paris/1823/2014) carries H275Y	Fold difference = os 446 zan 3	
A(H1N1)pdm09 virus (A/Hungary/366/2014) carries H275Y	Not tested	